Part 1 Proposal Forms and Documents

PREFACE

This Statement of Work (SOW) consists of five parts.

Part 1 Contains typical contractual forms, procedures, bidding instructions, bond information, clauses and wage decisions.

Part 2 Contains general contract administrative and execution requirements including, but not limited to safety, design criteria & process, quality control, security, schedule, invoicing, temporary facilities, and design and construction oversight processes.

Part 3 Lists the project requirements, specific scope items, background project information, references, and expected quality level above and beyond those outlined in Part 4. Part 3 will be provided with the specific Task Order. Within Part 3 section 5.0 ROOM REQUIREMENTS provides detailed requirements on a room by room basis that further defines requirements that are in addition to the ENGINEERING SYSTEMS REQUIREMENTS SECTION.

Part 4 Contains Performance Specifications and minimum quality requirements.

Part 5 Contains prescriptive specifications, contains necessary edited UFGS. The scheduling and safety spec are included here.

Part 6 includes all 100% full design attachments which are:

- DRAWINGS
- SPECIFICATIONS
- BASIS OF DESIGN
- REPORTS
- PSC AND FASCICOLO DELL'OPERA
- SPECIFIC THIRD PARTY AGENCIES APPROVALS DOCUMENTS;
- PDC (PERMIT TO BUILD-PERMESSO DI COSTRUIRE) APPROVED
- MOU AND LICENSE SIGNED DOCUMENTS
- UTILITIES LINES RIGHT OF WAY ACT BETWEEN COMUNE DI LONGARE AND PRIVATES FOR NEW TOC PIPE INSTALLATION
- LETTER OF WAIVER TO TAX PAYMENT BY US GOV AND BY CONTRACTOR WORKING FOR US GOV PROJECTS
- VI.ABILITA' APPROVAL FOR TOC CROSSING UNDER RIVIERA BERICA

TABLE OF CONTENTS (Parts 2-5)

PART 5 – PRESCRIPTIVE SPECIFICATIONS

Part 2 General Requirements

TABLE OF CONTENTS

00 00 00 GENERAL REQUIREMENTS

1.1 Definitions

As used throughout the SOW, the following terms have the meaning set forth below:

Contracting Officer (KO): The individual designated to administer the contract. Throughout this contract, this individual will be responsible and possess the authority to act on behalf of the Government with respect to the specific contract.

Contracting Officer’s Representative (COR): The individual designated by the Contracting Officer as the authorized representative of the KO. The COR is responsible for monitoring performance and technical management of the effort required and should be contacted regarding questions or problems of a technical nature.

Contract: Contract or task order.

Contractor: The term Contractor refers to both the Prime Contractor and all subcontractors, whether in contract with the Prime Contractor or other subcontractors at any tier, including the Designer of Record.

Designer of Record (DOR): Licensed architect/engineer working as subcontractor to or partner with Prime Contractor who provides design for this contract.

Quality Control (QC): Contractor's system to control the quality of design, material, equipment and construction.

Quality Assurance (QA) Program: Government's program to evaluate the effectiveness of the Contractor's quality control. The Government's QA Program is not a substitute for the Contractor's QC Program.

Holidays:

U.S. HOLIDAYS	DATE
New Year's Day	1st January
Martin Luther King Jr.'s Birthday	3rd Monday in January
President's Day	3rd Monday in February
Memorial Day	Last Monday in May
Emancipation Day	19th June
Independence Day	4th July
Labor Day	1st Monday in September
Columbus Day	2nd Monday in October
Veterans' Day	11th November
Thanksgiving Day	4th Thursday in November
Christmas Day	25th December

ITALIAN HOLIDAYS	DATE
New Year's Day	1 January
Epiphany Day	6 January
Easter Monday	Varies yearly
Liberation Day	25 April
Labor Day	1 May
Republic Day	2 June
Assumption Day	15 August
Local Holiday	8 September
San Francesco Day	4 October
All Saints' Day	1November
Immaculate Conception Day	8 December
Christmas Day	25 December
Saint Stephen's Day	26 December

1.2 Accessibility

Provide barrier-free design in accordance with UFC 1-200-01, DoD Building Code.

01 11 00 SUMMARY OF WORK

1.1 Project Description

Work will be provided in Part 3 as a separate Task Order.

1.2 Protection of Government Property

Take special care to protect Government Property. Return areas damaged as a result of construction under this contract to their original condition. In addition to FAR 52.236-9, Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements, perform the following:

a. Remove or alter existing work or facilities in such a manner as to prevent injury or damage to any portion of the existing work or facilities that remain.

b. Repair or replace portions of existing work altered during construction operations to match existing or adjoining work, as approved by the COR. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

c. Preserve the natural resources in accordance with the approved environmental protection plan prepared in accordance with UFGS 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

1.3 Government Furnished Material and Equipment

If applicable, the Government will furnish the materials and equipment for installation by the Contractor pursuant to contract clause FAR 52.245-2, Government Property (Fixed Price Contracts). Notify the COR in writing at least 15 calendar days before the materials and equipment are required. Pick up materials and equipment no later than 30 calendar days after such date. When materials and equipment are not picked up by the 30th day, the Contractor will be charged for storage at the prevailing rate. The COR will specify the location of materials and equipment and the delivery location.

01 14 00.05 20 WORK RESTRICTIONS

1.1 Existing Underground Utilities

The Contractor shall verify on-site utilities and have them marked out prior to the start of construction. Determine the elevations of existing utilities and underground obstructions indicated as existing to remain in locations to be traversed by new utilities and other work provided herein before new work is laid closer than the nearest manhole or other structure at which an adjustment in grade could be made. Where dig permits are required, obtain permits and notify the COR 14 calendar days prior to any excavation. Maintain all utility markings for the duration of the contract.

1.2 Hours of Operation

The hours of operation for routine work shall be 0800 thru 1700 hours Monday through Friday. Installation access hours 0600-2000 to allow contractor adequate time to access site to complete work during hours of operation Work shall not be performed on Italian and American holidays occurring during the normal work week unless prior approval by the KO or the COR. Work shall not be performed on local holidays occurring during the normal work week. When an American holiday occurs on a Saturday or a Sunday, the holiday is observed on the preceding Friday or following Monday, respectively. Work during other than normal hours shall not be performed without prior written approval of the COR/KO. The Contractor shall submit request for such approval to the COR and/or the KO at least 24 hours in advance. Major utility cutovers should occur after normal working hours or on Saturdays, Sundays, and holidays to the greatest extent possible.

1.3 Installation Access

The Contractor shall obtain Defense Biometric Identification System (DBIDS) installation passes for each employee. The Contractor shall acknowledge a minimum 30 days for processing and issuance of Installation Pass. The Contractor shall inform his personnel that they may be subjected to search when entering or leaving the installation at the discretion of the Installation Commander. Upon completion of the contract work or upon employee termination the Contractor shall return all passes to the issuing office.

All Contractor employees, including subcontractors, subcontractors' employees, suppliers, and suppliers' employees are required to comply with the Installation Security Requirements regarding personnel, vehicle, and equipment security passes and access the jobsite. Nothing in the contract is to be construed in any way to limit the authority of the Commanding Officer to prescribe new, or to enforce existing security regulations governing the admission or exclusion of persons and the conduct of persons while on the installation, including but not limited to, the rights to search of all persons or vehicles on the installation. Coordinate with the COR for specific security and access requirements.

Access to U.S. installations and controlled areas is limited to personnel who meet security criteria and are authorized by Host Nation law to work in that country. Failure to submit required information/data and obtain required documentation or clearances in accordance with AE Regulation 190-16, Installation Access Control, will be grounds for denying access to U.S. installations and controlled areas. The Contractor is responsible to ensure that any Subcontractor used in performance of this contract complies with these requirements and that all employees, of both the Contractor and any Subcontractor utilized by the contractor, are made aware of and comply with these requirements.

The Contractor is responsible for being aware of and complying with the requirements associated with Installation Access Control. The Government is not liable for any costs associated with performance delays due solely to a firm's failure to comply with Installation Access Control (IAC) processing requirements.

The Contractor is responsible for returning installation passes to the issuing Installation Access Control Office (IACO) when the contract is completed or when a contractor employee no longer requires access.

AE 190-16 can be found on the following website: https://www.aepubs.eur.army.mil/AE-Regulations/?smdsearch8659=AE%20Regulation%20190-16

Below is the responsible Organizational Sponsor & Installation Access Control Office for this contract:

Organizational Sponsor: DPW Construction Management Branch Chief

Installation Access Control Office:
Location: Installation Access Control Office (IACO) Caserma Ederle (BLDG 300A)
IACO Del Din Visitor Center (BLDG 112)
VMC IACO office email: usarmy.usag-italy.id-europe.list.access-controlbranch@army.mil

DSN Phone No: 646-5343 Commercial Phone No: 0444 717053

1) Required documents to be provided for Installation Pass included will be sent electronically to the designated individual. Information will be provided by the COR upon awarded.

   a. AE Form 190-16A Installation Pass Request (Forms are subject to update).
   b. Identification Card (Carta d'Identità in Italy)
   c. Sign-in Privilege Responsibilities (As Required)
      Prime Contractor will be allowed to have selected individuals with Sign-in Privileges. Individuals in key positions, Project Manager, Site Superintendent, Quality Control Manager and Site Safety Health Officer.
   d. AE Form 190-16F Access Roster (For access for activities less than 30 days no renewals will be accepted.)

2) Contractor employees must conduct themselves in a proper, efficient, courteous and businesslike manner. Remove from the site any individual whose continued employment is deemed by the COR to be contrary to the public interest or inconsistent with the best interests of National Security.

1.4 Access to Buildings/Occupied Buildings:

The Contractor may work in or around existing occupied buildings. The Contractor is responsible, via the COR, to obtain access to building and facilities and arrange for them to be opened and closed. Do not enter the building(s) without prior approval of the COR. Keep the existing buildings and their contents secure at all times. Provide temporary closures as required to maintain security. Provide dust covers or protective enclosures to protect existing work that remains, and Government material located in the building during the construction period. If required to perform work, the Contractor shall protect or otherwise move and replace the furniture in its original locations upon completion of the work. Contractor personnel will not be permitted in security-regulated buildings or areas unless cleared by the Security Officer.

1.5 AT Level 1 Training:

All contactor employees, to include subcontractor employees, requiring access to Army installations, facilities and controlled access areas shall complete AT Level 1 awareness training within 20 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee, to the COR or to the KO, if a COR is not assigned, within 30 calendar days after completion of training by all employees and subcontractor personnel. AT Level 1 awareness training is available at the following website: https://jkodirect.jten.mil/pdf/atl1/launch.html

1.6 iWATCH/iSALUTE Training:

The contractor and all associated subcontractors shall brief all employees on the local iWATCH/iSALUTE program (training standards provided by the requiring activity ATO). This locally developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training will be completed within 20 calendar days of contract award and within 30 calendar days of new employees commencing performance with the results reported to the COR no later than 20 days after contract award.

1.7 Security Requirements

All security requirements apply to all subcontractors and suppliers associated with this contract. Comply with the following:

a. Do not publicly disclose any information concerning any aspect of the materials or services relating to this contract, without prior written approval of the COR.

b. Do not disclose or cause to be disseminated any information concerning the operations of the activity's security or interrupt the continuity of its operations.

c. Do not disclose any information to any person not entitled to receive it. Failure to safeguard any classified information that may come to the Contractor or any person under his control, may subject the Contractor, his agents or employees to criminal liability under 18 U.S.C., Sections 793 and 798.

d. Direct to the Installation Security Officer, via the COR, for resolution all inquiries, comments or complaints arising from any matter observed, experienced, or learned as a result of or in connection with the performance of this contract, the resolution of which may require the dissemination of official information.

e. Coordinate photography requirements with the COR. Some areas restrict or prohibit photographing Government property.Military escorts may be required for certain facilities.

Deviations from or violations of any of the provisions of this paragraph, will, in addition to all other criminal and civil remedies provided by law, subject the Contractor to immediate termination for default and withdrawal of the Government's acceptance and approval of employment of the individuals involved.

1.AT Level I Training.

All contractor employees, including subcontractor employees, requiring access to Army installations, facilities, and controlled-access areas will complete AT Level I awareness training within 20 calendar days after the contract start-date or effective date of incorporation of this
requirement into the contract, whichever is applicable. The contractor will submit certificates of completion for each affected contractor employee and subcontractor employee to the COR or to the contracting officer, if a COR is not assigned, within 30 calendar days after training is completed by all employees and subcontractor personnel.
AT Level I awareness training is available online at
https://jkodirect.jten.mil/pdf/atl1/launch.html
Specific AOR training content may be directed by the USEUCOM Commander, with the USAREUR ATO being the local POC.

OPSEC Training: Per AR 530-1, Operations Security, the contractor employees must complete Level I OPSEC Awareness training. New employees must be trained within 30 calendar days of their reporting for duty and annually thereafter.

3. Access and General Protection/Security Policy and Procedures.

Contractor and all associated sub-contractors employees shall provide all information required for background checks to meet installation access requirements to be accomplished by installation provost marshal office, director of emergency services, or security office. Contractor workforce must comply with all personal identification verification requirements (FAR clause 52.204-9, Personal Identity Verification of Contractor Personnel) as directed by DOD, HQDA, and/or local policy. In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes.

3a. For contractors requiring common access cards (CACs).

Before CAC issuance, the contractor employee requires, at a minimum, a favorably adjudicated National Agency Check with Inquiries (NACI) or an equivalent or higher investigation in accordance with Army Directive 2014-05. The contractor employee will be issued a CAC only if duties involve one of the following: (1) Both physical access to a DOD facility and access, via logon, to DOD networks on-site or remotely; (2) Remote access, via logon, to a DOD network using DOD-approved remote access procedures; or (3) Physical access to multiple DOD facilities or multiple non-DOD federally controlled facilities on behalf of the DOD on a recurring basis for a period of 6 months or more. At the discretion of the sponsoring activity, an initial CAC may be issued based on a favorable review of the FBI fingerprint check and a successfully scheduled NACI at the Office of Personnel Management.

3b. For contractors that do not require CACs, but require access to a DOD facility or installation.

Contractor and all associated sub-contractors employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (TSDB) (Army Directive 2014-05/AR 190-13), applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by government representative), or at OCONUS locations, in accordance with status of forces agreements and other theater regulations.

01 20 00.05 20 PRICE AND PAYMENT PROCEDURES

1.1 Schedule of Prices

Submit a schedule of prices for approval. Include a detailed breakdown of the contract price with quantities and unit prices for each work activity or Definable Feature of Work (DFOW). Include general conditions, profit, and overhead costs within the unit prices. Break down into design and construction is required. No other requests for payment will be processed without an approved schedule of prices and baseline schedule.

1.2 Budget Management

The Contractor must be responsible for budget management throughout the entire project.

01 30 00.05 20 ADMINISTRATIVE REQUIREMENTS

1.1 Supervision

The Contractor must have a Project Superintendent fluent in English and host nation language on the job site during working hours. The Superintendent must have a minimum of 5 years of experience as a Superintendent on previous projects of similar size and complexity. The individual must be familiar with the requirements of EM 385-1-1 D.Lgs. 81/2008 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the Alternate Superintendent are the same as for the Project Superintendent. The KO may request proof of the Superintendent's qualifications at any point in the project if the performance of the Superintendent is in question. The Superintendent shall not serve as Quality Control Manager (QCM) or Site Safety and Health Officer (SSHO). The QCM may serve as SSHO unless otherwise specified in Part 3 Statement of Work / Project Program. Provide a Superintendent Resume prior to the Post Award Kick-off Meeting (PAK), for the proposed Superintendent describing experience with references and qualifications to the COR for approval. The COR reserves the right to interview the proposed Superintendent at any time in order to verify the submitted qualifications.

1.1.1 Duties

The Project Superintendent is primarily responsible for managing subcontractors and coordinating day-to-day production and schedule adherence on the project. The Superintendent is required to attend RedZone meetings, partnering meetings, and quality control meetings. The Superintendent or Alternate must be on site at all times during the performance of the contract until the work is completed and accepted.

1.1.2 Non-Compliance Actions

The Project Superintendent is subject to removal by the KO for non-compliance with requirements specified in the contract and for failure to manage the project to ensure timely completion. Furthermore, the KO may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

01 31 19.05 20 POST AWARD MEETINGS

1.1 Post Award Kick-off Meeting (PAK)

Prior to commencement of design, and within 30 calendar days from NTP, meet with the COR, and Government personnel. If mutually agreed upon by the Contractor and the COR, the PAK Meeting may be held concurrently with the Design Conference. The project team will develop a mutual understanding relative to the approved proposal, safety program, environmental permits and requirements, quality control procedures, and design and construction schedule. Contractor's personnel must attend including the Project Manager, Project Superintendent, Site Safety and Health Officer, Quality Control Manager, Design Quality Control Manager, Designer of Record, Design Safety

Coordinator, major subcontractors, and others as determined by the COR. If the Statement of Work includes work on any fire protection system, including fire alarm and mass notification systems, the Contractor and the DOR must meet with the Installation Fire Department to establish clear expectations of fire protection requirements of the project.

1.1.1 PAK Meeting Outcome

a. Integrate the Contractor and all client representatives into the project team.

b. Review the administrative requirements of the contract that are critical during the design phase.

c. Establish clear lines of communication and points of contact for Government and Contractor team members.

d. Review the project design schedule and design package requirement, design submittal packaging and preliminary construction schedule in accordance with Section 01 32 16.00
20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES. Discuss design milestones and events that will be included in the Quality Control Plan.

e. Establish clear expectations and schedules for facility turnover, providing DD Form 1354 asset management records, Third Party Certification, if applicable, and training of Government maintenance personnel

1.2 Informal Partnering

To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget and on schedule. The COR will organize initial and follow-up informal partnering sessions with project team key personnel, including Contractor's personnel and Government personnel. The initial session may be a part of the PAK Meeting.

1.3 Design Conferences

Contractor’s personnel is expected to attend Design Conferences. Conferences will be held remotely or in person as specified in Part 3 Statement of Work / Project Program. Contractor shall be responsible for providing meeting minutes in English to the COR within 7 days of each design conference. The initial design conference may require the Contractor to visit the site and conduct extensive interviews, and site surveying, conduct problem solving discussions with the individual users and installation personnel to acquire all necessary site information, review user operations and discuss user needs.

1.4 Pre-Construction Conference

Prior to construction or demolition work start, meet with COR to discuss and develop mutual understanding relative to administration of the safety programs, environmental issues, safety of building occupants and surrounding area, hazardous materials, waste disposal, construction QC procedures, construction schedule, labor provisions and other construction phase contract procedures. The final design must be approved before the preconstruction conference. Contractor shall acknowledge that construction mobilization is subject to the release of the "Notifica Preliminare". “Notifica Preliminare” is typically issued 30 days after submittal date.

1.5 Red Zone Meeting (RZ)

Participate in Red Zone (RZ) meetings with the COR to identify strategies to ensure the project is carried to expeditious closure and turnover to the client. Discuss any progress changes, critical work activities and their planned completion dates during regularly scheduled RZ Meetings beginning approximately at 75% stage of construction prior to project completion.

01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES

Develop and implement a design and construction schedule in accordance with Part 5 Section 01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULE.

01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES

1.1 Construction submittal approvals are listed below:

a. DOR Approval required for:
    1. Fire Protection related submittals;
    2. Earthwork and Geotechnical submittals;
    3. All components of the roof and building enclosure;
    4. All components of the interior doors, hardware, cabinets, fireproofing/firestopping;
    5. Structural engineering related submittals;
    6. HVAC Testing, Adjusting, and Balancing;
    7. Telecommunications;
    8. SID submittals and Furnishings, Fixtures, and Equipment (FF&E) packages.

b. Government Approval required for:
    1. Government approved submittals as required by UFGS specification sections edited by the DOR;
    2. Substations;
    3. Transformers;
    4. Medium Voltage Switchgear;
    5. Medium Voltage Cable;
    6. 400-Hz Converters;
    7. Emergency Generators;
    8. Automatic Transfer Switches;
    9. Uninterruptible Power Supplies;
    10. Electronic Security Systems;

11. Large mechanical equipment: Chillers, Cooling Towers, Air Handling Units;
12. Section 23 09 23.02 BACnet Direct Digital Control for HVAC and Other Building Control Systems;
13. Section 23 05 93 Testing, Adjusting, and Balancing for HVAC;
14. Railings, paint, finish materials/colors. Features shall be in continuity with the Installation Planning Standard (IPS);
15. Furniture, Fixture and Equipment (FF&E).

Government review or approval of any portion of any submittal does not relieve the Contractor from responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms to the requirements of the contract documents. Furthermore, Government review or approval of a submittal is not to be construed as a complete check.

1.2 Submit the following construction submittals, approved by the DOR, to the Government for surveillance:

a. Submit fire protection related submittals pertaining to spray-applied fire proofing and fire stopping, exterior fire alarm reporting systems, interior fire alarm & detection systems, and fire suppression systems including fire pumps and standpipe systems.

b. Submit geotechnical related submittals pertaining to the soils investigations (reports and soils analysis), foundations (shallow and deep), pavements structure design, test pile and production pile testing and installation.

c. Submit roofing submittals pertaining to materials and systems used to make up the roof system.

d. Submit telecommunications related shop drawings for coordination with the 509th Signal Battalion.

e. Submit Performance Verification and Acceptance Testing required by International Building Code (IBC), Local and Italian National Code.

f. For any pre-engineered buildings, submit shop drawings showing engineering data and complete building drawings, signed and sealed by a registered professional engineer.

g. Comply with sustainability submittals per UFGS section 01 33 29 Sustainability Requirements and Reporting.

h. Submit Structural Shop Drawings and structural product data.

i. Submit the design safety plan “Piano di Sicurezza e Coordinamento” as per D.Lgs. 81/2008.

1.3 Construction Submittal Process

The Contractor Quality Control Manager (QCM) and, when required, the Designer of Record (DOR), are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. The Contractor shall submit to the COR the design drawings, appropriate connection, fabrication, layout, and product specific drawings. Provide to the Government submittals as listed.

a. QC Plan, prior to the Initial Design Conference. Design and construction will be permitted to begin only after acceptance of the QC Plan or acceptance of an interim plan applicable to the particular feature of work to be started.

b. Pre-Construction submittals (SD-01), prior to construction, approved in accordance with QC Plan. The QCM is the approving authority for submittals unless otherwise indicated.

c. Sustainability Action Plan, and Sustainability Notebook when required, in accordance with UFGS section 01 33 29 Sustainability Requirements and Reporting.

d. DOR approved construction submittals identified for Government surveillance; typically Fire Protection system and Life Safety submittals. Stamp the submittals "FOR SURVEILLANCE ONLY." Submit Surveillance submittals to the Government prior to starting work for that item. Submittals required for surveillance will be returned only if corrective actions are required.

e. Safety Data Sheets (SDS) as applicable.

f. Material Safety Data Sheet (MSDS).

g. Baseline Schedule: due prior to PAK.

h. SSHO: prior to the PAK, provide proof that the Site Safety Health Officer meets the requirement of EM 385-1-1 or host nation safety requirements.

i. Environmental Protection Plan: due prior to start of the work.

j. Contractor Safety Self-Evaluation Checklist.

k. Accident Reports. Submit if incidence occurs.

l. Accident Prevention Plan and Activity Hazard Analysis. Submit prior to construction.

m. Schedule of Prices, initial is due at the PAK and updated schedules are due at each design submission. Final schedule of prices is due prior to construction.

n. Record drawings submittal requirements are listed in Part 5 Section 01 78 00 CLOSEOUT SUBMITTALS.

o. Operation and Maintenance Information submittal requirements are listed in Part 5 Section 01 78 23 OPERATION AND MAINTENANCE DATA.

p. Licenses and Permits: Per this section and Part 4.

q. Completed High Performance and Sustainable Building (HPSB) compliance checklists in accordance with UFC 1-200-02, High Performance and Sustainable

Building Requirements and UFGS Section 01 33 29 Sustainability Requirements and Reporting. Attach completed HPSB Checklist to Interim DD 1354.

r. DD Form 1354:

Assist the COR in preparing Draft and Interim DD Form 1354, TRANSFER AND ACCEPTANCE OF MILITARY REAL PROPERTY, by providing quantities, unit of measurements, and costs in accordance with UFC 1-300-08. Quantities and units of measure shall be in accordance with the current version of the Real Property Categorization System, available upon request from the Real Property Office.

1. Draft DD Form 1354: Assist the COR in determining applicable real property assets broken out by construction categories in order to submit a "Draft DD Form 1354" for Government review and approval. "Draft DD Form 1354" must include all quantities and units of measure, but does not require cost breakdown. Download the current blank editable DD Form 1354 in ADOBE (PDF) from the following web site: https://www.esd.whs.mil/Portals/54/Documents/DD/forms/dd/dd1354.pdf

2. Interim DD Form 1354: Assist the COR in providing any update to the Draft DD Form 1354 to include any additional assets, improvements, or alterations that occurred during construction. Use the Draft DD Form 1354 to identify costs. Submit Interim DD Form 1354 for Government review and approval 45 days prior to the Beneficial Occupancy Date (BOD). If modifications to the Interim DD Form 1354 are required by the Government, the corrected version must be submitted prior to the BOD.

   When documenting demolition work, the DD Form 1354 must list the quantitative data associated with this work in parenthesis to show the cost should be deleted and identified under each category code on the DD 1354. Coordinate with the Real Property Office to assist in determining the negative value for demolition work.

   Demolition portion of the project shall be accounted by indicating, as additional DD Form 1354 item numbers, all demolished real property facility numbers and category codes using negative numbers listed in parenthesis for units of measure, and in block 19.

   Facility is assigned by Real Property personnel and always after the DD 1354 is endorsed by RPAO. Contractor and COR turnover of the building is always to the Garrison via the Real Property Office and never to an individual tenant or Unit.

s. Construction Submittals:

1) In addition to the design submittals noted in Section 01 33 10.05 20, construction submittals will be required. The list of required submittals will be generated from the Contractor-prepared specifications and indicated in the submittal register.

2) Shop drawing submissions will be governed by UFGS 01 33 00.05 20 Construction Submittal Procedures, available on the Whole Building Design website www.wbdg.org.

Contractor must provide construction submittals to the COR and assume a 14 day review period. Submittals shall be submitted electronically only unless otherwise noted.

1.4 Operations and Maintenance Information (OMSI) SD-10

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel. Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item. Data incorporated in an operations and maintenance manual or control system. Provide the required operations and maintenance information in accordance with Part 5 Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.5 Record Drawings

The Contractor shall maintain at the job site one set of full-size prints of the contract drawings, accurately marked in red with adequate dimensions, to show all variations between the construction actually provided and that indicated or specified in the contract documents, including buried or concealed construction. Special attention shall be given to recording the horizontal and vertical location of all buried utilities that differ from the contract drawings. Existing utility lines and features revealed during the course of construction shall also be accurately located and dimensioned. Variations in the interior utility systems shall be clearly defined, dimensioned, and coordinated with exterior utility connections at the building five-foot line, where applicable. Existing topographic features which differ from those shown on the contract drawings shall also be accurately located and recorded. Where a choice of materials or methods is permitted herein, or where variations in scope or character of methods is permitted herein, or where variations in scope or character of work from that of the original contract are authorized, the drawings shall be marked to define the construction actually provided. The representations of such changes shall conform to standard drafting practice and shall include such supplementary notes, legends, and details as necessary to clearly portray the as-built construction.

These drawings shall be available for review by the Government at all times. Upon completion of the work, the set of the marked up prints shall be certified as correct, signed by the QCM, and delivered to the COR for approval before acceptance. Requests for partial payments will not be approved if the marked prints are not kept current. The asbuilt modifications must be accomplished by electronic drafting methods on the DWG design drawings to create a complete set of record drawings.

Provide a CAD drawing identical to each signed drawing that incorporate modifications to the as-built conditions. After all as-built conditions are recorded on the DWG files, produce a PDF file of record drawings in conformance with FC 1-300-09N. QCM electronic signatures or scanned signatures placed as images are required on record drawings. Request for final payment will not be approved until the record drawings are delivered to the Government requiring activity and accepted. Refer also to Part 5 Section 01 78 00 CLOSEOUT SUBMITTALS.

01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES

1.1 Summary

This section includes requirements for contractor originated design documents and design submittals. Refer to Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES for more specific requirements of this SOW.

1.2 Reference Standards

This SOW references published standards, the titles of which can be found in the Unified Master Reference List (UMRL) on the Whole Building Design Guide at the Unified Facilities Guide Specification (UFGS) Website. The publications referenced form a part of this specification to the extent referenced.

The advisory provisions of all codes, requirements, and standards are mandatory; substitute words such as "must" or "required" for words such as "shall", "should", "may", or "recommended," wherever they appear. The results of these wording substitutions incorporate these code and standard statements as requirements. Reference to the "authority having jurisdiction" is interpreted to mean KO or COR. Comply with the required and advisory portions of the current edition of the standard at the time of contract solicitation. Provide work in compliance with the following design standards and codes, as a minimum. Government standards listed in this SOW take precedence over industry standards. The following list of codes and standards is not comprehensive and is augmented by other codes and standards referenced and cross-referenced in the SOW. Refer to Parts 3 and 4 for specific requirements within other UFC's. Part 3 contains the project requirement, specific scope items, functional and performance requirements, background project information, and expected quality levels that exceed Part 4. Part 4 identifies design criteria, verification requirements, and performance and quality requirements of products.

a. U.S. Department of Defense (DoD) Unified Facilities Criteria (UFC)

UFC 1-200-01	DoD Building Code (General Building Requirements)
UFC 1-200-02	High Performance and Sustainable Building Requirements
UFC 1-300-08	Criteria for Transfer and Acceptance of DoD Real Property
FC 1-300-09N	Navy and Marine Corps Design Procedures
UFC 3-560-01	Electrical Safety, O&M
UFC 3-600-01	Fire Protection Engineering
UFC 4-010-06	Cybersecurity of Facility-Related Control Systems

UFC 1-200-01, DoD Building Code is a hub document that provides general building requirements and references other critical UFCs. A reference to UFC 1-200-01 requires compliance with the Tri-Service Core UFCs. Refer to the UFC 1-200-01 for a complete list of all applicable Tri-Service Core UFCs at website location http://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc. Contractor originated design documents must provide a project design that complies with the SOW, FC 1-300-09N, UFC 1-200-01, UFC 3-810-01N, the Core UFCs, and other UFC's listed above. UFC 1200-01 requires compliance with UFC 1-200-02, "High Performance and Sustainable Building Requirements".

01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS

Develop and implement a safety program in accordance with Part 5 Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENT.

01 45 00.10.20 QUALITY CONTROL FOR MINOR CONSTRUCTION

Establish and maintain a QC program as described in this section. The QC program consists of a QCM, a QC plan, a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, and QC certifications and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations which comply with the requirements of this contract. The QC program must cover on-site and off-site work and must be keyed to the work sequence. No work or testing may be performed unless the QCM is on the work site.

1.1. Quality Control Manager Qualifications

Appoint a Quality Control Manager (QCM) responsible for the QC program. The Superintendent shall not serve as the QCM. The QCM must have a minimum 5 years of combined experience as a superintendent, inspector, QCM, project manager, or construction manager on similar size and type construction contracts which included the major trades that are part of this contract. The individual must be familiar with the requirements of the EM 385-1-1, D.Lgs. 81/2008 and have experience in the areas of hazard identification and safety compliance. The QCM must have completed the course Construction Quality Management for Contractors and will have a current certificate. The QCM can serve as SSHO unless otherwise specified in Part 3. Provide qualification of the QCM prior to the PAK.

1.2. Quality Control Manager Responsibilities

a. Provide a QCM at the work site to implement and manage the QC program. The QCM is required to attend the Post Award Kick-off, Partnering, Design Development, Coordination and Mutual Understanding Meeting, conduct the QC meetings, perform the three phases of control, perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this contract. The QCM is responsible for managing and coordinating the three phases of control and documentation performed by others.

b. QC Meetings, after the start of construction, the QCM must conduct QC meetings once every two weeks at the work site with the Superintendent and the foreman responsible for the ongoing and upcoming work. The QCM must prepare the minutes of the meeting and provide a copy to the COR within two working days after the meeting.

c. If Commissioning is part of work, engage Commissioning Authority (CxA), coordinate their oversight of Contractor's work and verify CxA's performance in accordance with UFGS section 01 45 00.05 20.

1.3. Quality Control Plan

Submit a QC Plan for Government review and acceptance prior to the Initial Design Conference. Update the QC Plan as needed prior to the start of construction. Submit the QCM resume for approval prior to the PAK meeting. The QC plan must include the following:

a. QC ORGANIZATION: A chart showing the QC organizational structure and its relationship to the production side of the organization.

b. NAMES, QUALIFICATIONS and RESPONSIBILITIES: resume format, for each person in the QC organization. Include the CQM for Contractors course certification required by the paragraph entitled "Construction Quality Management Training". Design and Construction

c. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONAL: Of each person in the QC organization.

d. OUTSIDE ORGANIZATIONS: Outside organizations, including consulting architectural and engineering firms and a description of the services these firms will provide.

e. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QCM and Alternate QCM and stating that they are responsible for managing and implementing the QC program as described in this contract. Include in this letter the QCM authority to direct the removal and replacement of non-conforming work.

f. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER (DESIGN & CONSTRUCTION): Procedures for scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, designers of record, consultants, architect engineers A/E, offsite fabricators, suppliers, and purchasing agents. Include submittal reviewer, estimated date of delivery, and identify which design submittals require Government approval prior to construction, and which construction submittals require DOR or Government approval prior to construction.

g. TESTING LABORATORIES: Accredited laboratories as applicable.

h. TESTING PLAN AND LOG: Tests required, referenced by specification paragraph number requiring the test, frequency, and person responsible for each test.

i. PROCEDURES TO COMPLETE REWORK ITEMS: Procedures for tracking design and construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected.

j. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task, which is separate and distinct from other tasks, and has the same control requirements and work crews. The list must be cross-referenced to the Contractor's Construction Schedule and the specification sections.

k. COMMUNICATION PLAN: Provide a plan for key decisions and possible problems the Contractor and Government may encounter during the design phase of the project. Communication Plan must indicate the frequency of design meetings and what information is covered in those meetings, key design decision points tied to the Network Analysis Schedule and how the DOR plans to include the Government in those decisions, peer review procedures, interdisciplinary coordination, design review procedures, and comment resolution.

l. PROCEDURES FOR PERFORMING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each DFOW, a Preparatory and Initial phase checklist shall be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.

m. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.

n. SPECIAL INSPECTIONS: Provide a list of Special Inspections, testing, approvals, certifications, observations, and quality assurance plans as prescribed in Chapter 17 of IBC.

o. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the contract.

1.4. Design Quality Control Plan

The following additional requirements apply to the Design Quality Control (DQC) Plan:

a. Submit and maintain a DQC Plan as an effective quality control program which assures that all services required by this contract are performed and provided in a manner that meets professional architectural and engineering quality standards. As a minimum, all documents must be technically reviewed by competent, independent reviewers identified in the DQC Plan. Correct errors and deficiencies in the design documents prior to submitting them to the Government.

b. Include the design schedule in the master project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific Contract period. This should be at a detailed level of scheduling sufficient to identify all major design tasks, including those that control the flow of work. Include review and

correction periods associated with each item. This should be a forward planning as well as a project monitoring tool. The schedule reflects calendar days and not dates for each activity. If the schedule is changed, submit a revised schedule reflecting the change within 7 calendar days. Include in the DQC Plan the discipline specific checklists to be used during the design and quality control of each submittal. Submit at each design phase as part of the project documentation these completed disciplinespecific checklists. ER 1110-1-12 provides some useful information in developing checklists.

c. Implement the DQC Plan by a Design Quality Control Manager (DQCM) who has the responsibility of being cognizant of and assuring that all documents on the project have been coordinated. This individual must be a person who has verifiable engineering or architectural design experience and is a registered professional engineer or architect. DOR cannot serve as DQCM on the same project. Notify the COR, in writing, of the name of the individual assigned to the position. After DQC Plan acceptance, any changes proposed by the Contractor are subject to the acceptance of the COR.

1.5. Three Phases of Quality Control

Utilize the Three Phases of Quality Control process consisting of:

a. Preparatory Phase: Notify the COR at least two work days in advance of each preparatory phase. Conduct the preparatory phase with the Project Superintendent and the foreman responsible for the definable feature of work. Document the results of the preparatory phase actions in the daily CQC Report and in the QC checklist. Review all applicable documents for compliance with all applicable laws, codes, regulations, and the requirements of the contract, including contract drawings and specifications. Determine requirements for testing and certification. Review submittal approvals for materials, equipment, shop drawings, and applicable methods of construction and installation. Include all Preparatory Phase items, along with the Preparatory Phase Checklist in the QC Report.

b. Initial Phase: Notify the COR at least two work days in advance of each initial phase. Conduct the Initial Phase with the foreman responsible for that DFOW. Observe the initial segment of the work to ensure that it complies with contract requirements. Document the results of the Initial Phase in the daily CQC Report and in the QC checklist. Observe and inspect the initial portion of the work performed under a Definable Feature of Work to establish the quality of the workmanship, resolve conflicts in construction, ensure that testing is done and certified as required, and to check all work procedures to ascertain the work is in conformance with required safety requirements. Record and report nonconforming work and work not of acceptable quality and requiring correction or rework. Include all Initial Phase items, along with initial phase checklist and, in the QC Report.

c. Follow-Up Phase: Occurs at the completion of each DFOW. Ensure the work is in compliance with contract requirements, quality of workmanship for all work is maintained, and all work performed meets safety requirements. Include all Follow-Up Phase items, including date, in the QC Report.

d. Additional Preparatory and Initial Phases: Additional preparatory and initial phases must be conducted on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW is resumed after substantial period of inactivity, or if other problems develop.

1.6. Contractor Production Report

Prepare Weekly Contractor Production Reports. Submit reports to the COR by 10:00 am the following Monday. Reports must include: Form can be downloaded at: http://www.wbdg.org/ffc/dod/unified-facilitiesguide-specifications-ufgs/forms-graphicstables

a. Worker hours by classification, move-on and move-off of construction equipment furnished by the prime, subcontractor or the Government, and materials and equipment delivered to the site.

b. Safety meetings, checks and inspections.

c. Disposition of Construction Waste Material: Per Waste Management Plan and per Environmental Protection Plan.

1.7. Completion Inspections

a. Punch-Out Inspection. Conduct an inspection of the work by the CQC System Manager near the end of the work, or any increment of the work established by a time stated in FAR 52.211-10 Commencement, Prosecution, and Completion of Work, or by the specifications. Prepare and include in the CQC documentation a punch list of items which do not conform to the approved drawings and specifications. Include within the list of deficiencies the estimated date by which the deficiencies will be corrected. The CQC System Manager or staff make a second inspection to ascertain that all deficiencies have been corrected. Once this is accomplished, notify the Government that the facility is ready for the Government Pre-Final inspection.

b. Pre-Final Inspection. The Government will perform the pre-final inspection to verify that the facility is complete and ready to be occupied. A Government Pre-Final Punch List may be developed as a result of this inspection. Ensure that all items on this list have been corrected before notifying the Government, so that a Final inspection with the customer can be scheduled. Correct any items noted on the Pre-Final inspection in a timely manner. These inspections and any deficiency corrections required by this paragraph need to be accomplished within the time slated for completion of the entire work or any particular increment of the work if the project is divided into increments by separate completion dates.

c. Final Acceptance Inspection. The Contractor's Quality Control Inspection personnel, plus the Superintendent or other primary management person, and the COR is required to be in attendance at the final acceptance inspection. Additional Government personnel including, but not limited to Base/Post Facility Engineer user groups, and major commands may also be in attendance. The final acceptance inspection will be formally scheduled by the COR based upon results of the Pre-Final inspection. Notify the COR at least 14 days prior to the final acceptance inspection and include the Contractor's assurance that all specific items previously identified to the Contractor as being unacceptable, along with all remaining work performed under the Contract, will be complete and acceptable by the date scheduled for the final acceptance inspection. Failure of the Contractor to have all work acceptably complete for this inspection will be cause for the KO to bill the Contractor for the Government's additional inspection cost in accordance FAR 52.246-12 Inspection of Construction.

01 50 00 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

1.1. Construction Site Plan

At the 65% design phase on D-B projects or at the PAK for D-B-B project submit for Government approval a site plan showing the locations and dimensions of temporary facilities including layouts and details, equipment and material storage area, and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.2. Contractor Work Site

Limit use of the premises for work and for storage of material and equipment associated with the contract. Unless otherwise specified or separately agreed to, Government owned material handling equipment, transportation equipment or general tools will not be available for Contractor's use. Clean work area daily and after completion of the work, removing all loose debris and disposing of all non-permanent materials in accordance with the contractor's Waste Management Plan. Contractor shall have an emergency anti spillage equipment. In the event of accidental spillage, the Contractor shall promptly restore the area.

a. Temporary Facilities: The Contractor may provide his own office facilities; coordinate and obtain advance approval through the COR. Provide and maintain suitable sanitary facilities within the construction limits of the contract. Dispose of sanitary waste in accordance with the applicable laws, and local regulation.

b. Contractor-Furnished Equipment: Equipment is subject to the inspection and approval of the COR, prior to and during the life of the contract. All equipment and vehicles must display readily visible Contractor identification markings. Relocate stored Contractor equipment which may interfere with operations of the Government or with others on-site.

c. Contractor-Furnished Material: Protect and secure products stored at this site. All replacement units, parts, components, and materials to be used in the maintenance, repair and alteration of facilities and equipment must be new and compatible with the existing equipment on which it is to be used, and must comply with applicable Government, commercial, or industrial standards.

d. The entire site must be fenced. Fence must be a minimum of 6 feet high chain link with screen material. Keep fencing in a state of good repair and proper alignment. If the project requires phasing provide construction fence only to areas around the exterior of the facility that will be under construction.

1.3. Temporary Signage:

a. Furnish the construction project sign, maintain the sign during construction, and remove from the job site upon completion of the project. The construction project sign package consists of project identification, Project POC and safety performance of the contractor.

b. Italian Construction Sign "Cartello di Cantiere" and Preliminary Notification "Notifica Preliminare" must be posted.

    1. Provide an Italian Construction Sign in compliance with Italian law. This board must show in Italian the following:

        a. The nature of works carried out;
        b. The cost of the project;
        c. How the project will be implemented;
        d. The reference number of the authorization to build;
        e. The Procuring Entity (name and registered address);
        f. The Contractor(s) (name and registered address);
        g. The Subcontractor(s), if any (also for utilities);
        h. The name of the Architectural Designer;
        i. The name of the Structural Designer;
        j. The name of the Systems Designer;
        k. The name of the Work Director;
        l. The name of Chief Operating Officers, if any, or Construction Controllers;
        m. The name of the Design Safety Coordinator;
        n. The name of the Construction Safety Coordinator;
        o. The name of Construction Manager;
        p. The name of Construction Tester;
        q. The Business Operators of contractor(s);
        r. The Business Operators of subcontractor(s)

    2. Provide information pertaining to the Contractor organization required to prepare the Preliminary Notification. The Government will complete the notification and provide it to the appropriate authorities through the Italian Base Commander staff.

1.4. Temporary Utilities

a. The Government will provide water and power in reasonable quantities at no cost to the Contractor.

b. All labor, material, and equipment necessary to affect temporary utility tie-ins, including transformers if necessary, will be at the expense of the Contractor and under the surveillance of the COR.

c. The Contractor will be responsible for any damages to Government, private or public facilities and property that may result from the installation and removal of these temporary utility tie-ins. Corrections and repairs will be made at the Contractor's expense.

d. The actual location and installation of the temporary tie-in, together with any interruptions of utilities systems, must be identified and approved by the COR prior to execution. Notify the COR 15 calendar days prior to any tie-ins.

e. Permanent utility systems, when indicated, will be available for tie-in.

f. Telephone and Data Service: The Contractor shall be responsible for providing their own telephone and data services for daily, work related communications. The Contractor shall not be permitted to utilize any Government equipment to perform daily construction duties.

g. Contractor Computer Equipment: Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1. Operating System. The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

2. Anti-Malware Software. The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

3. Passwords and Passphrases. The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

4. Contractor Computer Cybersecurity Compliance Statements. Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at http://www.wbdg.org/ffc/dod/unifiedfacilities-guidespecifications-ufgs/forms-graphics-tables Each Statement must be signed by a cybersecurity representative for the relevant company.

h. Maintain utility services to existing facilities surrounding the site at all times during construction.

i. Contractor must install and certify back flow preventers on all connections to the potable water supply system.

1.5. Traffic Provisions

a. Conduct operations in a manner that will not close a thoroughfare or interfere with traffic except with written permission of the COR at least 14 calendar days prior to the proposed modification date, and provide a Traffic Control Plan for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with Italian road code. Contractor may move oversized and slow-moving vehicles to the worksite provided coordination with the Garrison has been completed.

b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the COR prior to starting any activity that will obstruct traffic.

c. Conduct work so as to minimize obstruction to parking. Parking areas that will be blocked must be identified in your traffic control plan and approved by the Garrison.

d. Provide, erect, and maintain, at Contractor's expense, lights, barriers, signals, passageways, detours, and other items, that may be required.

e. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic.

01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS

1.1 Unforeseen Hazardous Conditions

Do not disturb hazardous materials and report condition immediately, to the KO potentially hazardous conditions that are uncovered or the Contractor becomes aware of that have not been identified in Part 3 of the SOW. This includes hazardous components and materials, and contamination. This includes conditions that are not only hazardous to humans but wildlife, marine life and the environment. The Contractor will stop work in the event of questionable material or conditions until identification and direction is provided. In the event that asbestos containing materials (ACM) shall be disturbed or removed, the Contractor shall submit to the Garrison Asbestos Manager, through the COR, a plan/notification describing the procedure to safely perform the work in accordance with Art. 250 and 256 D.lgs. 81/08.

1.2 Environmental Protection Requirements

The DOR must edit and submit UFGS 01 57 19, Temporary Environmental Controls, and UFGS 01 57 19.01 20, Supplementary Temporary Environmental Controls. Comply with state and local regulations within the edited UFGS section.

In addition to submitting UFGS 01 57 19, the following general environmental conditions apply:

1. General: The Contractor shall be responsible for providing a plan for the prevention of environmental pollution during and as a result of construction operations under this contract. During the PAK, the construction contractor will meet with the COR to develop an understanding of the environmental pollution control program.

2. Protection of Land Resources: The land resources outside the limits of permanent work shall be preserved in its present condition, or restored to its original condition, after the completion of construction. The Contractor shall confine his construction activities to areas defined by the plans or specifications.

3. Prevention of Landscape Defacement: Except in areas shown on plans or specified to be cleared, the Contractor shall not deface, injure or destroy trees or shrubs, nor remove or cut them, without written authorization from the COR. Survey monuments or markers shall be protected before operations in the vicinity of the survey markers.

4. Restoration of Landscape Damage: Any landscape feature damaged by theContractor’s operations shall be restored to its original condition at the Contractor’s expense. The KO shall decide what method of restoration shall be used.

5. Temporary Excavations and Embankments: If the Contractor proposes to

construct temporary roads or embankments for plant and/or work areas, the Contractor shall submit for approval at least seven (7) calendar days prior to scheduled start of such temporary work:

a. A layout of all temporary roads, excavations and embankments to be constructed within the work area.

b. Details of the completed burrow excavations.

6. Post-Construction Cleanup: The Contractor shall remove all signs of temporary construction such as haul roads, work areas, structures, foundations of temporary structures, and stockpiles of materials.

7. Protection of Water Resources: The Contractor shall not pollute streams of drainage courses with fuels, oils, bitumen, calcium chloride, acids, construction wastes, or other harmful materials. The Contractor is responsible for investigating and complying with all applicable local, regional and national laws and directives concerning pollution. All work shall be performed so the objectionable conditions shall not be created.

8. Erosion Control: Prior to the start of construction, the Contractor shall submit for COR approval a plan showing proposed methods for controlling erosion and disposing of wastes. Fills and waste areas shall be constructed by selective placement to eliminate silts or clays on the surface that will erode and contaminate adjacent streams and reservoirs.

9. Spillage: Special measures shall be taken to prevent chemicals, fuels, oils, greases, bituminous materials, waste washings, herbicides and insecticides, cement and other contaminants from contaminating the soil and ground water, or from entering streams.

10. Disposal: Disposal of any materials, wastes, effluents, trash, garbage, oil, grease, chemicals, etc., shall be subject to the approval of the Contracting Officer. Ground which has been contaminated by the Contractor activities shall be excavated and disposed of as directed by the Contracting Officer, and replaced with suitable fill material, compacted and finished with top soil, at the Contractor’s expense.

01 74 19.05 20 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT FOR DESIGN-BUILD

Develop a Waste Management Plan that identifies all recyclable material and disposal methods for all material. Contractor must reduce, recycle or salvage as much waste material as possible with a goal of diverting at least 60% of construction waste from landfill. Address waste reduction, recycling and salvage as part of the waste management plan. Report volume or weight of disposed and recycled materials. Report destination of debris diverted from disposal. The Contractor is responsible for removing and disposing of all waste materials generated. Consider all material recyclable or reusable, unless clearly demonstrated the material requiring disposal is waste material.

Provide waste management plan according to UFGS section 01 74 19.05 20 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT FOR DESIGNBUILD.

Report:

a. With each application for progress and final payment, submit a summary of construction and demolition debris diversion and disposal including beginning and ending dates of period covered.

b. Quantify all materials diverted from landfill disposal through salvage or recycling during the period with the receiving parties, dates removed, transportation costs, weigh tickets, manifests, and invoices. Include the net total costs or savings for each salvaged or recycled material.

c. Quantify all materials disposed of during the period with the receiving parties, dates removed, transportation costs, weigh tickets, tipping fees, manifests, and invoices. Include the net total costs for each disposal.

Cleanup:

Provide final cleaning in accordance with ASTM E1971 and submit one copy of the listing of completed final clean-up items. Clean all areas impacted by the work under this project. This includes all rooms where work was completed and all circulation paths (corridors, vestibules, etc.) leading to work areas from building entrances. Provide all devices and tools required for the performance of the work to include all necessary additional scaffolding and ladders.

Leave premises "broom clean." Comply with GS-37 for general purpose cleaning and bathroom cleaning. Use only nonhazardous cleaning materials, including natural cleaning materials, in the final cleanup. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition.

Clean or otherwise replace filters of operating equipment and comply with the Indoor Air Quality (IAQ) Management Plan. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site. Recycle, salvage, and return construction and demolition waste from project in accordance with Section 01

57 19 TEMPORARY ENVIRONMENTAL CONTROLS, and 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

Part 3 Statement of Work / Project Program

TABLE OF CONTENTS

1.0 PROJECT DESCRIPTION

The purpose of this contract is to repair the existing old and failed storm water and sanitary sewer main systems of the entire Miotto base in accordance with the 100% design package enclosed under Part 6 of this Statement of Work (SOW). Major project elements include:

1. Repair the failed stormwater system and extend the system where necessary and upgrade the size and type of piping where currently undersized;
2. Current failed system conditions have created serious flooding, soil erosion and land slide problems during last year’s storm events. All these naturalistic issues will be fixed with these system repairs;
3. The sanitary sewer system is, in some areas of Miotto base, connected with the storm water system so mixing the storm and sanitary water. The intent of this project is to disconnect and divide the two systems to have them properly working and properly

discharging the water as per current regulations;

4. Also, currently, the existing storm water and sanitary sewer channeling systems are both discharging the rain water and the waste water into the main public mixed line (managed by Vi.Acqua) which runs along Riviera Berica-Via Colderuga and which is already critically undersized and not able to discharge all rain water coming from Miotto base and from the Longare community lands during heavy storm events (as the one of 23 September 2024). The intent of this project is to divide the Sanitary Sewer system from the Storm Water system and create a new piping direct connection between the internal repaired Miotto base storm water system and the Bisatto River. In this way, the main mixed public line along Riviera Berica will be alleviated from the storm water which previously was coming from the Miotto base. The new direct connection between Miotto base and the Bisatto river will be realized with the installation of an underground TOC pipe which shall traverse an existing public right-of-way across private properties. The right of way act for the perpetual passage of the new TOC, has been already prepared and signed by both parties (the Comune di Longare and all private owners of the land in which the new TOC pipe will traverse). The act is attached to this contract under Part 6. Moreover, US Army Corps of Engineers Real Estate has entered into a license with the Comune di Longare for US Government use of subject perpetual right-of-way. See attached under Part 6.
5. In order to meet all current regulations and standards, Genio Civile prescribed to laminate the storm water coming from Miotto Base before its entry into the Bisatto river. To do so, a lamination tank will be constructed under an existing Comune di Longare sportfield. Two documents, a Memorandum of Understanding (MOU) (signed by USAG Italy and Comune di Longare) and a License (signed by USACE Real Estate and Comune di Longare) have been executed, to facilitate subject lamination tank construction. See documents under Part 6 of this contract.
6. To realize the lamination tank will be necessary to use the area of the current football training sportfield of the Comune di Longare property. The training sportfield impacted by the lamination tank construction must be restored above the new tank installation. A synthetic surfaced training field will be installed on top of the tank.
7. Due to many floodings and land slide events, some naturalistic features have been included in the design to slow down the storm water velocity within Miotto prior to its discharge from the installation and entry into the lamination tank. Subject features shall be integrated within this storm water project in accordance with the 100% design (Part 6 of this SOW) to include sedimentation tanks and retention ponds/weir walls to facilitate slowing of storm water velocities during storm.

2.0 PROJECT OBJECTIVES

Existing storm water and sanitary sewer systems of Miotto Military installation are old and failed and require repair to bring these systems to current code and regulations. The goal of the project is to execute all works in accordance with the 100% design package enclosed under Part 6 of this Statement of Work (SOW).

Main primary items to be built under this contract are:

1. Excavation to facilitate removal of existing piping and manholes and disposal of these existing sanitary sewer and storm water system components;

2. Furnishing and installation of piping, manholes, ditches, channels, etc for sanitary sewer and storm water systems and backfilling piping trenches as required, with resolution of all interferences with existing other underground lines through crossing, bypassing etc;
3. Furnishing and installation of first flush water treatment/oil water separators and all special manholes, tanks and related accessories;
4. Furnishing and installation of storm water and debris control technical naturalistic solutions through weirs, gabions, wood walls, retention basins, wood poles fixings, geo-mesh, idroseedings etc…
5. Furnishing and installation of TOC (Trivellazione Orizzontale Controllata) underground line and related manholes with clapet valves at the beginning and at the end of the pipe;
6. Furnishing and installation of lamination tank and all associated lines, manholes and tanks which support it;
7. Furnishing and installation of football training synthetic turf sportfield, on top of the constructed lamination tank, and associated support attributes. Subject installation restores the existing training sportfield which is impacted by the lamination tank installation;
8. Furnishing and installation of sedimentation open pond to be installed near the cave/dog training area and all associated lines, manholes and tanks in its support;
9. Furnishing and installation of 2 new lifts stations for the pumping of stormwater where gravity flow is not possible. Works include all associated systems such as DDC, electrical, piping, etc.
10. Localized wood cutting where specified and authorized;
11. Perform mitigation planting of bushes and trees where specified and authorized;
12. Furnish and install wellpoint system during excavations
13. Perform new retaining walls and hillside reshaping;
14. Potential removal and disposal of asbestos piping if encountered during works. The contractor shall provide unit pricing under CLIN 003
15. Temporary perimeter fence construction to provide for a continuous secure perimeter, if works dictate the temporary removal of a section of existing perimeter fence.

The project is divided into separate CLINS which are indicated as follow:

1. CLIN 001 — Storm water system works
2. CLIN 002 — Sanitary sewer system works
3. CLIN 003 — Asbestos pipe removal for project works internal and external to the Base.

For CLIN 003, the contractor is required to provide a unit cost for each linear meter considering to safely and properly treat and then remove and dispose of any asbestos lined pipe discovered during project execution. Also costs for the SPISAL plan/notification procedure and package preparation have to be considered too.

To ensure for execution of all items of work in the 100% design package Contractors are being made aware of the following requirements and must:

1. Portion of works will be realized outside the Miotto base. Specifically, will be executed on Comune di Longare land (for the lamination tank and its associated piping) and also on privates properties with a perpetual public right-of-way (for the TOC pipe) so all these works have been preliminary authorized with a dedicated PDC (Permesso di Costruire) which must be fully respected and adhered to. Any changes/deviations to the approved PDC would generate a "Variante in Corso d'Opera" which will delay works. For works external to the installation the Comune di Longare will be inclined not to accept changes/deviations. If the need for a Variante does arise, it will be responsibility of Contractor to resubmit a new request and pay for it;
2. For all works to be executed outside Miotto base and so authorized through PDC, the "Dichiarazione Inizio Lavori" and the final "Dichiarazione Fine Lavori" will be properly prepared and submitted by the U.S. Government's DOR. Contractor will provide support to the U.S. Government DOR when specific contractor documentation is required (as in example all "Conformita' degli Impianti", etc);
3. For all structural works to be executed inside and outside Miotto base the "Deposito Strutture Ultimate" and the "Collaudo Finale" will be properly prepared and submitted by the U.S. Government DOR/final tester. The Contractor shall provide assistance to the U.S. Government DOR/final tester when specific contractor documentation is required;
4. Work execution will be divided into phases (which are clearly identified in the PSC). First Phase (Phase 1) will be mandatorily related to execution of all works outside Miotto base. As per agreements with Comune di Longare, works need to be started and completed between September 2026 and September 2027. All other phases shall be executed in a manner to ensure the use of all internal roads by, military vehicles, and emergency vehicles during the entire construction performance period. Contractor will perform work along roads in a manner that provides for a minimum space for an alternate one-way (senso unico alternato) flow of traffic. Contractor will provide for the use of construction site traffic lights with the presence of motion sensors to facilitate traffic flows during road works.
5. Construction contract overall period of performance is around 3 years (1010 days). Works external to the installation (within Comune di Longare) shall have a separate period of performance of the initial 12 months of the contract. The contractor can perform simultaneously works inside and outside the Miotto installation.
6. Miotto base is a special site in which many constraints are present. During design process all Third Party Agencies have been involved, and specific approval authorizations have been obtained. In particular, authorizations obtained from Italian Agencies regarding constraints in the Miotto installation are as follows:
    a. Autorizzazione Paesaggistica Ordinaria (Landscaping Request of Approval) as per DPR 31/2017;
    b. "Verifica dell'interesse Archeologico"(Archeological Request of Approval) from SABAP (Sovrintendenza Archeologica Belle Arti e Paesaggio)
    c. VPIA (Verifica Preventiva dell'Interesse Archeologico) from SABAP (Sovrintendenza Archeologica Belle Arti e Paesaggio).
    d. Autorizzazione Vincolo Idrogeologico e Forestale (Hydrogeologic and Forest Request of Approval) as per R.D. 16.05.1926 n. 1126, L.R. 52/1978 and related

e. VINCA (Valutazione di incidenza Ambientale) as per L.R. 12/2024, Regolamento Regionale n.4 del 9 Gennaio 2025 (BUR n.9 de 19 gennaio 2025) and related.

Contractor must adhere to the enclosed approved design package, the Authorization Letter and all prescriptions/ requirements contained within it. All modifications/deviations from the approved design will generate a "Variante all'Approvazione" for which a new approval would be required. If the need for a Variante arises, it will be the responsibility of Contractor to submit a new request and pay for it;

7. Since the risk of archeological findings is high as stated by SABAP, archeological assistance during all Contractor excavation works is required. A specialized Dott. Archeologo, who will oversee all excavation activities will be provided by the US Government. Procedure of archeological support will include following tasks to be performed by the US Government DOR:
   a) communicate to SABAP the date of work site start, specifying the date of excavation start and communicate the name of the engaged Dott. Archeologist;
   b) within 6 months from the end of archeological support, complete and submit to SABAP the final report (to be performed also in case of not archeological findings)
8. Based on type of excavations foreseen by this project, which could be considered performed on already treated subbases materials, and based on PSC statement, the UXO remediation is not required to be performed by Contractor within this contract.
9. Contractor will manage the "Gestione Terre e Rocce da scavo" procedure. Documentation for the soil excavations and proper disposal/moving will be prepared by contractor based on current DPR 120/2017 and D. Lgs.152/2006
10. For all works to be executed outside Miotto military base, the Contractor will be responsible for the official submission of the request for the occupation of public land (Occupazione Suolo Pubblico) and the payment of the related charges. The request must be submitted before the start of works. The US Government is exempt from the payment of any type of taxes; therefore, a waiver letter will be made available to the Contractor for their exemption from payment of the primary fee. Only "marche da bollo" will be requested to be paid by the Contractor. The waiver letter is attached in Part 6;
11. Before start of works outside Miotto military base, the Contractor will receive from Comune di Longare the "Ordinanza" which represents the official temporary consignment of the area to the Contractor for the time needed for the works. Without the "Ordinanza" the contractor is not authorized to execute works and occupy Comune land;
12. Official approval to build the new TOC underground pipe under the road Riviera Berica has been obtained from Vi.Abilita' Agency during design process. Contractor is responsible to strictly follow the approved design, the Authorization Letter and all the prescriptions/indications contained in it. All modifications/deviations from the approved design will generate a "Variante all'Approvazione" for which a new approval would be needed causing time delays. If need for a Variante arises, it will be responsibility of Contractor to resubmit a new request and pay for it.
13. Official approval to build the new TOC underground pipe and the bypass line along the bike path near main water pipe line has been obtained from Vi.Acqua Agency during design process. Contractor is responsible to strictly follow the approved design, the Authorization Letter and all the prescriptions/indications contained in it. All

modifications/deviations from the approved design will generate a "Variante all'Approvazione" for which a new approval will be needed causing time delays. If a need for a Variante arises, it will be responsibility of Contractor to resubmit a new request and pay for it.

14. The Contractor shall submit to the Comune di Longare the bank guarantee of the works to be carried outside the Base. Contractor is responsible to prepare the official court Italian translation of this document.
15. Contractor will build a temporary perimeter fence to provide for continuous perimeter security, if works along, near and under existing perimeter fence require temporary removal of the existing permanent fence. If removed, the existing perimeter fence must be restored to its original configuration.
16. Contractor will be responsible for coordination and preparation of minutes of each meeting which will be held on site during the construction process especially the coordination meeting which will happen with the presence of outside Third Party Agencies. Minutes will be prepared in Italian and English languages. Due to the complexity of the project, the contractor is not authorized to directly communicate with Third Party Agencies but all communications will be managed by the US Government PM -DPW design branch.
17. Excavation soil is supposed to be type A as per Legislative Decree 152/2006, Annex 5, Part IV, Table 1 classification. Type A is intended as sites for reuse as public or private green areas and residential zones
16. Contractor will provide wellpoint system during excavations in certain areas where indicated in the 100% design package.
18. Contractor will be responsible to provide, as closeout documentation, schemes of existing electrical panels impacted by this contract where AS BUILT or specific design is missing. Schemes will be required for the final "Conformita' degli Impianti" and for the closeout documentation.
19. Topographic survey Data: US Government recently performed a GIS/georadar survey of all utilities present at Miotto base. The DWG file generated has been used by US Government DOR to prepare the full 100% design package which is here attached as Part 6. The contractor will be responsible to perform their own survey as verification of existing conditions and as survey of all details necessary to properly prepare shop drawings.
20. All visible retaining wall façades shall be finished to graphically and aesthetically obtain stone wall effect. The effect will be realized using a polystyrene master sheet to be applied inside the formwork before the concrete cast in place;
21. Before start of works outside and inside Miotto military base, the CSC will prepare the "Notifica Preliminare". Contractor will support the CSC when specific contractor documentation will be required. The contractor is not authorized to execute works without having received notification from the COR that the "Notifica Preliminare" has been sent and received by SPISAL Third Party Agency.
22. Contractor is responsible to restore the road and the area affected by construction works and vehicles traffic in its "Status Ante Opera". If damages to the bike path will occur, the contractor will fix them at their cost
23. The TOC route has been already established and agreed in the utilities lines right of way Act signed by Comune di Longare and all privates owners who will be crossed by the new line. Contractor is responsible to verify on site with topographic instruments the

correct position already established and execute in the exact position.

24. During the performance of works outside Miotto base on the lamination tank, the adjacent Comune di Longare existing primary sportfield complex (palazzetto dello sport) will be simultaneously under renovation by Comune di Longare. The contractor shall coordinate with the adjacent contractor accordingly to identify and address any instances of interferences. The DSC, within the PSC, has taken care of interferences but also Contractor will be responsible to perform safely during the work activities and coordinate accordingly.

2.1 APPLICABLE CODES AND STANDARDS

In addition to the codes and standards listed in Part 4, the design and construction must be in accordance with the latest revision/edition of the following referenced codes and standards.

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

Industry Criteria:

ASHRAE 15	Safety Standard for Refrigeration Systems;
ASHRAE 62.1	Ventilation for Acceptable Indoor Air Quality;
ASHRAE 90.1	Energy Standard for Buildings Except Low-Rise Residential Buildings;
ASHRAE 189.1	Standard for the Design of High-Performance Green Buildings;
IBC	International Building Code;
IMC	International Mechanical Code;
IPC	International Plumbing Code.

National Fire Protection Association (NFPA):

NFPA 13	Standard for the Installation of Sprinkler Systems;
NFPA 70	National Electrical Code;
NFPA 72	National Fire Alarm Code;
NFPA 101	Life Safety Code.

Military Criteria:

Energy Policy Act of 2005 (Public Law 109-58);

UFC 1-200-01 General Building Requirements;

UFC 1-200-02 High Performance and Sustainable Building Requirements;

UFC 4-010-01 DoD Minimum Antiterrorism Standards for Buildings;

Technical Guide for Installation Information Infrastructure Architecture (I3A);

EM 385-1-1 Safety and Health Requirements Manual.

Italian Norms:

Law No. 1086	(May 11, 1971) Norms for Works in Reinforced Concrete and Steel and Application Instructions;
Law No. 64	(February 2, 1974) Measures for Buildings with Special Requirements for Seismic Zones;
Law No. 373	(April 30, 1976) Norms for the Containment of Energy Consumption for Thermal Uses in Buildings;
Law No. 13	(January 9, 1989) Provisions to Support the Overcoming and Removal of Architectural Barriers in Private Buildings;
D.P.R. No. 503	(July 24, 1996) Regulations for the Elimination of Architectural Barriers in Buildings, Public Spaces and Services;
Law No. 183	(May 18, 1989) Standards for the Organizational and Functional Restructuring of Land Protection;
Law No. 10/1991	(January 9, 1991) Norms for the Implementation of the National Energy Plan as Regards the National Energy Use, Energy Saving and Development of Renewable Sources of Energy;
D. Lgs. 192/2005	Legislative Decree n. 192 on Implementation of Directive 2002/91/EC Relating to the Energy Performance of Buildings;
D.M. No. 37	(January 22, 2008) Regulation on the Implementation of Article 11 Quaterdecies, Paragraph 13, Letter a of Law No. 248 of 2005 Concerning the Reorganization of the Provisions relating to the Installation of Systems in Buildings;
D.L. No. 227	(August 15, 1991) Implementation of Instructions n. 80/1107/CEE, n. 82/605/CEE, n. 83/447/CEE, n. 86/188/CEE, n. 88/642/CEE Regarding the Protection of Workers against the Risks Arising from Chemical and Biological Exposure during Work, as per Article 7 of

	Law dated 30 July 1990, n. 212;
D.L. No. 285	(April 30, 1992) New Highway Code and subsequent Amendments and Additions;
D.M. 5/11/2001	(November 5, 2011) Functional and Geometrical Norms for Roads' Construction.
Law No. 447	(October 26, 1995) Framework Law on Noise Pollution;
D.L. 02/05/01	(2001) Criteria for Selection and Use of Personal Protection Devices;
D.P.R. No. 380	(June 6, 2001) Law of Legislative and Regulatory Provisions on Buildings;
D.L. No. 42	(January 22, 2004) Code of Cultural Heritage and Landscape, in Accordance with Article 10 of the Law of 6 July 2002, n. 137;
D.P.R. No. 236	(November 15, 2012) Regulation of Activities of the Army Corps of Engineers in accordance with Article 196 of the Decree 12 April 2006 n. 163;
D.L. No. 139	(March 8, 2006) Reorganization of the Provisions relating to the Functions and Duties of the National Fire Brigade, in accordance with Article 11 of Law 29 July 2003, n. 229;
D.L. No. 152	(April 3, 2006) Environmental Regulations;
D.M. 11/10/2017	(11 October 2017) Minimum Environmental Criteria
Regional Law No. 1400	(August 29, 2017) Methodology on Rules on Environmental Assessment, V.Inc.A.;
Circolare No. 617	(February 2, 2009) Instructions for the Application of the New technical Standards for Construction of the D.M. 14 January 2008;
D.M. 17/01/2018	(January 17, 2018) Technical Standards for Construction;
D.Lgs. 81/2008	(2008) Unified Safety
D.P.R. 915	(September 10, 1982) Actuation of CEE n. 75/442 related to Waste and n. 78/319 related to Toxic and Noxious Wastes.
Law 357/97	(11 September 1997) Regulation Implementing Directive 92/43/EEC on the Conservation of Natural and Seminatural

	Habitats as well as Wild Flora and Fauna;
Regional Law 30/15	(March 19, 2015) Standards for the Conservation and Enhancement of the Regional Naturalistic-Environmental Heritage.
R.D. 1126	(May 16,1926) Rules relating to the management of the forestry and pastoral assets of the State, Municipalities and other Entities
L.R. 52	(13 September 1978) Regional Forestry law
L.R. 12	(27 May 2024) Regional regulations on strategic environmental assessment (VAS), environmental impact assessment (VIA), environmental impact assessment (VINCA) and integrated environmental authorization (AIA).
Regolamento Regionale 4	(09 January 2025) Implementing regulation on VINCA (Article 17 of regional law 27 May 2024, n. 12).
DPR 120	(13 June 2017) Regulation containing the simplified discipline of the management of excavated earth and rocks, pursuant to article 8 of the legislative decree of 12 September 2014, n. 133, converted, with amendments, by law of 11 November 2014, n. 164

Federal Facilities Environmental Stewardship & Compliance Assistance Center:

Final Governing Standards	(September 7, 2012) Italian Environmental Final Governing Standards.

It being understood that the work will be executed in compliance with the most modern of technology in use criteria, the state of the art with references to the laws, rules, regulations, provisions and applicable EU directives on national territory on buildings, systems, pollution, health and safety and accident prevent regulations during the construction works. The list of applicable references indicated herein is not all inclusive, therefore any valid regulations not specifically mentioned in this document are in any case prescribed and must be respected.

2.2 PUBLIC RELEASE OF INFORMATION

2.2.1 Prohibition

There shall be no public release of information or photographs concerning any aspect of the materials or services relating to this bid, contract, or purchase order or other documents resulting therefrom without the prior written approval of the KO.

2.2.2 Subcontracts and Purchase Orders
The Contractor agrees to insert the substance of above paragraph "Prohibition" in each subcontract and purchase order generated for this contract.

2.2.3 English Speaking Representative
At all times when any performance of the work at the site is being conducted by any employee of the Contractor or his subcontractors, the Contractor shall have a representative present on the site who is capable of explaining the work operations and receiving instructions in the English language. The KO shall have the right to determine without appeal of such decision, whether the proposed representative has sufficient technical and lingual capabilities and the Contractor shall immediately replace any individual not acceptable to the KO. The Contractor's Project Manager, Site Superintendent, Quality Control Manager, Site Safety and Health Officer shall be fluent (can read, speak, and write) in both English and Italian languages.

2.2.4 Norms and Laws
In case of differences between project specifications or the accompanying drawings and referenced norms and laws, norms and laws shall govern. In case of differences between project specifications and the accompanying drawings, the specifications shall govern.

2.2.5 Industry Standards
When both U.S. and European, or Italian, Industry Standards are applicable to this project and the standards are in conflict with each other, the most stringent of the industry shall govern.

2.2.6 Prohibited Items
Use of the following items in this construction project is prohibited:
a. Use of aluminum for electrical conductors.
b. Embedding aluminum conduit in concrete.
c. Use of fluorescent light ballasts and other products containing PCB's.
d. Use of urea-formaldehyde foam insulation products.
e. Use of any paint/coatings having a lead content of over 0.06 percent by weight of non-volatile content.
f. Use of ozone depleting chemicals.
g. Use of zinc-chromate.
h. Use of materials containing asbestos.

2.2.7 Proprietary Names
Manufacturer's proprietary names indicated for colors, textures and patterns of materials are for the purpose of color, texture and pattern selection only. Other manufacturers materials are acceptable and provided they closely approximate colors, textures and patterns indicated and provided they conform to all other requirements subject to COR approval.

2.2.8 Submittal of Proof of Qualifications and Experience
Where qualifications or experience requirements are set forth in the specifications with respect to equipment and equipment installers, written proof of such qualifications or experience must be provided within thirty (30) calendar days after Contract award, and before placing any order for such equipment or before dispatching equipment installers to

the project site.

2.2.9 Oral Modification
No oral statement of any person other than the KO as provided in the clause in this Contract entitled FAR 52.243-4 "Changes" (AUG 1987), shall in any manner or degree modify or otherwise affect the terms of this Contract.

2.2.10 Final Governing Standards (FGS)
Final Governing Standards (FGS) for environmental protection at DOD installations and facilities in Italy went into effect in June 1994. These standards were developed by comparing and adopting the more protective requirements of the Overseas Environmental Baseline Guidance Document (OEBGD), European Union and national, regional, and local environmental laws and regulations. The 2012 version of FGS, updated in 2015, are applicable to work under this SOW. FGS are available upon request.

2.2.11 Certification of Compliance with Italian Laws

2.2.11.1 CE marking
Construction products for which, at the date of this agreement, have been enacted mandatory harmonized standards under Directive 89/106/EEC, must comply with the rules and bear CE marking.

2.2.11.2 Certification of Mechanical and Electrical Systems
As a condition of final acceptance of the work, the Contractor shall submit to the COR a certificate of compliance for all contract work certifying that the work complies with Italian Law No. 37 and that the provided work is adequate and safe for the designated use. The certification shall be made by the responsible technician of the installation firm which shall be regularly registered on the “Registro delle Imprese” or on the “Albo Provinciale delle Imprese Artigiane”.

2.2.11.3 Certification
As a condition of final acceptance of the work, the Contractor shall submit to the COR a Certificate of Compliance for all contract work, certifying that the work complies with all applicable Italian or US laws and that the provided work is adequate and safe for the designated use.

2.2.12 Static Load and Final Testing
The Contractor shall refer to the final tester in accordance with Italian Law No. 1086. The tester shall be hired by US Government and his/her name shall be communicated to the Contractor prior to work commencement. The tester will be an enabled professional according to what is prescribed by the current laws. The tester will write the test certificate and transmit it to the COR. The Contractor shall provide the tester with the support, personnel and equipment as is deemed necessary for the tests at no additional cost for the US Government. The Contractor shall also be obliged to eventual restoration of any elements modified or damaged during the performance of the test, or any work found unsatisfactory or not certifiable by the tester. The Contractor shall comply with the pertinent articles of Italian Law No. 1086, D.P.R. No. 380 and Circolare No. 617. Because construction will be performed on property owned by the Italian State, the articles of the Law No. 1086 pertaining to notification and participation by civil authorities are not applicable. Article 2 of Italian Law No. 1086 requires the static load tests to be performed under the supervision of an engineer or architect registered in the National Professional

Rolls of Italy. Since GENIODIFE intends to participate in static load and final testing, the Contractor shall notify the COR at least 21 calendar days prior to any static load or final test to enable the COR to notify GENIODIFE. The COR will submit one original certification to GENIODIFE.

2.2.12.1 Static Load

The load tests, when deemed necessary by the tester must identify the correspondence of theoretical and experimental behavior. The materials of the elements subjected to testing must have reached the resistance provided for their operation in the final exercise. The test program established by the tester, with an indication of the loading procedures and performance expectations must be submitted to the COR and made know to the Contractor. The load tests must be performed in the manner specified by the tester who assumes full responsibility, while for their material implementation is responsible the Contractor. The load tests are tests of behavior works under the action of exercise. These must be in general such as to introduce the maximum operating stresses for characteristics combinations (rare). Depending on the type of the structure and nature of load tests can be conveniently protracted or repeated over several cycles. The decision on the outcome of the test is the responsibility of the tester and is judged on the basis of the following:

a. Deformations increase roughly proportionally to the loads;
b. During the test have not produced fractures, cracks, deformations or disruptions that compromise the safety or preservation of the work;
c. The residual deformation after the first application of the maximum load does not exceed a proportionate share of the total displacement due to the predictable initial inelastic settlement of structure under test. In the case instead that this limit is exceeded, successive load tests should indicate that the structure tends to an elastic behavior;
d. The elastic deformation appears not greater than that calculated. The static test, in the opinion of the tester and in relation to the importance of the work, may be supplemented by dynamic tests and tests to failure of structural elements.

2.2.12.2 Final Testing

The work may not be placed in service prior to the execution of the static test. The static testing must include the following tasks:

a. Control of the requirements for the works to be carried out with materials regulated by DPR 380, Law No.1086/71 and Law No. 64/74;
b. Inspection of the work in the various phases of construction of structural elements with particular regard to the most important structural parts. The inspection work will be performed at the presence of the COR and the Contractor. The tester will check out the implementation of the design specifications and the performing of the experimental controls;
c. Examination of the certificates of the materials testing, including:
   1. Check the number of withdrawals and compliance with the requirement of Chapter 11 of the D.M. 14.01.2008;
   2. The control that the tests results are in compliance with the acceptance criteria set out in that Chapter 11 D.M. 14.01.2008.
d. Examination of the certificates referred to controls at the factory and in the

production cycle, provided in Chapter 11 of the D.M. 14.01.2008;
e. Check of the records and results of any load test made by the COR. The tester must also:
1. Examine the proposed work, the structural and geotechnical design approach, the structural scheme and the actions used in the design;
2. Review the investigations carried out during the design and construction.
f. Finally, at its discretion, the tester may ask to perform all these investigations, studies, survey, experiments and research useful in order to form the belief in the safety, durability, and testability of the work including the following:
1. Load Tests;
2 Testing of material used, even by non-destructive methods.

2.2.13 Other Tests and Certifications
As a condition of final acceptance of the work the Contractor shall provide with all certifications and testing required by Italian Law. The Contractor is also responsible for all permits required by Italian Law (See Contract Clause FAR 52.236-7, "Permits and Responsibilities"). The Contractor shall submit to the COR or to the Quality Management Specialist all certifications required by Italian Law. Such certifications may include but not be limited by the following as applicable to the project:

a. Certificate of conformity for electrical and electronic system in accordance with D.M. No. 37. Include a separate specific certificate of compliance for each electrical panel in accordance with EN 61439-1 and EN 61439-2;
b. Certificate of compliance for heating system in accordance with D.M. No. 37;
c. Certificate of compliance for fire prevention system in accordance with Italian Decree 22 January 2008 No. 37 and NFPA 13 and NFPA;
d. Certificate of compliance for elevator system in accordance with D.M. No. 37, including CE Declaration of Conformity and technical data sheets of systems along with the following list of items as per D.P.R. 162/99 and Elevator Directive 95/16/CE. The Contractor/Installer shall provide the following documentation:
1. Certificate of compliance CE of product and installation;
2. Register manual for the overall system;
3. Instruction manual which contains drawings and layouts necessary for normal use, maintenance, inspections, repair, periodic verifications and emergency maneuvers;
4. A manual where eventual repairs and periodic verifications can be noted.
Contractor/Installer shall register the elevator to the municipality and perform recurring maintenance during the entire warranty period.
e. Certificate of actual building energy consumption in accordance with Italian D.P.R. No. 59;
f. Certificate of compliance for mechanical systems in accordance with Italian D.M. 01/12/75 and D.M. No. 37.
g. Certificate of compliance for mechanical systems in accordance with ISPESL requirements including the following documentation by the Contractor/Installer:
1. Declaration of the correct installation of safety components, in relation to the indications of the constructor and-/or applicable norms;
2. Certification of producer of the technical characteristics and certification of the

materials and components utilized;
3. Declaration that the adopted solutions meet the design intent;
4. Instruction manual containing drawings and layouts necessary for normal use, maintenance, inspections, repair, periodic verifications and emergency maneuvers;
5. A manual where eventual repairs, normal use, and periodic verifications can be noted.

h. INAIL certification for equipment system prior to DD1354 approval.

i. All the electrical works shall be performed by qualified personnel in possession at least of the "PES" qualification (Persona Esperta), or of the "PEI" qualification (Persona Idonea), as required by Norma CEI 11-27. Copy of the "PES" or "PEI" Nomination Letters of the electrical Subcontractor's personnel shall be provided by the Contractor to the COR. The COR will deliver the Nomination Letters to Mr. Thomas Raffaello, officially appointed as "RI" (Responsabile Impianto), in compliance with Dlgs. 81/08, CEI 0-10, CEI EN 50110 1-2 and CEI 11-27 Norms.

2.2.14 Italian Safety Law Compliance

2.2.14.1 Required Compliance with Italian Law 81/2008 subsequent Additions and Amendments.

The Contractor is responsible for conformance with the requirements of Italian law 81/2008 in effect at the time of award of the project. The Contractor shall complete and provide any information in a timely manner when requested by the Government to carry out its responsibilities under Law 81/2008. The Government will provide the Contractor with a scope of work. The Contractor will be responsible for fully preparing any and all documents, to include the appointment of a Design Safety Coordinator (DSC) “Coordinatore per la Sicurezza in fase di Progettazione” and the preparation of the Design Safety Plan “Piano di Sicurezza e Coordinamento”, the Facility Maintenance Handbook “Fascicolo dell’ Opera” and fulfilling all conditions related to Italian safety requirements for the planning and design phase of the project, even if the Contractor is not providing any architectural/engineering services on the SOW.

2.2.14.2 Construction Safety Coordinator (CSC) “Coordinatore per la Sicurezza in fase di Esecuzione”

The Government will provide the name of the CSC to the Contractor. The CSC shall perform the duties specified in Italian law 81/2008 during the construction phase of work. No construction shall begin until all appropriate documents have been submitted, reviewed and accepted, and the CSC has confirmed all requirements of Italian safety law have been met.

2.2.14.3 Responsibility

Any delays and/or damages resulting from erroneous, incorrect or inadequate information submitted become the sole responsibility of the Contractor.
Under no circumstances shall any direction from the CSC be subject to a change of the SOW or subject to a potential claim.

2.2.15 Environmental Policy and Restrictions

2.2.15.1 Final Governing Standards (FGS)

The Contractor is required to comply with the requirements of the Final Governing Standards (FGS) for environmental protection for all work performed on Department of Defense (DOD) installations in Italy.

2.2.15.2 Environmental Impact Study
Within this contract, the Contractor is required to comply and strictly follow the VINCA approval already obtained during the design process.
2.2.15.3 Recycling and Reuse Requirements
In accordance with USAG Italy policies, the Contractor is required to reduce, recycle, or salvage as much construction waste material as possible with a goal of diverting at least 60% from landfills. The Contractor is required to provide a Waste Management Plan (WMP) prior to the start of construction. The WMP shall indicate the anticipated types and quantities of demolition, site-clearing, and construction waste to be generated by the Contractors operations. Include estimated quantities and assumptions for estimates. For each type of waste, indicate whether it will be salvaged, recycled, or disposed of in landfill or incinerator.
Each month during construction, the Contractor shall submit a Waste Reduction Progress Report identifying, for each type of material, the total quantity (in tons) of waste (a) generated, (b) salvaged (estimated and actual), (c) recycled (estimated and actual), and (d) recovered (salvaged plus recycled). The amount recovered shall also be reported as a percentage of total waste.

2.2.15.4 Cultural Resources
Cultural resources such as archaeological remains or villa remains may be uncovered during construction excavation operations. If during excavations, suspected cultural remains are found, excavation operations shall immediately cease and Contractor shall notify the COR.

If required, when the project intervention area falls within a Landscape constrained area, the Contractor shall conduct a landscape assessment in order to understand if it is necessary to submit a Landscape authorization request to the Italian authorities. In this case, the Contractor shall prepare all the documentation required for the Landscape authorization procedure. All documents must be provided in English and Italian language and submitted to the COR and the installation Environmental Office. Within this contract, the Contractor is required only to comply and strictly follow the Landscape approval already obtained during the design process.

When the project intervention area falls within an archaeological constrained area, the Contractor shall conduct a preliminary assessment in order to understand if it is necessary to submit a request to the Italian authorities for their binding opinion and if a preliminary archaeological survey is required. In this case, the Contractor shall prepare all the documentation required for the procedure. All documents must be provided in English and Italian language and submitted to the COR and the installation Environmental Office. Within this contract, the Contractor is required only to comply and strictly follow the SABAP approval already obtained during the design process.

The Contractor shall prepare a Vi.Arch. report including design package. The Contractor shall also prepare, prior to start excavation, the "Comunicazione Lavori" which shall include all specific work site data (e.g. prime contractor name and address, name of the archeologist on that specific work site, contract awarded amount, period of performance). Within this contract, the Contractor is required only to comply and strictly follow the SABAP approval already obtained during the design process.

The Contractor will be required to incorporate any required changes or restrictions into their design and construction activities. Should any changes, restrictions, or other events associated with this process have an impact on the time to perform the work, the Contractor must show through a Time Impact Analysis and substantiate the impact to the critical path. No adjustment in the contract cost will be allowed or made for complying with any restrictions or requirements."

In this contract the Contractor is required to perform all excavations with archeological assistance (as prescribed by SABAP). The archeologist will be provided by the US Government and name of him/her will be provided to Contractor prior to start of excavation activities.

2.2.16 UXO Remediation
UXO remediation has not been prescribed by the Design Safety Coordinator (DSC) within the PSC (Piano Sicurezza e Coordinamento) as per Art.28 D.Lgs.81/2008.

2.2.17 Commissioning Report Provide a Commissioning Report if required by UFC 1-200-02 Chapter 2.

2.2.18 Energy Compliance Analysis
Provide an Energy Compliance Analysis if required by UFC 1-200-02 Chapter 4.

2.3 DESIGN SUBMITTALS

A full 100% design package has been already prepared through an AE contract and is attached in Part 6. Contractor is responsible to strictly follow and execute it.

2.4 SUSTAINABLE DESIGN

N/A

2.5 CYBERSECURITY

All Facility Related Control Systems (FRCS) that comprise a "system", an IT network that leaves a facility, is connected to other buildings, servers, workstations, or the internet through switches, fiber optics, copper, or wireless is considered an IT System and FRCS.

The current systems include; Fire Alarm Reporting System (FARS), Utility Monitoring and Control System (UMCS), Electrical Distribution System (EDS), and Access Control Systems (PACS, keycards), plus security systems such as Electronic Surveillance System (ESS), Closed Circuit Television (CCTV) and Intrusion Detection System (IDS).

All stations within USAG Italy are required to obtain, and currently have an Authority to Operate (ATO) under DOD Risk Management Framework (RMF) for all Facility Related Control Systems (FRCS). To maintain compliance with the ATO, all FRCS modifications, changes, or additions that occur during a building renovation, equipment replacement, or expansion of the system during installation of new equipment must follow the following process.

No FRCS changes are authorized without working through the System Owner, this includes installation and demolition. UMCS system connections already exist in the building mechanical room. No equipment not already included in the ATO, Network Diagram, and Hardware/Software (HW/SW) list is authorized without specific prior approval. All IT equipment must be specifically approved in writing in advance. IT equipment is anything that has an IP address, meaning it is connected to a switch or a controller through a fiber or Cat 6 (RJ 45) copper cable. Any item with an IP address is a Level 1 device and will be listed in the RMF documentation, including the network diagram and hardware/software list.

The system owners are DPW and DES for the Garrison. There is a full CCB process that will review and approve changes to the configuration. The Contractor shall submit their design and approach for approval both of the design and the changes to the FRCS. IT equipment must be procured through a process as well, including purchased by the Government through Army CHESS, or through a waiver approval process ITAS (Acquisition Supply) waiver. Additionally, any IT device connected to the FRCS will require technical compatibility approval through an ITTV (Technical Validation) if the item is not already present in the FRCS.

Due to the process noted above, the Contactor shall plan for purchase, waiver decisions and approvals at least 45 days prior to installation and system integration.

Items connected to the FRCS that are not IT equipment, such as a room or water temperature sensor, a smoke detector, or a motion detector, again, items without an IP address, are considered Level 0. These items are not required to be proprietary, nor included in the HW/SW list, and therefore not specific to the system ATO. The only requirement for Level 0 equipment is that it work properly with the specific IT equipment that is required by the system ATO.

Any building that is connected to the UMCS systems, which includes all of Del Din, most of Ederle, non-Housing buildings at Villaggio, and much of Darby and Depot needs to have the UMCS connected equipment considered as part of any repair and replacement of equipment, thermostats, and input sensors. The ABB Electrical Distribution System, Notifire Fire Alarm Annunciator System, Personnel Access Control System (Del Din) are handled similarly.

Contactors shall coordinate with the systems owner, prior performing any repairs, installation or removal of equipment tied to the Siemens system / sensors. All new equipment to be installed and connected to the UMCS shall be submitted for Government approval prior to installation. Submittals shall include connection information and compatible sensors. Installation and integration of all new equipment shall be coordinated and scheduled in advance.

To comply with Cybersecurity and to maintain configuration control and standardization of the UMCS, only authorized persons are allowed access to the UMCS. No UMCS data or files will be shared with any vendors or Contractors. No vendor or Contractor will be allowed to program the UMCS without prior authorization from the UMCS Engineer, and only then in the presence and direct supervision of an authorized Siemens technician.

Unauthorized access is a Cybersecurity violation that may result in disciplinary action to the employee and a negative report against the vendor or Contractor.

Examples:

- Replacement of a hot water storage tank. Prior notification of the work. Siemens technician removes the tank temperature sensor. Contractor can then remove the tank and reinstall a new tank, being provided with the necessary location and type of sensor to be connected. Contractor or Siemens connect the sensor and verify function in Desigo.

- Contractor renovates a few rooms of a building connected to the UMCS. Design and procurement of the thermostats, fan coils, and air handler must be coordinated in advance to ensure they will work properly, connecting to the building controller and programming in the Desigo must be done by persons authorized by the UMCS Engineer to operate on the UMCS and any software modification must be screened and approved by the S-6 system administrators.

The Contractor shall provide any concerns to the COR, through the formal RFI submittal process, preferably during the design phase, but not before procurement and installation of the work.

3.0 SITE ANALYSIS

Existing condition of site areas interested by this contract are within and outside Miotto military base in Longare. Work areas are inside wood areas and on sloped hillside and near and along roads so they will be accessible by construction site vehicles, but in some cases by small construction vehicles only. Items for particular consideration are noted below:

- All naturalistic interventions which will be executed in sloped areas and near perimeter fencing;
- Some work site areas are accessible by only unpaved roads.
- New retaining walls and manholes will be executed on sloped areas thereby increasing degree of access and work difficulty.
- During this work the access for emergency vehicles, to all Miotto buildings and infrastructure, shall remain open at all times.
- New maintenance roads will be created for future maintenance access to the new infrastructures.
- For the exterior works, all works need to have all preliminary authorization required by DPR 380/2001. Also, works will be performed in a consolidated residential area so all necessary considerations, such as noise and dust control, must be addressed by the contractor.
- During work outside Miotto base, another adjacent work site will be in progress for the renovation of the Comune di Longare sportfield complex. The Contractor l collaborate with the Comune di Longare’s Contractor to address any interferences that may arise.

Project works are within constraint areas so, the contractor shall take into account this aspect especially in regards to vegetation and trees which, before being removed, will require a prior authorization from Regione Veneto-Forest Agency.

3.1 EXISTING SITE CONDITIONS

Work areas are numerous and are within and outside Miotto military base, to include areas of landslides; roadways and other locations for storm water channel, piping, and lift station works; lamination tank, synthetic sport field, and TOC piping on properties outside of the installation. All works shall be accomplished in accordance with the 100% design (Part 6 of this SOW).

The Contractor will be responsible for providing all investigation, computation, labor, analysis, load test that he/she deems necessary for the execution of the project and shall perform all and any operations necessary and incidental to the project execution. Existing underground utilities system shall be investigated and confirmed by performing manhole openings, topographic surveys, and georeferencing of lines and manholes. Surveying shall be accomplished within and adjacent to areas of contract works, as necessary, to fully define existing site conditions.

Topographic survey Data: US Government performed a survey of all utilities present at Miotto base. Subject DWG file has been used by the US Government DOR during the design process. The contractor will be responsible to perform their own survey as verification of existing conditions and survey all details necessary to properly prepare shop drawings as required. Specifically, for works along Comune di Longare bike path and near/around the existing Vi. Acqua main public water pipe, coordination meetings with Vi.Acqua representatives and hand dug explorative excavations (saggi di scavo) will be required to be performed by Contractor.

4.0 BUILDING REQUIREMENTS

N/A. Anyway, if building will be impacted by works, they shall be finished in colors similar to the existing ones. Construction materials are anticipated to be traditional for the area.

4.1 REAL PROPERTY EQUIPMENT AND MATERIALS DATA

No real property equipment will be furnished by US GOV within this contract.

4.2 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide O&M data package 1-2-3-4-5 in accordance with Part 5 Section 01 78 23, except where specified otherwise.

4.3 OPERATION AND MAINTENANCE DATA

Provide operation and maintenance data in accordance with Part 5 Section 01 78 23, except where specified otherwise.

O&M Database;

Training Plan;

Training Outline;

Training Content;

Training Video Recording;

Validation of Training Completion;

4.4 CLOSEOUT SUBMITTALS

Provide closeout submittals in accordance with Part 5 Section 01 78 00, except where specified otherwise.

5.0 ROOM REQUIREMENTS

N/A

6.0 ENGINEERING SYSTEMS REQUIREMENTS (ESR)

A10 Foundations
A20 Basement Construction
B10 Superstructure
B20 Exterior Closure
B30 Roofing
C10 Interior Construction
C20 Stairs
C30 Interior Finishes
D20 Plumbing
D30 HVAC
D40 Fire Protection Systems
D50 Electrical
E10 Equipment
E20 Furnishings
F10 Special Construction
F20 Selective Building Demo
G10 Site Preparations
G20 Site Improvements
G30 Site Civil/Mechanical Utilities
G40 Site Electrical Utilities

A10 FOUNDATION

See 100% design package attached in Part 6

A20 BASEMENT CONSTRUCTION

N/A

B10 SUPERSTRUCTURE

N/A

B20 EXTERIOR ENCLOSURE

Work on exterior enclosure are not applicable to the project, however any alteration and damage must be repaired. Remove dirt, loose particles, grease, mold, rust, and other foreign matter and substances deleterious to the restoration of the areas affected by the project. Cracking, flaking and peeling or other deteriorated painting shall be removed. Painting and plaster damaged by, but not limited to, moisture, holes and cracks shall be removed including adjacent adherent painting and plaster. Restore the areas with suitable material to match nearby undamaged areas. Surfaces damaged by rust, fungus and mold shall be cleaned.

B30 ROOFING

N/A

C10 INTERIOR CONSTRUCTION

N/A

C20 STAIRS

N/A

C30 INTERIOR FINISHES

N/A

D20 PLUMBING

N/A

D30 HVAC

N/A

D40 FIRE PROTECTION SYSTEMS

N/A

D50 ELECTRICAL

N/A

E10 EQUIPMENT

N/A

E20 FURNISHINGS

N/A

F10 SPECIAL CONSTRUCTION

See 100% design package attached in Part 6

F20 SELECTIVE BUILDING DEMOLITION

N/A

F2020 HAZARDOUS COMPONENT ABATEMENT

See 100% design package attached in Part 6

G10 SITE PREPARATION

See 100% design package attached in Part 6

G20 SITE IMPROVEMENTS

See 100% design package attached in Part 6

G30 SITE CIVIL/MECHANICAL UTILITIES

See 100% design package attached in Part 6

G40 SITE ELECTRICAL UTILITIES

See 100% design package attached in Part 6

Part 4 Minimum Materials, Engineering and Construction Requirements
1.0 GENERAL REQUIREMENTS

The requirements indicated here are minimum performance requirements. More specific project functional and performance requirements, scope items and expected quality levels over and above the standards in Part 4 are identified in Part 3 of the Request for Proposal or Basic Ordering Agreement. The Contractor is encouraged to exceed the minimum requirements. The Contractor's performance evaluation will be based in part on enhancements to materials, engineering, design and construction provided for the contract that exceed minimum requirements.

Part 4 is a general section. Not all items in Part 4 will be required for this project. See Part 3 for project-specific requirements.

In general, unless otherwise indicated, provide all labor, equipment and materials necessary to complete the work required for the contract. All work must be in conformance with all applicable referenced criteria, construction standards, laws and regulations, including applicable building fire and life safety codes.

Recycled Materials Considerations:

An Affirmative Procurement Program has been established within the Federal government to promote the purchase of products containing recovered materials. This program promotes the purchase of products containing materials recovered from the solid waste stream. The intent is to conserve resources and reduce solid waste by developing markets for recycled products and encouraging manufacturers to produce quality recycled content products. Use products that meet or exceed the EPA guideline standards for recovered content as required by the Federal Acquisition Regulations (FAR). Availability lists of manufacturers and EPA research on product usage are on the Construction Criteria Base (CCB) at www.wbdg.org/ccb/ccb.php under Documents Library, NAVFAC Criteria. A partial list of products containing recycled materials for possible use is as follows:

- Rock Wool Insulation
- Fiberglass Insulation
- Cellulose Insulation
- Structural Fiberboard and Laminated Paperboard
- Cement and Concrete - Coal Fly Ash
- Carpet including backings and cushions

- Floor Tiles
- Reprocessed and Consolidated Latex Paint
- Crushed Concrete Aggregate for new asphalt, concrete or subgrade
- Recycled glass for terrazzo aggregate
- Acoustical Ceiling Tile
- Gypsum Wallboard
- Steel wall studs
- Cellulose spray applied fireproofing
- HDPE Toilet Partitions

1.1 MATERIALS AND METHODS OF CONSTRUCTION

Only new materials and equipment are to be installed in the work. All materials, equipment and appliances must be of the current manufacturers' products. Do not use obsolete or discontinued materials, equipment and appliances, except that construction materials containing recycled content as described in Paragraph 1 of this Part that completely comply with all materials specifications found elsewhere in this Part may be used.

1.2 APPLICABLE CODES AND STANDARDS

The design and construction must be in accordance with established construction practices, and the latest revision/edition of the following referenced codes and standards. The term "Latest Revision/Edition" is defined as the version as of the project award date. References are available at www.wbdg.org/ndbm/ . The advisory provisions of all codes and standards is mandatory, as though the word "must" had been substituted for "should" wherever it appears. Reference to the "authority having jurisdiction" is to be construed to mean "Contracting Officer". Comply with the required and advisory portions of the current edition of the standard at the time of contract award. All work must comply with UFC 1-200-01, General Building Requirements, and IBC 2009 or later edition as modified by applicable NFPA Standard as well as codes and standards listed in RFP Part 2.

1.3 LOCATION-SPECIFIC CODES AND STANDARDS

See Part 3.

1.4 DISCREPANCIES

When discrepancies in the referenced standards and the contract requirements occur, the more stringent requirements govern. The word "should" in all NFPA publications is to be interpreted as a requirement. The Authority Having Jurisdiction in the interpretation of the codes and standards, and approving the exceptions allowed in the referenced standards, is the Contracting Officer, and the parties designated by the Contracting Officer.

2.0 PERFORMANCE TECHNICAL SPECIFICATIONS

Note: The paragraph numbers used correspond with the numbers used in UNIFORMAT II/Work Breakdown Structures (WBS) as listed in the Whole Building Design Guide, Navy Design Build Master, accessible at this website: www.wbdg.org/ndbm .

SECTION A. SUBSTRUCTURE

A10 FOUNDATION

Foundations must be reinforced concrete slabs-on-grade with continuous strip footings or isolated spread footings. Concrete slabs must not be less than 4 inches in thickness and footings must not be less than 18 inches below the lowest adjacent grade. Design and construct foundations of reinforced concrete. Comply with IBC, applicable UFGS and with applicable requirements in Section B Shell. For the purposes of interpreting IBC Chapter 18, the "Owner" and "Building Official" is interpreted to mean the "Government", and the "Applicant" is interpreted to mean the "Contractor/Designer of Record".

1. Contractor-Foundation Design: The Designer of Record must evaluate the RFP data, and obtain and evaluate all additional data as required to support the design and construction.
2. Geotechnical Site Data required in Design Drawings: The Contractor's final design drawings must include:
    a. Notes identifying the soil allowable bearing capacity used in design./li>
    b. Subsurface soil information, be it Government provided or Contractor obtained, that represents subsurface conditions existing on the project site (such as boring logs, test pits, laboratory test results and groundwater observations). The locations of all borings must be indicated on the drawings.
3. Performance Verification and Acceptance Testing: Verification of satisfactory construction and system performance is required to be via Performance Verification Testing, as detailed in this part of the RFP.
    a. Earthwork: Perform quality assurance for earthwork in accordance with IBC Chapter 17 and applicable UFGS. See Section G1030.

A20 BASEMENT CONSTRUCTION

[Provide basement construction in accordance with UFC 3-301-01, Structural Engineering. Basement walls include exterior walls below the ground floor level of the building, including walls that are below grade, elevator pits and other pits. Provide basement walls constructed of [cast-in-place concrete] [precast concrete] [or] [masonry].

Provide waterproofing [and insulation] of basement walls.] If basement construction is required in Part 3, refer to Standard Design-Build template PTS Section A20 BASEMENT CONSTRUCTION.

SECTION B. SHELL

B10 SUPERSTRUCTURES

Superstructure work includes structural frames, bearing walls, floors, roofs, roof canopies, and balcony construction. Unless otherwise specified in Part 3, superstructures may be designed and constructed using any materials or combination of different materials allowed by applicable codes and standards. Comply with IBC and applicable UFGS. Special inspection, testing, approvals, certifications, observations and quality assurance plans as prescribed in Chapter 17 of the IBC are required.

1. Concrete: All concrete must be constructed in accordance with ACI 301. Concrete must have a 28-day minimum compressive strength of 3,000 psi. Slump must be between 2 and 4 inches in accordance with ASTM C143. Provide joints as required to minimize cracking. All concrete must be reinforced. Provide joints as required by applicable ACI standards. Unless otherwise specified in Part 3 or as indicated by the contracting officer, provide steel trowel finish for all exposed floor surfaces. Exterior surfaces must be a broom finish.

2. Masonry:

    a. All concrete masonry must be constructed in accordance with ACI 530.1. Concrete masonry must have a minimum 28-day compressive strength of 1500 psi. Concrete masonry units must conform to ASTM C90, grade A1. Broken blocks are not allowed. Use only standard size and shape blocks. Block may be cut when necessary. Mortar must be Type S.

    b. When used, brick must conform to ASTM C216. In exposed construction, broken brick is not be allowed. Standard size brick may be cut to fit job condition. Use Type S mortar.

    c. Provide metal anchors for masonry and brick, including veneer construction as required by IBC.

3. Structural Steel: Structural steel exposed to weathering must be adequately protected to prevent corrosion.

4. Steel deck: Steel form deck must have a G90 galvanized finish, and must have a minimum 26-gage thickness. All other steel deck must have a G90 galvanized finish, and must have a minimum 20-gage thickness.

5. Cold-formed metal framing: Cold-formed steel studs, joists and track must be galvanized with a minimum thickness of 20-gage.

6. Wood framing: Wood framing members must be new lumber, unless otherwise allowed by Part 3. Timber can be Douglas Fir, Douglas Fir-Larch, Hem-Fir, Southern Pine or other structurally competent species allowed by applicable codes and standards. Wood framing must meet the following minimum grading requirements:

   a. Studs - #2

   b. Joists and rafters- #2

   c. Beams, 4x and larger - #1

   d. Posts, 4x and larger - #1

   e. Blocking - #3

   f. Fascia, trim - #1

   g. Wood Structural Panel Sheathing (Exterior Glue)

   h. Roof - APA rated with span index of 24/0 - minimum thickness 1/2 inch

   i. Walls - APA rated with span index of 32/16 - minimum thickness 1/2 inch

   j. Flooring- APA rated with span index of 48/24 - minimum thickness 3/4 inch

B20 EXTERIOR ENCLOSURE

B2010 EXTERIOR WALLS

1. Exterior Wall Performance:

   a. Vapor Transmission Analysis: Perform a job specific vapor transmission analysis in accordance with ASHRAE 90.1 or WUFI. The conclusion of the analysis must indicate the appropriate locations of needed vapor retarders, air barriers, and anticipated dew-point locations in the exterior enclosure during different critical times of the year.

b. Maximum Air Infiltration: The air leak flow rate must not exceed 0.25 CFM at 75 Pa per square foot (0.076 cm 75 Pa per square meter) of building envelope area including roof or ceiling, walls and floor as provided by the DOR.

Where required in RFP Part 3, provide air barrier testing. Perform testing as required by UFGS 07 08 27.00 10, Building Air Barrier System Testing for Commissioning. DOR must edit this section and incorporate into the project specification. Repair leaks and repeat testing until prescribed maximum air leak flow rate is achieved. Provide intermediate and final reports.

c. Wind Loads: Provide wind load calculations for exterior cladding in accordance with ASCE-7 with comparative analysis of the cladding system to be provided.

d. Water Penetration: No water penetration is acceptable at a pressure of 39 Kg/m2 (8 psf) of fixed area when tested in accordance with ASTM E 331.

e. Insulating Value: Provide complete thermal envelope in accordance with ASHRAE 90.1, Chapter 5 with improvements required to meet project energy goals.

Where required in RFP Part 3, provide infrared thermal envelope performance testing. Test the building envelope using Infrared Thermography in accordance with the requirements of ASTM C1060 (latest edition) and ISO 6781. The Contracting Officer will witness the testing. Provide thermography test report including thermographs in color and a color temperature scale to define the temperature indicated by the various colors. The report must identify the high temperature reading, the outdoor air temperature, the building indoor air temperature, and the wind speed and direction. Report to note any areas of compromise in the building envelope, and note all actions required and taken to correct those areas. Repair and repeat testing until discrepancies are demonstrated to be resolved.

2. Masonry Veneer Exterior Wall Closure Components: Masonry veneer includes load bearing and non-load bearing exterior walls of the structure, and must include colored mortar, special shapes such as sills, headers, trim units and copings of brick masonry, precast concrete, concrete masonry units, or other approved material. Utilize BIA Technical Notes to design, detail, and construct brick masonry walls. Substitute directive language in the place of BIA suggestive language. The results of these wording substitutions change this document to required procedures. Tie the veneer to the backup wall system with a system that allows the veneer to move independently of the backup wall system, while being

structurally supported. The masonry veneer must allow for expansion and contraction of the veneer without cracking the exterior material.

a. Masonry Veneer Installation: Conform to ACI 530.1 for masonry veneer installation, including cold weather construction. Antifreeze admixtures are not to be used.

b. Mortar: Provide factory-tinted colored mortar conforming to ASTM C270, unless DOR directs otherwise.

c. Expansion/Control Joints: Locate expansion/control joints and seal with proper backing material and ASTM C 920 polyurethane sealant, or preformed foam or rubberized expansion joint closure. Conform to UFC 3-101-01, Architectureand BIA Technotes 18, 18A.

d. Brick: Meet ASTM C216, Grade SW, type FBS, or type FBX for detail work. ASTM C67 test rating shall be "Not effloresced". Use FBA brick only for special architectural effects requiring a non-uniform size.

e. Split Faced or Ground Faced Masonry: ASTM C 90

f. Cast Stone Trim Units: Cast Stone must meet or exceed the requirements of ASTM C 1364.

g. Wall Cavity: shall Comply with the and BIA Technical Notes 21A, 21B, 21C, 28B

h. Through-Wall Flashing Components: Through-wall flashing with weep holes must be incorporated in cavity wall construction. Flashing must be 7 ounce copper flashing with a 3 ounce bituminous coating on each side or a fiberglass fabric bonded on each side of the copper sheet; 16-ounce uncoated copper, 28 gauge Type 302 or 304 stainless steel is also acceptable. 'Flexible membrane flashing, plastic or PVC-based membrane flashing is prohibited.

i. Reinforcing in Veneer Layer: Reinforcing in the veneer layer must be galvanized in accordance with ASTM A 123/A123M, ASTM A153/A153M, or ASTM A653/A653M, Z275 (G90) coating, and be of sufficient size to eliminate damage to the veneer layer from wind and other live and dead loads imposed on the veneer layer.

j. Masonry Cleaning: Clean the masonry in accordance with manufacturer's instructions and BIA Technote 20.

3. Metal Wall Panel Exterior Closure

Panels must have factory applied, baked coating to the exterior and interior of metal wall panels and metal accessories. Exterior finish topcoat must be of 70 percent polyvinylidene fluoride (PVDF) resin with not less than 0.8 mil dry film thickness (DFT). Exterior primer must be standard with panel manufacturer with not less than 0.8 mil dry film thickness (DFT).

Wall system and attachments must resist wind loads as determined by ASCE 7, with a factor of safety appropriate for the material holding the anchor. Maximum deflection due to wind on aluminum wall panels must be 1/60. Maximum deflection due to wind on steel wall panels and girts behind aluminum or steel wall panels must be limited to 1/120 of their respective spans, except that when interior finishes are used the maximum allowable deflection must be limited to 1/180 of their respective spans.

Conformations - Non-insulated steel or aluminum wall panels must have configurations for overlapping adjacent sheets or interlocking ribs for securing adjacent sheets and must be fastened to framework using concealed fasteners, or choose the option for exposed fasteners when exposed fasteners are acceptable at the installation. Length of sheets must be sufficient to cover the entire height of any unbroken wall surface.

a. Steel Wall Panels:

1) Material and Coating: Form sheets from steel conforming to ASTM A 653/A 653M, Structural Grade 40, galvanized coating conforming to ASTM A 924/A 924M, Class G-90; aluminum-coated steel conforming to SAE AMS 5036; or steel-coated with aluminum-zinc alloy conforming to ASTM A 792/A 792M, except that coating chemical composition must be approximately 55 percent aluminum, 1.6 percent silicon, and 43.4 percent zinc with minimum coating weight of 0.5 ounce per square foot.

2) Gage: Minimum 22 U.S. Standard Gage for wall panels, but in no case lighter than required to meet maximum deflection requirements specified.

b. Aluminum Wall Panels:

1) Material and Coating - Form sheets of Alloy 3004 or Alclad 3004 conforming to ASTM B 209 having proper temper to suit respective forming operations.

2) Thickness - Minimum 0.81 mm (0.032 inch) nominal, but in no case thinner than that required to meet maximum deflection requirements specified.

c. Insulated Aluminum or Steel Wall Panels: Insulated wall panels must be steel or aluminum factory-fabricated units with insulating core between metal face sheets securely fastened together and uniformly separated with rigid spacers. Panels must have a factory color finish. Wall panels must have edge configurations with interlocking ribs for securing adjacent panels. System must utilize factory fabricated corners and trim pieces at intersections with other materials. Insulated wall panels must be fastened to framework using concealed fasteners.

1) Insulated Steel Panels - Zinc-coated steel conforming to ASTM A 653/A 653M; or Aluminum-zinc alloy coated steel conforming to ASTM A 792/A 792M, AZ 55 coating. Uncoated wall panels must be 0.61 mm (0.024 inch) thick minimum.

2) Insulated Aluminum Panels - Alloy conforming to ASTM B209, temper as required for the forming operation, minimum 0.81 mm (0.032 inch) thick.

4. Stucco Exterior Wall Closure

a. Portland Cement Plaster: ASTM C150, gray Portland cement Type II with 13 mm (1/2 inch) maximum chopped alkali resistant fiberglass strands, minimum 1.5 percent by weight to cement; .68 kg (1 1/2 pounds) per sack of cement. Lime must conform to ASTM C206, Type S. System must utilize stainless steel or zinc corner beads, J-beads and other accessories. Unless specifically deleted, the system must utilize an acrylic admixture or coating to give additional moisture suppression to control fungus growth.

b. Exterior Insulation and Finish System (EIFS): EIMA TM 101 and 01 EIMA TM 101.86. EIFS must be used as the non-primary or the primary exterior finish material only for projects where it is necessary to match existing EIFS.

5. Precast Concrete Wall Panels: ACI 211.1 and ACI 301. PCI MNL-116 or PCI MNL-117. Concrete must have a minimum 28-day compressive strength of 281 Kg/cm2 (4000 psi). Joints must include properly sized and placed backing material and fully loaded and tooled sealant joint of no less than 1/4 inch sealant material thickness.

6. Other Wall Finish Systems

a. Horizontal Wood Siding: Horizontal Wood Siding: DOC PS 20, exterior, lap type, 6 inches wide, maximum practicable lengths, 11 mm (7/16 inch) thick, smooth face. All surfaces of wood siding and trim must be shop coated with an alkyd primer.

Species and Grades 1. Grade 1 Common spruce-pine-fir; NELMA, NLGA, WCLIB, or WWPA. 2. Grade Prime or D finish, pressurepreservative-treated hem-fir; NLGA, WCLIB, or WWPA.; 3. Grade D Select (Quality) eastern white pine, eastern hemlock-balsam fir-tamarack, eastern spruce, or white woods; NELMA, NLGA, WCLIB, or WWPA. 4. Grade D Select northern white cedar; NELMA or NLGA. 5. Grade B & B, pressure-preservative-treated southern pine; SPIB.

b. Vinyl Siding System: Integrally colored, vinyl siding complying with ASTM D 3679.

c. Manufactured Faced Panels Systems Exterior Wall Siding: Glass Fiber Reinforced Cementitious Panels System: Siding made from fibercement board that does not contain asbestos fibers; complies with ASTM C 1186, Type A, Grade II; horizontal or vertical pattern in plain or beaded-edge style. Texture: Rough sawn or smooth, factory primed.

7. Exterior Wall Backup Construction

a. Concrete Unit Masonry: Provide concrete unit masonry to comply with ACI 530.1. Load-bearing units: ASTM C90, Non-load bearing- units: ASTM C129, Type I or II. Provide ground face units, split-faced units, ground-faced units, or split-ribbed units for exposed exterior walls. Provide water repellent admixture to masonry units where the exterior face of the units will not receive a waterproof coating such as paint

b. Dampproofing: Dampproof the cavity-facing wythe of the backup masonry using asphaltic primer according to ASTM D 41, if dampproofing is not provided by a sprayed on foam or other DORapproved membrane insulation system.

8. Load-Bearing Metal Framing System
If permitted, provide load-bearing metal framing including top and bottom tracks, bracing, fastenings, and other accessories necessary for complete installation. Framing members must have the structural properties indicated. Where physical structural properties are not indicated, they must be as necessary to withstand all imposed loads. Design framing in accordance with AISI SG-673. Install in accordance with DOR-approved shop drawings and manufacturer's installation instructions.

9. Exterior Studs:

Max. Deflection Criteria	Exterior Finish
L/360	Cement Plaster, Wood Veneer, Synthetic Plaster, Metal Panels

L/600 Brick Veneer, Stone Panels

Wall deflections must be computed on the basis that studs withstand all lateral forces independent of any composite action from sheathing materials. Studs abutting windows or louvers must also be designed not to exceed 1/4-inch maximum deflection and as required in UFC 4-010-01, DoD Minimum Antiterrorism Standards for Buildings.

1) Studs - ASTM A 1003/ASTM A 1003M, Structural Grade 50, Type H minimum; provide Z180 (G60) galvanized coating in accordance with ASTM A 653/ASTM A 653M. Do not expose studs to direct moisture contact2) Bracing - Provide horizontal bracing in accordance with design calculations and AISI SG673, consisting of, as a minimum, runner channel cut to fit between and welded to the studs.3) Sheathing - Provide sheathing to withstand structural loads imposed on the wall structure. Cover sheathing with either a 15 pound asphaltimpregnated building paper, or air barrier as required by the wall moisture analysis. Sheathing must be one of the following:a) Plywood: C-D Grade, Exposure 1;b) Structural-Use and OSB Panels;c) Gypsum: ASTM C 79/C

79M and ASTM C 1177/C 1177M, 13 mm (1/2 inch) thick fire retardant (Type X) 15 mm (5/8 inch) thick; 1.2 meters (4 feet) wide with square edge for supports 400 mm (16 inches) o.c. with or without corner bracing of framing. Gypsum sheathing must be faced with materials capable of resisting six months of weathering exposure without degradation of the covering or the gypsum. Seal all joints as recommended by the manufacturer.

10. Wood Framing System: All materials shall be kiln-dried lumber complying with DOC PS 20. Installation must be in accordance with AF&PA T11. Use preservative pressure treated lumber at sill plates and other members in contact with concrete and masonry surfaces.

    a. Species and Grades: Provide species and grades listed: 1) Grade 2 Common spruce-pine-fir; NELMA, NLGA, WCLIB, or WWPA: 2) Grade 2 Common, hem-fir; Douglas-fir; NLGA, WCLIB, or WWPA; 3) Grade 2 Common, southern pine; SPIB.

    b. Sheathing: Sheathing must withstand structural loads imposed on the wall structure. Cover sheathing with either a 15 pound asphalt-impregnated building paper, or air barrier as required by the wall moisture analysis. Sheathing must be as for Metal Studs.

11. Cast-in-place Concrete System: Concrete construction must be in accordance with ACI 301.

12. Insulation and Vapor Retarder: Insulation, Vapor Retarders, and Air Barrier Systems in or on Exterior Enclosure must include: insulation, liquid, sheet or

continuous film materials installed separately in or on wall assemblies to provide resistance to heat loss/gain, and vapor penetration.

a. Vapor retarder: Comply with ASTM C755. Incorporate in the exterior wall system where required by vapor transmission calculations or dew point analysis indicates the need or in conditions of high moisture exposure.

b. Bituminous Dampproofing: Bituminous Dampproofing must be ASTM D449, Type I or Type II bituminous dampproofing on the exterior surface of the interior wythe of masonry in a cavity wall (back-up wall for masonry veneer).

c. Building Paper: FS UU-B-790, Type I, Grade D, Style 1.

d. Air Barrier: Building wrap consisting of air barrier sheeting complying with ASTM E 1677, Type 1, not less than 3 mils thick with a permeance of not less than 575 ng/Pa x s x sq.m. (10 perms). Building wrap must have a flame spread index of less than 25 in accordance with ASTM E 84. Provide building wrap over sheathing of wood or metal framed construction to reduce air penetration and airborne vapor penetration. Provide building wrap tape as recommended by the manufacturer for sealing all joints in the building wrap. Install in accordance with manufacturer's instructions. Air barrier installation at windows must be in accordance with ASTM E 2112.

e. Insulation Systems: Vertical and horizontal polystyrene insulation conforming to ASTM C578 or rigid polyisocyanurate board wall insulating products conforming to ASTM C591 or mineral-fiber blanket insulation conforming to ASTM C 665 must be provided.

13. Parapets: Avoid parapets when possible, but when necessary, provide parapets with the same materials as the exterior wall construction. Provide scuppers and wall edge according to SMACNA.

14. Exterior Louvers and Screens: If required, provide louvers for Screened Equipment Enclosure or as louvers for exterior doors.

    Storm shutters must comply with ASTM E 1996-03.

15. Balcony Walls and Handrails: Balcony walls to match exterior construction. Handrails to comply with the IBC and OSHA.

16. Exterior Soffits: Exterior soffit system.

17. Exterior Painting and Special Finishes; All painting and coating materials must be low VOC. Painting practices must comply with applicable federal, state and local laws enacted to insure compliance with Federal Clean Air Standards. Apply

coating materials in accordance with SSPC PA 1. SSPC PA 1 methods are applicable to all substrates.

All paint must be in accordance with the Master Painters Institute (MPI) standards for the exterior architectural surface being finished. The current MPI, "Approved Product List" which lists paint by brand, label, product name and product code as of the date of contract award, will be used to determine compliance with the submittal requirements of this specification. Provide paint systems tested to "Detailed Performance Level" standard as defined by MPI.

18. Exterior Joint Sealant: Sealant joint design, priming, tooling, masking, cleaning and application must be in accordance with the general requirements of Sealants: A Professionals' Guide from the Sealant, Waterproofing & Restoration Institute (SWRI). All sealant must conform to ASTM C 920.

19. Sun Control Devices: Sun control devices must be manufactured devices to provide sun control on exterior windows and storefronts. Sun control devices must be designed and installed to withstand the wind loads prevailing at the project site.

B2020 EXTERIOR WINDOWS

All windows and doors in new or existing buildings, which are subject to Anti-terrorism Standards, must be blast-resistant as prescribed in UFC 4-010-01, DoD Minimum

Antiterrorism Standards for Buildings.

Unless otherwise allowed by Part 3, windows for new facilities must be aluminum. In building additions or renovations windows must match existing window materials, except when all windows are to be replaced. If all windows are to be replaced, they shall be aluminum. Exterior windows design, dimension, and construction must meet or exceed the requirements for Anti Terrorism Force Protection requirements. In addition, exterior windows must meet or exceed Energy Star requirements. The design and placement of exterior windows must take into considerations view, natural light, privacy, and protection for the occupants of the facilities. Provide operable hardware and insect screen for exterior windows. Windows must be fabricated by manufacturers normally involved in the manufacturing of windows and be of the current make and model. Do not use obsolete or discontinued windows. Provide weather stripping, STC and IIC rating, commensurate with the intended use of the facility. Submit catalog information and manufacturer's specifications for approval by Contracting Officer prior to purchase of windows.

All window assemblies must meet performance grade CW 60 minimum tested in accordance with AAMA/WDMA/CSA 101/I.S.2/A440-08 or most current edition of this standard.

Where windows separate conditioned spaces from non-conditioned spaces, provide windows bearing NFRC energy label indicating window exceeds current EnergyStar criteria. For storefront or curtainwall systems, provide thermally broken framing and insulating glazing with whole-assembly U-value of 0.40 or less. Provide windows exceeding requirements of ASHRAE 90.1, Table 5.5 for project climate zone.

Windows must consist of fixed and operable sash used singly and in multiples. Provide operable sash in spaces occupied by people as a minimum. Include operating hardware, non-corroding framed metal screens for operable sash, integrated blinds set between glass panels and security grilles. Provide jamb support for larger windows where recommended by manufacturer.

1. Metal Windows: All windows must conform to ANSI/AAMA/WDMA 101. Metal windows with insulating glass must have thermally broken frames and sash. Factory finish aluminum windows and provide with aluminum frame screens with aluminum mesh at operable sash,

2. Storefronts: Provide one-story storefront system fabricated from formed and extruded aluminum and glass components for exterior use. Utilize the specific section of the Standard Design-Build Performance Technical Specifications Section B202002 for the storefront to be provided. Storefront framing must meet or exceed the structural requirements, as measured in accordance with ANSI/ASTM E330: Design system to withstand this as a minimum and comply with design pressure established within the required ASCE 7-05 Wind Speed Calculations determined by the overall average opening within the project.

3. Glazing: All exterior glazing must be insulating glass.

    a. Clear Glass - Type I, Class 1 (clear), Quality q4 (A);

    b. Heat-Absorbing Glass - ASTM 1036, Type I, Class 2 Quality q3 (select) ray frames;

    c. Wire Glass - Type II, Class 1, Form 1, Quality q8 Mesh m1 or Form 2, Quality q7;

    d. Laminated Glass - ASTM 1172, total thickness shall be nominally 6 mm (1/4 inch);

    e. Insulating Glass Units - Typically ASTM C 1036, Type I, Class 1, Quality q4, minimum 6 mm;

    f. Tempered Glass - ASTM C 1048, Kind FT (fully tempered);

    g. Patterned Glass - ASTM 1036, Type II, Class 1 (translucent), Form 3 (patterned), Quality q7 (decorative), Finish f1 (patterned one side), Pattern p2 (geometric) 5.55 mm (7/32 inch) thick.

B2030 EXTERIOR DOORS

Exterior doors must be heavy duty insulated steel doors and frames for service access. Provide door frames with welded corners. Use heavy-duty overhead holder and closer to protect doors from wind damage. Steel must have G60 galvanized coating in accordance with ASTM A 924/A 924M and ASTM A 653/A 653M when the job site is located within 300 feet from a body of salt water. Provide commercial quality, coating Class A zinc coating in accordance with ASTM A591 for other steel or steel skin hollow metal doors at other locations. Provide kickplates on the inside face of all exterior doors. Weatherprotect all exterior doors and related construction with low infiltration weatherstripping and sealants. Provide threshold with offset to stop water penetration while maintaining accessibility compliance. Conform to the design criteria of ASCE 7. See the hardware schedule for door hardware requirements.

Where doors separate conditioned spaces from non-conditioned spaces, provide doors exceeding current EnergyStar criteria. For storefront entrances, provide thermally broken door framing and insulating glazing with whole-assembly U-value of 0.67 or less. Provide doors exceeding requirements of ASHRAE 90.1, Table 5.5 for project climate zone.

1. Steel Doors: Exterior doors must comply with ANSI A250.8-1998 (SDI-100). Hardware preparation must be in accordance with ANSI A250.6. Doors must be hung in accordance with ANSI A115.16.

   a. Doors Required:

1) Standard Duty Doors - Level 1, MSG # 20 (IP 0.032", 0.8 mm), physical performance Level C, Model 1 or 2.2) Heavy Duty Doors - MSG # 18 (IP

0,042", 1 mm), physical performance Level B, Model 1 or 2.3) Extra Heavy Duty Doors - Level 3, MSG #16, (0.053", 1.3 mm) physical performance Level A, Model 1, 2, or 3.4) Maximum Duty Doors - Level 4 (IP 0.067", 1.6 mm), physical performance Level A, Model 1 or 2.

   b. Insulated steel doors and frames are required for entrances to dwelling units, and may also be specified as a Contractor's option to Level 1 standard hollow metal doors. Do not use wood doors for exterior doors, unless they are fully protected from the elements, an exterior grade species, and specially finished. If wood doors are used, provide in accordance with Standard Design-Build Performance Technical Specification Paragraph B203001 2.

2. Standard Steel Frames: ANSI A 250.8. Form frames with welded corners for installation in exterior walls. Form stops and beads of 20 gage steel. Frames must be set in accordance with ASTM A250.11. Anchor all frames with a

minimum of three jamb anchors and base steel anchors per frame, zinc-coated or painted with rust-inhibitive paint, not lighter than 18 gage. Mortar infill frames in masonry walls, and infill with gypsum board compound at each jamb anchor in metal frame walls. Only use surface exposed bolted anchors in concrete walls.

3. Door and Frame Finishes: a) Exterior Doors, Factory-Primed and Field Painted Finish - Doors and frames must be factory primed with a rust inhibitive coating as specified in ANSI A250.8. Factory prime doors on six sides of the door; b) Exterior Doors Galvanized Finish -- Must be Commercial Quality, Coating Class A, zinc coating in accordance with ASTM A 591 when facility is located further than 91 meters (300 feet) from the ocean. When facility is located within 91 meters (300 feet) of the ocean, provide G60 galvanized coating in accordance with ASTM A 924/A 924M and ASTM A 653/A 653M.

4. Upward Acting Doors: Upward acting doors must be capable of withstanding the design wind loading of ASCE 7. Provide galvanized steel tracks not lighter than 14 gage for 50 mm (2 inch) tracks and not lighter than 12 gage for 75 mm (3 inch) track. Provide a positive locking device and cylinder lock with two keys on manually operated doors.

5. Overhead and Roll-up Doors: Large exterior overhead and roll-up doors system must consist of manual or automatic exterior doors and door assemblies.

6. Rolling Service Doors and Grilles: Coiling overhead doors must have minimum 22 gage thermal insulated slats. Electric operators must have three-button switches conforming to NEMA MG 1, NEMA ICS 1, and NEMA ICS 2, and auxiliary hand chain operation, weather-stripping and wind-locks. Doors must be capable of withstanding the design wind loading of ASCE 7 and still operate normally. Finish of the door must be hot-dipped galvanized with a painted finish.

7. Sectional Overhead Doors: Sectional overhead doors must conform to NAGDM 102, Residential or Commercial or Industrial door standards. If doors

are electrically operated, pushbuttons must be full-guarded to prevent accidental operation, and include limit switches to automatically stop doors at the fully open and closed positions. Limit switch positions must be readily adjustable.

8. Hardware: Provide the services of a Certified Door Hardware Consultant to prepare the door hardware schedule.

Provide all new hardware with satin chrome finish throughout. Hardware must be commercial grade, suitable for the operational requirements and in compliance with life safety code and handicapped accessibility requirements, similar in quality to the hardware shown in C1020 Interior Doors and Hardware below.

Coordination: Provide a master keying system compatible with the existing base system. Provide an emergency access key box for exterior door fireman key access. Coordinate with the local authority and the Contracting Officer to determine the local requirements for hardware, keying and master keying.

B30 ROOFING

For repair of existing roofing, the cutting of the existing roof shall be kept to a minimum and, where necessary, must be made in a clean and orderly manner to prevent the appearance of a patch.

Repair all damage to existing and new roofing caused by the work of this Contract at no additional cost to the Government. The work must be executed in such a manner as to maintain the integrity of the existing roofing manufacturer's warranty.

1. Pre-Roofing Conference: Prior to beginning roofing work, hold a Pre-Roofing Conference with the personnel directly responsible for the roofing systems work, as well as the roofing manufacturer's technical representative.

2. Roof Design Assurance: If the roofing project is significant (Significant Roof - A single or group of buildings greater than 1,400 m^2 (15,000 sf)), or where extenuating circumstances of the roof project such as building use, content, safety, or visibility require a roofing consultant, provide the services of a Registered Roof Consultant (RRC) certified by the Roof Consultant Institute, or a Registered Professional architect or Engineer who specializes in roofing, to approve the roof design. The roof consultant must be engaged in roofing design and roofing construction as his primary endeavor. The roof consultant must verify in writing that the design for the project is in accordance with the current edition of NRCA Roofing and Waterproofing Manual, UFC's, and RFP, and standard industry practices and building codes.

   If a Roof Design Assurance Consultant is needed, consider using a Registered Roof Observer as a QC specialist.

B3010 ROOF COVERINGS

Roof coverings and procedures must comply with the requirements of UFC 3-110-03, Roofing, and NRCA, Roofing and Waterproofing Manualfound at http://www.nrca.net/rp/technical/manual/manual.aspx as the primary NAVFAC roofing criteria. Roof selection must comply with UFC 3-110-03, Roofing. Determine wind uplift using wind speed in accordance with ASCE-7.

1. STEEP SLOPE ROOF SYSTEMS: Steep slope systems bare roofs with a pitch greater than 3 in 12. Steep Slope Systems are slate roofing, Asphalt Shingles, Roof Tiles, Foam Set Tiles, Metal Roof Panels (Architectural Standing Seam

Metal Roofs on supported substrate), and Structural Standing Seam Metal Roof (SSSMR). Asphalt shingles can only be used for residential construction and light commercial construction.

2. LOW SLOPE ROOF SYSTEMS: Low slope systems are roofs with a pitch 3 in 12 or less. Low slope roofing systems must be built-up asphalt roofing (aggregate surfaced, with modified bituminous components), modified bituminous membrane roofing of a minimum of 3 plies with aggregate surface or granular surface modified bitumen cap sheet, or structural standing seam metal roofing. Use EPDM systems only to match existing construction.

3. ROOF COMPONENTS:

   a. Insulation: For existing structures, provide insulation in accordance with ASHRAE 90.1. For new construction, provide R-30 insulation in the ceilings, attic spaces and soffit areas for interior spaces. Injected polyurethane and Urea Formaldehyde Foam field applied is not permitted. Provide acoustical insulation above walls separating bathroom/restrooms and corridor and adjacent occupied spaces, and between offices and corridors. Insulation must have a minimum sound attenuation rating of STC-55.

   Insulation must be Polyisocyanurate Rigid Board Insulation, Mineral Fiber Blanket Insulation to conform to ASTM C 991, with Glass Mat Gypsum Roof Board for use above the deck or insulation conforming to ASTM C 1177/C 1177M, where necessary.

   Only on portions of the roof where the sloping of structure does not allow the minimum slopes, provide a factory tapered roof insulation system to provide positive drainage of roof system, and to include drainage around curbs, penetrations, and projections through the roof plane.

   Provide Glass Mat Protection Board meeting ASTM C 1177 for use as a thermal barrier (underlayment) or protection board for hot-mopped applications.

   b. Vapor Retarder: Determine the need and location in the roof assembly

   for a vapor retarder. Where the mean January temperature is 40 degrees Fahrenheit or less, and the expected interior relative humidity is 45% or greater, use a vapor retarder. Otherwise, use ASHRAE 90.1 for the determination.

   1) Vapor Retarders as Integral Facing - Alloy conforming to ASTM B 209, or Vapor Retarders Separate from Insulation - Vapor retarder material must be 10 mil polyethylene sheeting conforming to ASTM D 4397.

2) A slip sheet is required to separate the roofing panels from the insulation facing where the facing would be in direct contact with the roofing panels. If a slip sheet is necessary for use with a vapor retarder, use a 5 lb. per 100 square feet rosin-sized, unsaturated building paper.

c. EPDM Rubber Boots: Flashing devices around pipe penetrations must be flexible, one-piece devices molded from weather-resistant EPDM rubber.

d. Prefabricated Curbs and Equipment Support: Provide Prefabricated curbs and equipment supports of structural quality, hot-dipped galvanized or galvanized sheet steel, factory primed and prepared for painting with mitered and welded joints. Provide integral base plates and water diverter crickets. Minimum height of curb must be 8 inches above finish roof.

e. Fasteners: Provide fasteners that meet all requirements of the NRCA and Factory Mutual.

f. Wood Nailers: Wood nailers shall be pressure-preservative-treated in accordance with AWPA M2 Standards, permanently marked or branded, and installed flush with the top of the adjacent insulation board.

g. Flashing and Sheet Metal: Provide flashing and sheet metal work including scuppers, splash pans, and sheet metal roofing. Flashing and sheet metal must be provided in accordance with roof manufacturer's printed installation instructions and in compliance with NRCA and SMACNA recommendations. Fabricate Flashing and sheet metal components from Copper, Lead-Coated Copper sheet, Steel Sheet, ZincCoated (Galvanized) - ASTM A 653/ A 653M, Stainless Steel - ASTM A 167, Type 302 or 304, 2D finish, or Pre-Finished Aluminum.

h. Gutters and Downspouts: Provide gutters and downspouts compatible with roofing material and finish. Concealed (interior) gutters and downspouts are prohibited. Provide splash guards at points of discharge.

i. Roof Openings and Supports: Provide flashings for roof openings and supports as recommended by the NRCA. Assure all penetration flashings extend minimum 200 mm (8 inches) above the finished roof surface.

j. Roof Hatches: Provide roof hatch where required by OSHA, HN norms, and as access to roof when roof mounted equipment is used or other routine roof maintenance is required.

k. Glazed Roof Openings: Skylights and other glazed roof openings shall be used only to supplement interior lighting levels (generally in steep slope or vertical applications), and otherwise, are discouraged from use.

l. Guards: Provide rails or guards as required by the OSHA, the International Building Code, HN norms, or other applicable safety standards.

m. Traffic Pads: Provide on roof system to protect roof from foot traffic. Provide traffic pads around roof mounted mechanical equipment and underneath removable mechanical equipment access panels. Traffic pads must be of compatible material to roof.

4. OTHER ROOFING

a. Lightning Protection: Lightning protection component penetrations and attachments must be sealed and flashed and anchored in a permanent manner and in a manner to avoid the degradation of the watertight integrity of the roof system.

b. Roof Drains (Existing): Where existing roof drains are to be reused in roof replacement construction, the contractor must provide new, compatible flashing materials, a new drain clamping ring and new bolts for anchorage. Reuse of existing clamping ring and bolts is unacceptable.

SECTION C. INTERIORS

C10 INTERIOR CONSTRUCTION

C1010 PARTITIONS

1. Fixed Partitions: wood frame; light gage steel frame; concrete masonry complying with ACI 530.1/ASCE 6/TMS 602 and associated ASTM Standards; or cast-in-place concrete complying with UFC 1-200-01, General Building Requirements, ACI 117 and ACI 301/301M. In addition, interior partitions must comply with tables for sound isolation and noise reduction in Chapter 1, "Architectural Graphic Standards". Include a statement of adherence to the applicable criteria.

Gypsum board/stud partitions may be standard gypsum board, moisture resistant, or impact resistant. Use cement board in showers and other wet areas. Reinforce points where doorknobs can strike a wall and anchorage points for wall mounted equipment.

2. Demountable or Removable Partitions: Must be of materials allowed by code and shall be anchored firmly to the structure to carry their own weight as well as impact forces and seismic lateral forces. Sound Transmission Class (STC) rating and Impact Isolation Class (IIC) rating shall be in accordance with ASTM E 90 or ASTM E 413 for frequency data, and shall meet the requirements of the intended use in Part 3. The majority of the components and hardware must be provided by a single manufacturer and on the manufacturer's current GSA price list. The product must be included on the NAVFAC Selection Tool for Movable Walls.

3. Glazed Partitions and Interior Windows: Must be of the materials allowed by IBC, and must comply with fire and smoke separation requirements. Provide safety glazing and fire resistant rating where they are required.

C1020 INTERIOR DOORS

1. Wood Doors: Stile and rail wood doors must be WDMA I.S.6A-01, premium or custom grade, heavy duty or extra heavy duty. Flush wood doors must be WDMA I.S.1A-04, premium or custom grade, heavy duty or extra heavy duty; or WDMA I.S.-97 (PC-5 5-ply particleboard core or SCLC-5 5-ply structural composite lumber core). Doors adjacent to paneling or millwork must comply with corresponding AWI millwork grade. Provide interior fire doors in rated walls.

2. Steel doors: Must be ANSI A 250.8, Level 1, (occasional use, low abuse types such as closet doors without locks); Level 2, (low use, moderate abuse types such as office/storeroom doors); Level 3, (moderate use, high abuse types such as BEQ sleeping room doors); Level 4, (high use, high abuse types such as corridors, stairways, assembly spaces, and main entry doors), with a physical performance level of 'A'. Maximum door undercut must not exceed 19 mm (3/4 inch).

3. Sound Insulated Doors and Frames: Utilize Sound Insulated Doors and Frames with sound control weatherstripping in rooms requiring wall assemblies to be sound insulated with a Sound Transmission Class (STC) rating as required. The STC rating for the door and frame assembly must be not less than the wall assembly STC rating.

4. Aluminum Doors and Frames: Provide swing-type aluminum doors and frames complete with framing members, transoms, side-lites, and accessories. Fabricate of ASTM B 221, Alloy 6063-TS for extrusions.

5. Steel Door Frames: ANSI A 250.8. Form frames with welded corners for installation in masonry partitions and knock-down field assembled corners for installation in metal stud and GWB partitions. Install frames in accordance with SDI 105. Form stops and beads with 20 gauge steel.

Provide a minimum of three jamb anchors and base steel anchors per frame, zinc-coated or painted with rust-inhibitive paint, not lighter the 18 gauge. Secure frames to previously installed concrete or masonry with expansion bolts in accordance with SDI 11-F. Provide mortar infill of frames in masonry walls, and gypsum board compound infill at each jamb anchor in metal frame walls.

6. Fire doors: Provide in conformance with NFPA 80 an NFPA 105. Fire doors and frames must bear the label of UL, FM or WHI attesting to the rating required. Door and frame assemblies must be tested for conformance per NFPA 252 or UL 10B (for neutral pressure) or UL 10C (for positive pressure). Wood fire doors must also comply with ASTM E 152.

Provide stainless steel astragals complying with NFPA 80 for fire-rated assemblies and NFPA 105 for smoke control assemblies.

7. Interior Door Hardware: Provide the services of a certified door hardware consultant to prepare the door hardware schedule. Unless otherwise noted, interior doors include latch, hinges, door stops and door silencers. Provide closers and kick plates for fire-rated, corridor, stairway and high-use non-residential doors.

a. Hinges - BHMA A156.1, Grade 1, 108 x 108 mm (4 1/2 x 4 1/2 inches) with nonremovable pin or anti-friction bearing hinges.

b. Locks and Latches - For non-residential buildings use Series 1000, Operational Grade 1, Security Grade 2 for stairways, building entrances, corridors, assembly spaces, and other high use interior doors. Use Series 4000, Grade 1 for non-residential locations not using Series 1000 hardware. For residential buildings use Series 4000, Grade 2 for interior doors. a) Mortise Locks and Latches - BHMA A 156.13, Series 1000, Operation Grade 1, Security Grade 2. b) Bored Locks and Latches - BHMA A 156.2, Series 4000, Grade 1, or Grade 2.

c. Exit Devices - BHMA A 156.3, Grade 1. Provide touch bars in lieu of conventional crossbars and arms. Use manufacturer's integral touch bars in aluminum storefront doors.

d. Card Key Access - Provide card key type access units for specialized entries. Provide lithium battery powered, magnetic stripe keycard locksets that are ANSI/BHMA A156.13, Series 1000, Grade 1, mortise or ANSI/BHMA A156.2, Series 4000, Grade 1, cylindrical locks, tamper resistant, UL listed with 25 mm (1 inch) throw deadbolt, 19 mm (3/4-inch) throw latch bolt, auxiliary dead-locking latch, and 68.75 mm (2-3/4 inch) backset.

Provide hardware keying compatible with the existing base-wide keying system. Replacement interchangeable cores must be compatible with the Best Lock system.

e. Key Cabinet: Provide a Key Cabinet with 30% over capacity.

C1030 SPECIALTIES

1. Compartments, Cubicles, & Toilet Partitions: FS A-A-60003. Provide toilet compartments at multi-fixture toilet rooms of Type I, Style B-Ceiling Hung, C-Overhead Braced, or F-Overhead braced-alcove. Reinforce panels to receive partition-mounted accessories. Urinal screens must be FS A-A-60003. Type III, Style A, floor supported and wall hung or Style D, wall hung. Wall hung urinal screens must be secured with continuous flanges to urinal screen and wall. Steel and Plastic toilet partitions must have a recovered materials content of 20 to 30 percent. Chrome-plated or stainless steel door latches and coat hooks. Provide one coat hook per compartment door. Latches and hinges for handicapped compartments must comply with ABA Accessibility Standards.

2. Toilet and Bath Accessories: Provide toilet and bath accessories and install per ABA Accessibility Standards and manufacturers' requirements.

3. Marker Boards and Tack Boards: Provide porcelain enamel marker boards fused to a nominal 28 gauge steel sheet and tack boards of cork, with a tensile strength of at least 40 psi when tested according to ASTM F 152, with woven or vinyl covering.

4. Signage: All doors must have an identifying device. All handicap accessible facilities must utilize signage which meets current ABA Accessibility Standards requirements with regard to Braille, raised characters, finishes (contrast), size and mounting height. If room names are subject to frequent change, provide an interchangeable strip to be utilized to facilitate removal and replacement.

5. Lockers: Provide lockers to meet FS AA-L-00486 (Rev J), enameled steel with special bases.

6. Shelving: Provide steel shelving.

7. Counters: Provide solid plastic or plastic laminate counter tops and back splashes, AWI Custom grade.

8. Cabinets: Provide cabinetry and millwork items with associated accessories and hardware. Cabinetry must be AWI premium or custom grade and have concealed hinges with adjustable standards for shelves.

9. Casework: Provide all built-in premanufactured metal cabinetry for specialized functions such as laboratories, libraries, medical and dental facilities. Casework must comply with Mil Std 1691.

10. Closets: Provide premanufactured or millwork closets or prefabricated coat closets.

11. Fire Extinguisher Cabinets: Provide fire extinguisher cabinets. Size and locate fire extinguisher cabinets to encase extinguisher as required by NFPA 10 & 101. Fire extinguishers will be provided by the Customer.

12. Firestopping Penetrations: Provide all sleeves, caulking, and flashing for firestopping penetrations.

13. Entrance Floor Grilles and Mats: Provide recessed pan or surface floor mats at main entrance only or all building entrances.

14. Ornamental Metal Work: Provide ornamental metalwork.

15. Other Interior Specialties: Motorized projection screen must be wall or ceiling or above-ceiling mounting. Pull-down projection screens must be provided in lieu of motorized projection screens as approved by the Activity.

C20 STAIR CONSTRUCTION

Provide interior and exterior stair construction. Stair design, materials and construction muost comply with IBC, and applicable codes and standards, including NFPA 101. Provide refuge area at top of stair in accordance with applicable Americans with Disability Act Design Guide requirements.

C30 INTERIOR FINISHES

C3010 WALL FINISHES

Unless otherwise noted in the RFP, primary wall finish is painted gypsum wall board. Provide fire resistive construction and finishes for fire separation between areas of the building in accordance with the latest adopted version of the IBC, and NFPA 101. Provide water resistant cementitious board at floors and walls of tubs and showers.

1. Ceramic Tile: Provide ceramic tile wall systems as defined in the Tile Council of America (TCA) handbook for ceramic tile installation and materials for the service requirements listed. Provide installation and materials in accordance with ANSI A108/A118 series standards, except do not use organic adhesives. Provide manufacturer's full range of colors and styles. Tile must be a minimum of one grade above base grade. Coordinate with ceramic bath accessories for modularity. Include all trim pieces, caps, stops, and returns to complete installation.
2. Wallcovering: Vinyl wallcovering must conform to ASTM F793, Category V Type II, 371 g to 624 g (13.1 to 22 ounces) total weight per square yard and width of 1370 mm (54 inches). Provide ASTM F793, Category VI, Type III, 624 g (22 ounces) and above to cover rough textured walls such as masonry. High performance fabric wallcovering must be woven or non-woven Class A, fire resistive material, a minimum of 1219 mm (48 inches) wide, with a soil repellent finish and a minimum of 340 g (12 ounces) per square yard exclusive of backing. "Tackable" wall covering must be "self-healing" from tack penetration through the covering into the substrate. Acoustical wallcovering must be textured, woven or non-woven, Class A fire resistive material with an acrylic backing, a minimum of 1219 mm (48 inches) wide and a minimum of 454 g (16 ounces) per square yard. The material must have an NRC rating of .15 on gypsum board in accordance with ASTM C423. Do not install wall covering on interior face of exterior walls.

C3020 FLOOR FINISHES

Provide new flooring materials as required. All flooring materials, adhesives, finish coats, sealers and mortar materials must meet or exceed EPA requirements for toxic substance content restrictions and air quality requirements; and meet or exceed fire protection requirements, such as smoke and flame spread requirements. When laying broadloom carpets and resilient flooring, use the widest sheet materials available to avoid or minimize the number and extent of seams. When seams are required, locate seams at infrequent traffic areas. Contractor is required to submit seam layout to Contracting Officer for approval prior to installation.

1. Ceramic Tile: Provide ceramic tile floor systems as defined in the Tile Council of America (TCA) handbook for ceramic tile installation and materials for the service requirements listed or applicable HN entity. Provide installation and materials in accordance with ANSI A108/A118 series standards, except do not use organic adhesives. Provide manufacturer's full range of colors and styles. Tile must be a minimum of one grade above base grade. Provide ceramic or porcelain tile with a minimum breaking strength of 202kg (300 pounds), ASTM C648, and a maximum absorption rate of 0.5%, ASTM C373. Tile must have a minimum coefficient of friction (wet and dry) of 0.6, ASTM C1028.

2. Resilient Flooring: Meet or exceed applicable ABA Accessibility Standards horizontal requirements. Install flooring per manufacturer's recommended methods and adhesives. Provide manufacturers full line of color and pattern selections, including multi-color patterns. Linoleum Sheet or Tile Flooring must be 2.5 mm (0.10 inch) gage; minimum 250 psi static load limit, ASTM F970; and with multi-color pattern and color extending throughout thickness, ASTM F2034, Type I. Resilient homogeneous vinyl sheet flooring must be commercial quality, 2.0 mm (0.080 inch) overall gage, with minimum 1.6 mm (.066 inch) thick wear layer, protective urethane finish, ASTM F1303, Type II, Grade 1, Class A. Resilient vinyl composition tile must be commercial grade, 3 mm (.125 inch) gage, ASTM F1066, Comp. 1, Class 2, through pattern.

3. Carpet: Carpet manufacturer and installer must be experienced, established and in good standing with the industry. Carpet, broadloom or tile must be installed per the Carpet & Rug Institute's recommendations. Carpet must be tufted, textured loop, cut/loop or tip sheared, a minimum of 26 oz. face weight, minimum density of 6600, 100% premium branded yarn- or solution-dyed, Type 6 or 6.6 continuous hollow filament nylon. Carpet must be multi-color and patterned for soil and wear hiding properties. Carpet must have high performance backing warranted against zippering, edge raveling and delamination, be anti-static and anti-microbial. Carpet must meet Flammability ratings; generate less than a 450 rating, ASTM E662; meet the Critical Radiant Flux Classification of not less than 0.45 W/sq. cm., ASTM E648. Where indicated in the room requirements, provide attached polyurethane cushion or separate polyurethane cushion for double stick pad installations, ASTM 1667 and ASTM 3676.

4. Wall Base: Provide porcelain or ceramic tile base for porcelain or ceramic tile floor. Provide solid, through color preformed rubber or vinyl base for carpeted/resilient flooring areas. Provide a sealant between base and floor finish in all wet areas.

C3030 CEILING FINISHES

Unless otherwise noted in the room requirements, acoustical ceiling panels must be 24 inch by 24 inch, with a minimum light reflectance of .75, Class A, flame spread 25 or less and smoke development of 50 or less, ASTM E84. Acoustical ceiling panels must have minimum 60% recycled content and conform to ASTM E1264. Panels must have a factory-applied standard washable painted finish or Type IV with factory-applied plastic membrane-faced vinyl, Form: 1, 2 or 3. Provide square edge except as noted.

Unless otherwise noted in the room requirements for entrance lobby, restrooms and showers, provide a painted, suspended gypsum board ceiling. Exposed structural systems shall be painted.

C3040 PAINTING

All painting and coating materials must be low VOC, comply with local air quality control laws and, regulations; and conform to the Master's Painters Institute's (MPI) Architectural, Interior Systems Manual and the MPI's Maintenance and Repainting Manual recommendations for paint systems, surface preparation and applications.

Provide minimum of one prime coat and two finish coats. The prime coat must not be combined with texture or other coatings. Seal and prime all surfaces to cover underlying stains or discoloration that may affect finish paint. Finish coats must provide full coverage of undercoats and substrates. All walls and ceilings in wet area must have semi-gloss paint. All wood or metal cased openings, door trims and casings, window trims and casing, and other finish trim must have semi-gloss paint. All interior walls and ceilings must have satin or eggshell finish. For previously painted surfaces, prime all surfaces to ensure compatibility of finish coats. Do not paint prefinished surfaces except as noted.

Provide Institutional Low Odor/Low VOC Latex paint or High Performance Architectural Latex systems as defined and approved by the MPI Systems Manual for the various substrates required to be painted.

Paint/Color Selection: Provide paint systems tested to "Detailed Performance Level" standard as defined by MPI. Paints must be readily available for purchase in standard colors.

SECTION D. SERVICES

D10 CONVEYING Elevators and Escalators - Not used

D20 PLUMBING

Provide plumbing fixtures, appliances, and equipment complete and usable as required by Part 3. All plumbing fixtures, appliances and equipment, piping, valves, accessories, and appurtenances must comply with International Plumbing Code (IPC) and all other applicable codes and standards, including energy, water conservation, and local activity regulations and standards. Provide all plumbing fixtures to meet current criteria of EPA Watersense program http://www.epa.gov/watersense

1. Domestic Water: Provide ASTM B 88 Type K or L copper tubing and fittings for pipe sizes 4 inches or smaller. Provide Type L tubing above ground with solder fittings. For buried piping, use Type K tubing with solder fittings, or Chlorinated polyvinyl chloride (CPVC) Plastic pipe, fittings, and solvent cement per ASTM D 2846/D 2846M for sizes 4 inches and smaller.

   Provide mineral fiber insulation with vapor barrier on domestic water (hot and cold) supply and recirculation piping. Provide re-circulating pumps or instantaneous water heaters for hot water systems with fixtures greater than 100 ft from hot water source. Provide water hammer arrestors per PDI STD WH-210 as required for rapid water shut off scenarios. All water valves except for fixture shut off valves must be ANSI B16.18 brass, full port ball type. All plumbing fixtures must have separate shut off valves. All piping must be concealed in walls, attic spaces, or in crawl spaces under floors. Provide access panels for valves behind walls. No under slab water piping is allowed. Fittings for annealed copper tubing must conform to ANSI Bl6.22. Solder and flux must be lead free. Exposed exterior piping is prohibited unless otherwise not practical. Provide identification for piping and equipment.

2. Wall Penetrations: Piping which penetrates fire rated walls must be completely sealed to maintain fire resistance integrity as required by Code. Penetrations through walls that are not fire rated must be adequately supported and sealed. Pipe penetrations through exterior walls must be sleeved, caulked with weatherproof sealant and provided with finish trim.

D2010 PLUMBING FIXTURES

Provide fixtures complete with fittings, and chromium-plated, or nickel-plated brass (polished bright or satin surface) trim. All fixtures, fittings, and trim, must be from the same manufacturer and must have the same finish. Access panels must be provided for all bathtubs and showers, except at exterior and party walls and where tub or showers are back to back. Provide cleanouts in accordance with the plumbing code. Rotate or extend cleanouts required to facilitate maintenance and clearing of blockage in waste piping.

1. Faucets: All faucets must be brass construction, washerless type, with seals and seats combined in one replaceable ceramic disk valve cartridge designed to be interchangeable with all lavatories, bathtubs and kitchen sinks, or having

replaceable seals and seats removable either as a seat insert or as a part of a replaceable valve unit. Faucets provided must be of the same type and manufacturer throughout the facility, unless otherwise noted. Lavatory faucets must be U.S. Environmental Protection Agency (EPA) Watersense® certified and labeled bathroom sink faucets or EU applicable certification.

2. Water Closets: Water closets must be in accordance with ANSI A112.19.2, with trim conforming to A112.19.5. Water closets must be vitreous china and have an elongated bowl with trip lever, unlined tank, close coupled siphon jet, floor outlet with wax gasket, flange and an anti-siphon float valve. Provide white closed front seat and cover for private toilets and open front seat cover for public facilities. Water consumption must be no greater than 1.6 gallon maximum per complete flushing cycle. Provide self-closing metering type flush valve on flush valve type water closets, unless electronic control is specified in Part 3. Maximum flush volume must not exceed 1.28 gallon per flush (GPF) (4.8 Liter per flush (LPF)) for single function flush valves. Dual function flush valves must provide a flush of 0.8 to 1.6 GPF (3.0 to 6.0 LPF) or 1.28 GPF (4.8 LPF) average for 2 low volume flushes and one high volume flush. Tank type water closets must be U.S. Environmental Protection Agency (EPA) Watersense® certified and labeled toilets.

3. Urinals: Provide U.S. Environmental Protection Agency (EPA) Watersense® certified and labeled ceramic-type urinals.

4. Lavatories: Unless otherwise specified by Part 3, lavatories must be integral to the vanity countertops. Each lavatory must be provided with hot and cold water tempered by means of a mixing valve or combination faucet.

5. Sinks: ASME/ANSI A112.19.3M sink, 20 gage stainless steel with integral mounting rim, minimum dimensions of 840 mm (33 inches) wide for two compartment or 560 mm (21 inches) wide for one compartment by 560 mm (21 inches) front to rear, with ledge back and undersides coated with sound dampening material.

6. Water Coolers: ARI 1010, wall-mounted, bubbler style, air-cooled condensing unit, 4.20 mL per second (4.0 gph) minimum capacity, stainless steel splash receptor, double wall heat exchanger, and all stainless steel cabinet. Install in accordance with the manufacturers instructions.

7. Showers: Provide U.S. Environmental Protection Agency (EPA) Watersense® certified and labeled showerheads connected to concealed pipe connected to copper alloy single control type mixing valve with front access integral screwdriver stops. Anchor the mixing valves and the pipe to each showerhead in wall to prevent movement. Unless otherwise specified by Part 3, showers must be supplied with water at a temperature no more than 110Â°F by means of a pressure balance, tempering or mixing valve.

8. Service sinks: ASME A112.19.1M, white enameled cast-iron or ASME A112.19.2M white vitreous china, wall mounted and floor supported by wall outlet cast-iron P-trap, minimum dimensions of 560 mm (22 inches) wide by 457 mm (18 inches) front to rear with 230 mm (9 inch) splashback, and stainless steel rim guard. Provide ASME A112.18.1M copper alloy back-mounted combination faucets with vacuum breaker and 20 mm (0.75 inch) external hose threads

9. Mop Sinks: Pre-cast terrazzo or ASME A112.19.2M white vitreous china floormounted mop sink, 914 mm x 914 mm x 305 mm (36 inches x 36 inches x 12 inches). Terrazzo must be made of marble chips cast in white Portland cement to a compressive strength of not less than 25 mPa (3625 PSI) 7 days after casting. Provide brass body drains with nickel bronze strainers cast integral with mop sink. Provide stainless steel rim guard for mop sink. Provide chrome-plated exposed hot and cold water faucets ASME A112.15.M wall-mounted copper alloy faucets swing spout with 20 mm (3/4 inch) hose connection, vacuum breaker, and pail hook. Provide mop hanger on wall above sink suitable for four mops.

10. Laundry Sinks: ANSI Z124.1, plastic, two compartment, minimum dimensions of 1016 mm wide by 533 mm (40 inches wide by 21 inches) front to rear, with floor-supported steel mounting frame secured to wall. Provide ASME A112.18.1M copper alloy centerset faucets, swing spout with aerator, and stainless steel drain outlets with cup strainers, and 40 mm (1.5 inch) adjustable Ptrap with drain piping to vertical vent stack.

11. Emergency Eyewash: ANSI Z358.1, wall-mounted self-cleaning, non-clogging eye and face wash with quick opening, full-flow valves, stainless steel eye and face wash receptor. Provide copper alloy control valves. Pressure-compensated tempering valve is required for emergency fixtures, with leaving water temperature setpoint adjustable throughout the range 15.5 and 35 degrees C (60 to 95 degrees F) unless cold water supply meets temperature criteria.

D2020 DOMESTIC WATER DISTRIBUTION

1. Natural Gas or Propane Fired Storage Water Heaters: Provide high efficiency storage type natural gas or propane fired water heaters per ANSI Z21.10.1 or ANSI Z21.10.3 meeting AGA requirements. For California, unit efficiency must meet or exceed that listed in the Title-24 Standards. Equipment efficiency must be in accordance with Energy Star or FEMP designated products list, http://energy.gov/eere/femp/find?product?categories?coveredefficiency?programs For gas water heaters use Energy Star labeled products. For products not listed by Energy Star or FEMP provide products with efficiency rated in the top 25% of available products.Water heaters must be equipped with glass-lined steel tanks, minimum R-15 polyurethane foam insulation, replaceable anodes, and adjustable range thermostat to allow hot water settings between 43 and 71 degrees C (110 and 160 degrees F). Water heater warranty must be a minimum of 10 years. Provide vent in accordance with NFPA 54. Provide low NOx burners that meet SCAQMD requirements. Install in accordance with manufacturer's instructions and the code. Where

earthquake loads are applicable, water heater supports must be designed and installed for seismic forces in accordance with the International Building Code.

2. Electric Water Heaters: Provide electric water heaters with double heating element per UL 174. Unit efficiency must meet or exceed that listed for FEMP or ENERGYSTAR, or as listed in ASHRAE 90.1, whichever is greatest. Water heaters must be equipped with glass-lined steel tanks, high efficiency type, insulated with polyurethane foam insulation, replaceable anodes, and adjustable range thermostat to allow hot water settings between 43 and 71 degrees

C.

3. Domestic Water Boilers: Boilers must be designed, tested, and installed per ASME CSD-1 (Controls and Safety Devices) and ASME BPVC (Boiler and Pressure Vessel Code). The boiler must meet the requirements of the UL 795, NFPA 85, ANSI Z83.3, and ASME CSD. Boilers must be certified by Naval Personnel or a contractor approved by the Contracting Officer.

D2030 SANITARY WASTE & VENT

All new sewers below concrete slab must be solid core, minimum schedule 40 (DWV Type), ABS in accordance with ASTM 2661. New waste and vent piping above floor must be Schedule 40 PVC (DWV Type) ASTM 2665 or ABS ASTM 2661. Use of ABS plastic pipe must conform to the IBC and IPC. Provide pipe sizing, configurations, and cleanouts as required by the IPC. Cellular core plastic pipe is not allowed. SOVENT systems are not allowed.

D2040 RAINWATER DRAINAGE

Below concrete slab must be solid core, minimum schedule 40 (DWV Type), ABS in accordance with ASTM 2661. Above floor must be cast iron hubless, or hub and spigot, or Schedule 40 PVC (DWV Type) ASTM 2665 or ABS ASTM 2661 as indicated in Part 3. Pipe materials must conform to the IBC and IPC. Provide pipe sizing, configurations, and cleanouts as required by applicable codes and standards.

D2090 OTHER PLUMBING SYSTEMS

Natural Gas Piping Systems: Exterior above grade natural gas piping must be schedule 40 galvanized steel pipe with threaded fittings and joints. Underground exterior gas piping must be polyethylene pipe that satisfies the requirements of NFPA 54, ASTM D2513-01, and ASME B31-8. Provide warning tape at 12 inches below grade directly above buried gas pipes. Below grade metal gas piping is prohibited. Interior gas piping must be ASTM A 53, schedule 40 black steel with ASME B16.3 threaded fittings and joints. The use of semi-rigid tubing and flexible connectors for gas equipment and appliances is prohibited except for final connections to the equipment and appliances where they must be provided. Provide flexible gas connections in accordance with ANSI Z21.45 and not more than 40 inches long. Provide accessible gas service with shutoff valve for all equipment. Gas piping must conform to NFPA 54 and must be pressure tested in accordance therewith. Gas piping is considered a fragile utility in the content of UFC 4-010-01, DOD Minimum Antiterrorism Standards for Buildings.

D30 HEATING, VENTILATION AND AIR CONDITIONING (HVAC) SYSTEMS

The HVAC systems must comply with the latest edition of the International Mechanical Code, International Plumbing Code, ASHRAE Standards, National Electrical Code, National Fire Protection Association Publications, International Building Code, and California Title 24 or ASHRAE 90.1 energy efficiency standards and UNI norms unless otherwise specified in Part 3. All equipment, appliances, ductwork and accessories must comply with applicable codes and standards. For projects located in California, also comply with California Energy Commission (CEC) efficiency rating requirements as stated in Ca. AB 970 Title 24. The Contractor must certify that the installation is in conformance with the applicable codes and standards at the completion of the contract, prior to final invoice being processed and final acceptance. Provide Energy Star rated equipment where available. Provide equipment with performance in excess of Energy Star requirements where specified.

1. Equipment Clearance: Provide working space around all equipment. Provide all required fittings, connections and accessories required for a complete and usable system. All equipment shall be installed per the manufacturer's recommendations. Where the word "should" is used in manufacturer's instructions, substitute the word "shall".

2. Material and Equipment Qualifications: All materials and equipment must have been in satisfactory commercial or industrial use for 2 years prior to the bid opening. The 2-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures during the 2-year period.

3. Motors: Single-phase fractional-horsepower alternating-current motors must be high efficiency types corresponding to the applications listed in NEMA MG 11. Select polyphase motors based on high efficiency characteristics relative to the applications as listed in NEMA MG 10. Additionally, all polyphase squirrelcage medium induction motors with continuous ratings must meet or exceed energy efficient ratings per Table 12-10 of NEMA MG 1. Provide controllers for 3-phase motors rated 0.75 kW (1 hp) and above with phase voltage monitors designed to protect motors from phase loss and over/under-voltage. Provide means to prevent automatic restart by a time adjustable restart relay. For packaged equipment, the

manufacturer must provide controllers including the required monitors and timed restart. Provide reduced voltage starters for all motors 25 hp and larger.

4. Equipment Support: Provide housekeeping pads and vibration isolators under all floor-mounted equipment.

5. Coatings: When required in Part 3, provide chiller and air handler coils with copper tube/copper fin coil construction or immersion applied, baked phenolic or other approved coating. Field applied coatings are not acceptable. Mechanical equipment casings must have painted finishes that pass a salt-spray test conducted

per ASTM B117 for duration of at least 500 hours.

6. Equipment Insulation: Provide insulation on all chilled water equipment. Insulate hot and chilled water pumps and equipment as suitable for the temperature and service in rigid block, semi-rigid board, or flexible unicellular insulation to fit as closely as possible to equipment. Provide vapor retarder for chilled water applications.

7. Acoustical considerations: Noise levels in all areas served (supply, return, and exhaust) by a mechanical system must comply with ASHRAE Design Guidelines for HVAC related background sound in rooms as indicated in the lasted ASHRAE Fundamentals Handbook. The RC-rating method must be utilized.

D3020 HEAT GENERATING SYSTEMS

1. Boilers: Boilers must be designed, tested, and installed per ASME CSD-1 (Controls and Safety Devices) and ASME BPVC (Boiler and Pressure Vessel Code). The boiler must meet the requirements of the UL 795, NFPA 85, ANSI Z83.3, and ASME CSD. Do not provide watertube boiler(s) for hydronic heating when size permits otherwise. Provide insulated boiler stack in accordance with manufacturer's recommendations and conform to NFPA 211 or provide premanufactured, multi-wall stacks complying with NFPA 54 or NFPA 58 and ULlisted. Low pressure boilers must be equipped with one or more pressure relieving devices, adjusted and sealed to discharge at a pressure not to exceed the maximum allowable working pressure of the boiler. The combined capacity of these devices must be such that with the fuel burning equipment installed and operating at maximum capacity, the pressure cannot rise more than 5 psi for steam boilers or 10% for water boilers above the maximum allowable working pressure of the boiler. Pressure relieving devices must be installed as required by the referenced code, be ASME stamped and rated, and must be installed with the valve spindle in the vertical position. Provide with manual lifting device for periodic testing. Boilers must comply with the local air quality regulations. Boilers must be equipped with pressure and temperature gauges as required for proper maintenance and operation. Thermometers must also be provided at the inlet and exit of the boiler, and be visible to the operator from the operating area.

2. Furnaces: UL-listed, factory assembled, self contained, forced circulation, furnace. Provide electronic ignition system. Unit must be design certified by AGA, GAMA efficiency rating certified, for gas furnaces and NFPA 31 for oil furnaces. Provide with cooling coil as necessary. Furnaces must comply with the local air quality regulations.

D3030 COOLING GENERATING SYSTEMS

1. Chillers: Air-cooled chillers must be type indicated in Part 3 and meet the requirements of ARI 550/590-98. Provide control panel with the manufacturers' standard controls and protection circuits. If DDC system is required in project, provide a control interface for remote monitoring of the chiller's operating parameters, functions and alarms from the DDC control system central workstation. Provide complete start-up and operational testing of chiller equipment.

2. Direct expansion systems: Provide units factory assembled, designed, tested, with ducted air distribution and rated in accordance with ARI 210/240 or ARI 340/360. Refrigerant piping size must be per the manufacturer's recommendations. Insulate refrigerant piping suction lines and condensate drain.

3. Refrigerants: The use of Ozone Depleting Substances (ODS) in new equipment is banned, as per EU Regulation (EC) No 1005/2009 and following amendments and modifications. The use of F-gases in new equipment is restricted, as per EU Regulation (EC) No 517/2014 and following amendments and modifications. All the HFC-refrigerants with a global warming potential (GWP) of 150 or more are not allowed to be used in movable air conditioning equipment. Starting from 2025, all the F-gases with a global warming potential (GWP) of 750 or more are not allowed to be used in □Single split□ systems that contain less than 3kg of refrigerant (a system with one cooling coil connected to a remote condensing unit).□

4. Coils: If coatings are indicated in Part 3, provide with copper tube/copper fin construction or immersion applied, baked phenolic or other approved coating that passes the 3000 hour salt spray resistance test using ASTM B117 procedure. Field applied coatings are not acceptable.

5. Variable Refrigerant Flow (VRF) systems: The system must consist of VRF heat pump units, branch circuit controllers, VRF fan coil units, and associated controls. The system must be designed to provide the facility with simultaneous heating and cooling utilizing hot gas refrigerant or sub-cooled liquid. Provide system with heat recovery. The heat pump units must be inverter driven and must utilize R410A refrigerant. Total capacity of the branch controllers must be between 50% and 150% of the rated capacity. All refrigerant piping must be sized and installed in strict compliance with the manufacturer□s requirements. Refrigerant piping

must be clean, dry, and leak free. Prior to installation all refrigerant pipes must remain sealed. During installation and prior to filling, nitrogen must be used to maintain cleanliness and prevent oxidation and scaling while brazing.

D3040 DISTRIBUTION SYSTEMS

1. Ductwork: All ductwork must be provided in accordance with the latest SMACNA guidelines. Flexible duct lengths must not exceed 5 feet. Provide galvanized sheet metal ducts except for special exhaust systems and the following:

a. For fume hood exhaust, kitchen hood exhaust, and dishwasher exhaust, provide stainless steel ductwork.

b. For shower area exhausts, provide aluminum or stainless steel ductwork and sloped to drain provisions. After the shower exhaust is mixed with a volume of general exhaust air equal to 200% of the shower exhaust rate, standard galvanized construction may be used.

c. Internal insulation-lined ductwork is prohibited in all areas. For ductwork located exterior to the building, provide externally insulated systems with sheet metal cladding. Provide external thermal insulation for all ductwork. Insulate ductwork in concealed spaces with blanket flexible mineral fiber. Insulate ductwork in Mechanical Rooms and exposed locations with rigid mineral fiber insulation. Provide insulation with factory applied all-purpose jacket with integral vapor retarder. In exposed locations, provide a jacket with white surface suitable for painting. Flame spread/smoke developed rating for all insulation shall not exceed 25/50. Minimum insulation thickness must be the minimum thickness required by ASHRAE 90.1. Insulate the backs of all supply air diffusers with blanket flexible mineral fiber insulation.

d. The ductwork must be sealed with an approved duct sealer and in accordance with SMACNA standards. If leakage testing is indicated in part 3, the duct leakage must not exceed 2%.

e. Provide manual volume dampers in each branch take-off from the main duct to control air quantity. Dampers must conform to SMACNA DCS. Dampers must be installed in accessible locations.

2. Fire Dampers: Fire dampers must be rated per UL 555. Fire dampers must be dynamic type rated for closure against a moving airstream. Provide fire dampers that do not intrude into the air stream when in the open position.

3. Piping:

a. Provide insulated, steel piping for sizes 4 inches and larger and insulated copper piping for sizes less than 4 inches for water supply and return piping to serve the HVAC equipment throughout the facility.

b. Provide system flushing and start-up for water systems.

c. Oil piping: ANSI/ASTM A53 or A106 piping with associated ASME fittings or ASTM B88, type L or M copper tubing with ASME B16.26 flared fittings or compression type fittings.

4. Exhaust Fans And Ducts:

a. General: Exhaust fans must be sized to move the volume of air required to comply with International Mechanical Code for the areas requiring exhaust.

b. Bathroom, restrooms and Utility Room Exhaust Fans: Exhaust fans must be sized to give not less than 10 air changes per hour in the space to be ventilated. Fans must have a maximum sound level of 3 sones and be separately switched from light.

c. Flues: When required, provide new Type B, U.L. listed, double wall flues. Flue installation must be in accordance with the International Mechanical Code.

5. Air handling units: Modular construction, double wall air handling units with minimum of 25 mm (1 inch) casing insulation. Provide ARI 430 certified fans and ARI certified coils. Provide stainless steel, positive draining condensate drain pan. For 100% outside air units provide capability for cooling, heating, dehumidification and reheat.

D3050 TERMINAL AND PACKAGE UNITS

1. Unit ventilators: Unit must be factory assembled unit ventilator capable of up to 100% outdoor air ventilation and UL-listed.

2. Unit heaters: ANSI Z83.8 and AGA label. Equip each heater with individually adjustable package discharge louver. Provide with thermostat.

3. Fan coil units: UL-Listed, factory assembled and tested fan coils, ARI 440 and ARI certified.

4. Packaged units:

Factory packaged rooftop units in accordance with ARI 430 and suitable for outdoor installation. Provide with manufacturer's roof curb.

Packaged through wall units must be factory assembled air conditioner or heat pump and rated in accordance with ARI 310 or ARI 380 and ARI certified. Unit must include heat and operate under the standard unit controls. Units must be designed to allow ease of maintenance by use of a wall sleeve. Units must have internal condensate removal (condensate must not be externally drained).

D3060 CONTROLS AND INSTRUMENTATION

1. General: Provide stand-alone or distributed direct digital controls, as required in Part 3.

2. Distributed Direct Digital Controls (DDC): DDC hardware must be UL-916 rated. Use controllers in a distributed control manner. Controllers must be stand alone with an internal clock and modem. The total number of I/O hardware points shall not exceed 48 in any controller. Provide sufficient memory for each controller to support required control, communication, trends, alarms, and messages. Provide communications ports for controller to controller interface and on-site interface. When providing a partial DDC system or connecting to an existing DDC system, provide a laptop computer with all necessary software for user interface.

D3070 SYSTEMS TESTING AND BALANCING

All HVAC water and air systems, both new and retrofit, must be TABed in accordance with NEBB or AABC standards. As part of any TAB air balancing effort, acceptable air quantity variations must be 0 to -10% for exhaust systems and 0 to +10% for supply air systems.

D40 FIRE PROTECTION

Provide new or extend existing Automatic Fire Sprinkler systems, Smoke and Heat detection systems, Fire Alarm and Mass Notification systems as required. The work for fire sprinklers, fire alarm, smoke detection, and heat detection must be provided by contractors licensed to perform such work.

Project Requirements: Prior to the start of design, the Designer of Record must meet with the Government's Fire Protection Engineer to determine the extent and types of fire protection required.

D4010 FIRE ALARM AND DETECTION

Fire alarm system includes manual stations, system smoke detectors, duct smoke detectors, heat detectors, audio/visual alarms, connection to basewide fire alarm monitoring, electrical supervision of fire pump controllers, and electrical supervision of all sprinkler system alarm and supervisory devices as required.

D4020 FIRE SUPPRESSION WATER SUPPLY AND EQUIPMENT

The water supply information is provided for bidding purposes. The design point of connection to existing water supply will require the approval of the Contracting Officer. The FPE DOR must conduct additional flow tests after contract award prior to any design submissions. Tests must be coordinated through the Contracting Officer.

D4040 SPRINKLERS

Areas subject to freezing must be provided with a dry pipe system.

D50 ELECTRICAL

D5010 ELECTRICAL SERVICE & DISTRIBUTION

Provide interior electrical wiring, fixtures, switches, outlets, and apparatus in accordance with applicable codes and standards. The electrical system must conform to NFPA 70 and CEI 64/8. Power and lighting circuits must be separate.

1. Wiring: All wiring must be in electrical metal conduits and must be concealed except in the industrial spaces and at locations indicated in Part 3. No conductors are permitted to be smaller than No. 12 AWG, copper wires. Wiring below slab or underground must be in Schedule 40 PVC with ground wire. Exposed conduits on the exterior of the building are prohibited. Provide a ground conductor for each circuit; conduits must not be used as the sole means of grounding. Cable assemblies Types AC, MC, or MI and flat conductors may only be used in locations allowed by UFC 3-520-01. Circuit breakers must be bolt-on type. Series rated circuit breakers and fusible panelboards must not be used.

2. Outlet Circuits Lighting and convenience outlets must be on separate circuits. Install GFI protected receptacles at all wet or damp areas. Location of outlets must be as required by applicable codes and standards. Provide extra outlets for maintenance and service staffs in spaces such as corridors, hallways and other public spaces as identified below. All exterior outlets must be on separate circuits, be GFI protected, and equipped with a cover to prevent accidental water infiltration into the devices.

   In addition to the location requirements specified by NFPA 70, locate general purpose and dedicated outlets in accordance with the following:

a. Mechanical equipment: Provide receptacle within 7.6 m (25 ft) of mechanical equipment on the interior and exterior of buildings.

b. Office, staff support spaces, and other workstation locations: One receptacle for each workstation with a minimum of one for every 3050 mm (10 ft) of wall space. When less than 3500 mm (10 ft) of wall at the floor line, provide a minimum of two receptacles spaced appropriately to anticipate furniture relocations. Limit loads to a maximum of four workstations per 20 amp circuit. See Appendix C, Table C1 for workstation load data.

c. Conference rooms and training rooms: One for every 3.6 m (12 ft) of wall space at the floor line. Ensure one receptacle is located next to each voice/data outlet. Provide one receptacle above the ceiling to support video projection device. Extend circuit to wall location for connection to motorized screen. When it is expected that a conference room table will be specifically dedicated to floor space in a conference room, locate a floor-mounted receptacle under the table. This receptacle may be part of combination power/communications outlet.

d. Provide power outlets throughout the building to serve all proposed equipment, including government furnished equipment, and allow for

future reconfiguration of equipment layout. Provide power connections to all ancillary office equipment such as printers, faxes, plotters, and shredders. Provide dedicated circuits where warranted.

e. In each telecommunications room provide a dedicated 16 amp circuit with a receptacle adjacent to each rack or backboard for each of the following:

1) CCTV for training systems

2) CCSTV for security systems

3) CATV 4) Voice systems

5) Data systems.

f. Provide dedicated receptacles as required throughout the facility for television monitors. These outlets will typically be located at the ceiling level for wall mounted television monitors.

g. Provide dedicated receptacles as required throughout the facility for tape players and disc players.

h. Corridors: One every 15 m (50 ft) maximum with a minimum of one per corridor.

i. Janitor's closet and toilet rooms: One GFI receptacle per closet. Provide GFI receptacles at counter height for each counter in toilets such that there is a minimum of one outlet for each two sinks.

j. Space with counter tops: One for every 1.200 m (4 ft) of countertop, with a minimum of one outlet. Provide GFI protection of outlets when located within 1.8 m (6 ft) of plumbing fixtures.

k. Building exterior: One for each wall, GFI protected and weatherproof.

l. Kitchen non-residential: One for each 3.05 m (10 ft) of wall space at the floor line. Provide GFI protection when located within 1.8 m (6 ft) of plumbing fixture.

m. Child occupied spaces (including toilets): One for every 3.6 m (12 ft) of wall space. Use child safety type such as those that require rotating an integral surface cover plate to access current. Removable caps and plugs are not acceptable.

n. All other rooms: One for every 7.6 m (25 ft) of wall space at the floor line. When 7.6 m (25 ft) or less of wall at the floor line exists in a room, provide a minimum of two receptacles spaced appropriately to anticipate furniture relocations.

o. Special purpose receptacles: Designer of Record must coordinate with the user to provide any special purpose outlets required. Provide outlets to allow connection of equipment in special use rooms.

3. Service Entrance Equipment: When a switchboard or switchgear is required, the Designer of Record must utilize UFGS Section 26 23 00, Low Voltage Switchgear or UFGS 26 24 13 Switchboards, for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

D5020 LIGHTING & BRANCH WIRING

1. Lighting Fixtures: All lighting fixtures must be energy conservation compact fluorescent or Light Emitting Diode (LED)except where indicated by Part 3.

a. Fixtures for Administrative and Commercial Spaces: For offices, commercial and administrative spaces and facilities provide high efficiency ballast, and instant or rapid start recessed fluorescent fixtures or LED fixtures.

b. Three-Way and Four-Way Switches: Provide three-way or four-way switching of light fixtures as necessary to facilitate movement between adjacent spaces to allow efficient energy management.

2. Exterior Lighting Fixtures for Large Open Areas: Exterior lighting fixtures for large open areas such as parking lots, streets and playgrounds must be energy efficient High-Intensity Discharge (HID) or compact fluorescent fixtures and must comply with local regulations regarding low lighting levels to avoid light pollution.

a. Photocell Overriding Switch: Provide photocell-overriding switch for all outdoor light fixtures.

D5030 COMMUNICATIONS & SECURITY

1. Telecommunications Systems: Provide a horizontal distribution system including, but not necessarily limited to, all wiring, pathway systems, connector blocks, protectors for all copper service entrance pairs, terminators for all fiber optic cables, outlet boxes, telephone jacks, and data jacks cover plates in accordance with EIA/TIA standards. Provide Category 6 UTP telephone premise wiring where telephones are required.
2. Public Address Systems: Provide a Public Address system with speakers in all locations identified in Part 3.
3. Intercommunications Systems: Provide an Intercommunication System to allow two-way communications between all locations identified in Part 3.
4. Television Systems: Provide television systems to the extent specified in Part 3. Coordinate with the local Cable Company, Local users and Local Authority at the Activity for other specific requirements. The interior cable outlets and wiring must be complete and ready for use. Wiring must not be run exposed on any surface of the building.
5. Security Systems: Provide an Intrusion Detection System (IDS) to sense all perimeter doors and windows and the interior volume in at least two locations. System must have a minimum 90-minute battery back-up and annunciate both locally and at the Base Security Office via a telephone dialer. System must have entry/exit timer. Provide wall mounted keypad control at two locations.

D5090 OTHER ELECTRICAL SERVICES

1. Surge Protective Device (SPD): Provide SPD in accordance with UFC 3-50101, Electrical Engineering and CEI EN 62305.

2. Variable Frequency Drives: When variable frequency drives are required, the Designer of Record must utilize UFGS Section 26 29 23 20 for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

3. Emergency Generators: When an emergency generator is required, the Designer of Record must utilize UFGS Section 26 32 13.00 20 for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

4. Automatic Transfer and Bypass/Isolation Switches: When an Automatic Transfer Switch is required, the Designer of Record must utilize UFGS Section 26 36 23.00 20 for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

5. Uninterruptible Power Supply (UPS) System: When a UPS system is required, the Designer of Record must utilize UFGS Section 26 33 53.00 20 and must submit the edited specification section as a part of the design submittal for the project.

6. 400 Hertz Systems: The Designer of Record must utilize UFGS Section 26 32 26 or 26 35 43 for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

7. Lightning Protection: When lightning protection is required, the Designer of Record must utilize UFGS Section 26 41 00.00 20 for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

8. Building Photovoltaic System: When a PV system is required, the Designer of Record must utilize UFGS Section 26 31 00 for the project specification, and must submit the edited specification section as part of the design submittal for the project.

SECTION E. EQUIPMENT AND FURNISHINGS

E10 EQUIPMENT

Equipment and Appliances: Provide appliances and equipment to fulfill the work for Part 3. Whenever possible, all appliances and equipment provided for the facilities in the contract must be by the same manufacturer and must be the current model available at the time of proposals. Discontinued makes and models are prohibited. All appliances and equipment must comply with applicable Energy Star efficiency rating requirements and must be rated as high efficiency models. Appliances and equipment on California projects must comply with California Title 24 and be rated as high efficiency. All appliances must be of the same manufacturer and shall be the same, or similar in color. Submit catalog information for approval by the Contracting Officer prior to purchasing, delivery and installation of the appliances at the job site. Equipment and appliances such as dishwashers, ice machines with drains, garbage disposers, and ovens/ranges are not considered FF&E.

E20 FURNISHINGS

E20 1.1 FURNISHINGS

The Contractor must have an Interior Designer, certified by the National Council for Interior Design Qualification (NCIDQ) or a state and/or jurisdiction Certified, Registered, or Licensed Interior Designer, or applicable HN entity prepare both the FF&E and the SID Package and participate in any design charrette to develop the building floor plan. As required, the Contractor must obtain services of equipment specialists to specify the audiovisual, shop, or specialty equipment. The Interior Designer and any specialists must not be affiliated with any furniture dealership/vendor or manufacturer. The Government Interior Designer reserves the right to approve/disapprove the qualifications of the Contractor's Interior Designer.

Systems furnishings installers must be the systems furniture manufacturer's approved dealer of record. In addition, installation dealers must be located within a 100 mile radius of the project site unless approved by the government Interior Designer.

For renovation projects, contractor to re-purpose/recycle existing furniture if not relocated by the government. Contractor to provide verification that the existing furniture was not disposed of at the landfill.

WINDOW TREATMENTS

Provide interior window coverings, associated hardware and controls at each exterior window and at any interior view window where privacy may be required. Refer to the Project Program for size, pattern and style of window treatments. At a minimum, functional window coverings are required on all projects.

Provide energy efficient solar shading systems for exterior windows. The system shall maintain visibility while reducing glare, solar heat gain during the summer and heat loss during the winter. Openness configuration shall be no more than 5% for most areas. The system fabrics and components must be dimensionally stable and must be manufactured to withstand fading, fire, mildew, and soiling. Product must have a minimum 10 year commercial warranty.

AUDITORIUM, LECTURE AND CLASSROOM SEATING

The system must permit the standards to be installed on radial lines from a common center for which concentric circles are determined with each row of units utilizing common middle standards. Standards in each row must be placed laterally so the aisle-end standards will be in alignment as indicated on seating layout drawing. The angle of inclination of backs must be adjusted for variations in sightlines. Mechanical attachment of components must be of sufficient flexibility so that when permanently assembled they will compensate for the changing dimensions laterally between standards caused by convergence toward the center. Seat and back attachments must absorb inaccuracies in lateral spacing of standards at point of attachment caused by unevenness of floor. Varying lateral dimensions of backs and seats must be in accordance with approved seating layout. Minimum width of seating unit must be 20 inches and may be used only to complete a specific row dimension.

TABLES FOR AUDITORIUM, LECTURE AND CLASSROOMS

Provide worksurfaces mounted to floor standards of tubular steel, sheet steel, or cast iron. The standards must be formed to fit the floor incline so that the standards will be vertical. The feet must be formed to eliminate tripping hazards and must have holes for bolt attachment to the floor. Provide riser standards, cantilevered standards and aisle and end standards as required. Provide communications, data and power routing as required. Provide a high-pressure plastic laminate over medium density particleboard for the worksurfaces with coordinating vinyl or resin edge detail.

MOVABLE FURNISHINGS

Furnishings, Fixtures, and Equipment (FF&E) includes furniture, shop equipment, audiovisual equipment, and specialty equipment. Weapon racks, drying cages, and lockers are not considered FF&E. FF&E must be fully integrated with the building systems and finishes. FF&E may also include specialty items for which the customer activity will be responsible for specifying.

Design and provide as required FF&E for all areas as developed during client programming. Design an FF&E package and prepare supporting plans and procurement data in accordance with the general interior design requirements in UFC 3-120-10.

SECTION F. SPECIAL CONSTRUCTION AND DEMOLITION

F10 SPECIAL CONSTRUCTION AND DEMOLITION

F1010 SPECIAL STRUCTURES

1. Pre-Engineered Buildings

Provide the design and installation in accordance with the UFC 3-101-01, Architecture and UFC 3-301-01, Structural Engineering.

a. Design Requirements - The metal building manufacturer must be accredited by the International Accreditation Services (IAS) AC472. The Metal Building System design must be in accordance with the Metal Building Manufacturers Association (MBMA) Metal Building Systems Manual. All structural design must comply with the provisions of Section B10, "Superstructures", above.

b. Additional Roof Design Requirements - Roof Decking - In addition to any other load requirements, roof decking must be designed to support a 91 kg (200-pound) concentrated load at mid-span on a 300 mm (12-inch) wide section of deck.

c. Deflection - the maximum deflection for -

1) Structural Members - main framing members must be L/240.

2) Purlins and Roof Panels: The deflection due to live, snow, or wind must not exceed L/180.

3) For buildings with masonry infill, limit frame sway to L/600th of building eave height.

4) Wall Panels - Maximum deflection of wall panels due to wind loads must be limited to L/120th of their respective spans except that when interior finishes are used the maximum allowable deflection must be limited to L/180th of their respective spans.

d. Wall and Roof materials -

1) Alum/Zinc-Coated Steel Sheet: ASTM A792/ A792M, AZ 55.

2) Aluminum Sheet: Alloy 3004 Alclad conforming to ASTM B209.

3) Framing and Structural Members - Steel - ASTM A992 / A992M, ASTM A529/ A529M, ASTM A572/ A572M, or ASTM A588/ A588M.

4) Framing and Structural Members, Aluminum: ASTM B221 or ASTM C308

e. Structural Tube: ASTM A500 or ASTM B221.

f. Fasteners - Must be compatible with the materials they are fastening to, be gasketed when exposed to weather to prevent leaks, and provide both shear and tensile strengths not less than 3,336 N (750 pounds) per fastener.

The main fastening system must use concealed fasteners, however, when exposed fasteners are needed, color fasteners must be color coated to match wall/roof panels.

g. Light Transmitting Roof Panels (Non-Insulated): ASTM D3841, Type II, Grade 1.

h. Insulation: Blanket-type fiberglass insulation with a factory applied facing on one side and having a permeance rating of 0.05 or less in accordance with ASTM E96. Flame Spread Rating 75 or less, and a Smoke Developed Rating of 150 or less when tested in accordance with ASTM E84.

i. Panel Finish - Factory Color Finish - Provide factory applied baked coatings to the exterior and interior of metal wall panels and metal accessories. Provide exterior primer standard with panel manufacturer but not less than 0.8 mil dry film thickness (DFT). Provide PVDF exterior finish top coat of 70 percent inorganic pigments with 0.8 mil DFT. Provide factory-applied clear finish over the color topcoat and edge coating for projects within 91 meters of a water shoreline or industrial environments. Field apply 70 percent PVDF clear coat to unfinished panel edges or field cut panels. Interior finish exposed to sun or rain must be the same coating and DFT as the exterior coating. Interior finish protected from sun or rain exposure must receive 1.0 mil DFT coating of siliconized polyester (SMP) resin coating with organic or blended pigments and manufacturer's standard primer.

F20 SELECTIVE BUILDING DEMOLITION

In general terms, demolition work must include the removal and effective management and disposition of existing construction and or structures. Take care to prevent damage to existing utilities and construction that are not scheduled for demolition. If damage occurs, make repairs to the satisfaction of the Contracting Officer and at no cost to the Government. Comply with local Activity and Installation local requirements regarding demolition and removal. Unless otherwise specified in Part 3, all demolished materials and equipment must be removed from government property in accordance with applicable laws and regulations, including local Activity or Installation regulations. Selling of demolished or salvaged materials and equipment on government properties is prohibited.

Demolition Plan: No demolition work is permitted to take place until a Demolition Plan has been submitted to, and approved by, the Contracting Officer. The Plans must take into consideration, and indicate method of removal, disposition, and abatement of environmentally hazardous materials. Demolition, disposition, and abatement of hazardous materials must comply with all applicable Local, State, and Federal regulations and laws. If destructive investigation is to occur, the Contracting Officer shall approve a Destructive Investigation Plan.

When hazardous materials such as asbestos, lead paint, PCB and other hazardous materials are involved in the performance of the work, the contractor must abate, remove and manage the hazardous materials in construction and finish materials such as insulation, flooring, wall materials, ceiling materials, roofing materials, heating, plumbing, ventilation, air conditioning equipment and installations in accordance with National as well as local Environmental Protection Laws and Regulations.

F2020 HAZARDOUS COMPONENT ABATEMENT

1. Asbestos in Schools and Child Occupied Facilities: For projects that require removal or disturbance of asbestos containing materials within schools and child occupied facilities, perform work in accordance with the edited UFGS 02 82 16, Asbestos Remediation.

2. Asbestos Materials: Asbestos must be removed, transported and managed in accordance with the edited UFGS 02 82 16, Asbestos Remediation.

3. Lead Based Paint in Target Housing and Child Occupied Facilities: Perform work for projects that require removal or disturbance of painted surfaces within target housing and a child occupied facilities in accordance with the edited UFGS 02 82 33, Lead Remediation.

4. Paint Related Work: Perform work which requires the disturbance of paint that have been determined to contain all or any of the following: lead, cadmium and chromium in accordance with the edited UFGS 02 83 13, Lead Remediation.

5. LLR Components: Perform work which requires removal of mercury and LLR components in accordance with the edited UFGS 01 57 19, Temporary Environmental Controls.

6. PCBs: Perform work which requires removal of PCB containing components or materials in accordance with the edited UFGS 02 84 33, Removal and Disposal of Polychlorinated Biphenyls (PCBs).]

7. Animal Droppings: Perform work which requires removal of animal droppings in accordance with the edit UFGS 01 57 19, Temporary Environmental Controls.

8. Mold and Spores: Perform work which requires removal, disposal and remediation of mold contaminated areas in accordance with the edit UFGS 02 85 00, Mold Remediation.

9. Radon: Perform work which involves implementation of a radon mitigation system in accordance with the edited UFGS 31 21 13, Radon Mitigation.

10. Mercury: Perform work which requires the removal of mercury containing equipment in accordance with the edited UFGS 02 84 16, Handling of Lighting Ballasts and Lamps Containing PCBs and Mercury.

11. Hazardous Materials Reporting:a. Daily Report: Notify the Contracting Officer of work involving hazardous materials abatement and removal, including the quantities involved, on daily reports.b. Hazardous Material Inventory Report: The Contractor must provide a list of all hazardous materials used on the site.

SECTION G. BUILDING SITEWORK

G10 SITE PREPARATIONS

1. General Requirements: Building site work includes site preparation, site improvements, site civil/mechanical utilities, site electrical utilities, exterior furnishings, landscaping, and irrigation. Provide site work in accordance with UFC 3-201-01, Civil Engineering.

2. Project Limitations: Prior to the start of design, determine the exact limit-ofwork line for the project periphery, considering items such as, but not limited to, utility work, landscape re-vegetation of disturbed areas, and lay down areas. The Civil Engineer of Record in coordination with other design team disciplines must determine limit-of-work lines. Minimize the impact of construction activity on operations and neighboring facilities.

3. Geotechnical Data: A geotechnical engineer must conduct the subsurface exploration, investigation/evaluation, testing, and analysis that the Designer of Record deems necessary for the design and construction of the proposed facilities, including building pad, structure, pavement sections, repairs, overlays, stormwater management facilities, utility structure foundations, septic systems, and other features requiring soil support.

3. Temporary Erosion & Sediment Control: Develop and implement temporary erosion and sediment control measures and other Best Management Practices (BMPs) prior to or in conjunction with commencement of earthwork in accordance with the state Erosion and Sediment Control Laws and Regulations. Remove all non-permanent erosion control measures after vegetation is fully established. Maintain temporary erosion control measures in accordance with state Erosion and Sediment Control Laws and Regulations throughout the project until areas are fully stabilized.

G1010 SITE CLEARING

1. Existing Utilities: When the Contractor is to work at a site that has existing utilities, the contractor is responsible for coordination with Contracting Officer and utility companies for staking out, capping, connection and relocation of any existing utility systems or traffic interruption. Notify utility locator service for area where Project is located before site clearing.

2. Interruption: All interruption to the existing utilities and traffic must be coordinated with and approved by the Contracting Officer not less than 14 calendar days in advance of such interruption.

G1020 SITE DEMOLITION & RELOCATIONS

Abandon utility systems in-place conforming to applicable codes and regulations, removing their presence from the ground surface and clearly indicating that they have been abandoned. Remove utilities underneath or within 3.0 m (10 feet) of any new facilities. Fill abandoned gravity systems with flowable fill. Fill abandoned utility system piping under pavements subject to potential vehicle loading with flowable fill. Remove existing utility structures to 900 mm (3 feet) below existing or new adjacent grade, whichever is greater. Break up bases to permit drainage. Fill with clean sand. Comply with the requirements of the utility provider concerning utility relocation.

G1030 SITE EARTHWORK

The DOR must utilize UFGS Section 31 23 00.00 20 for the project specification and must submit the edited section as a part of the design submittal. Perform quality assurance for earthwork in accordance with UFGS Section 31 23 00.00 20. If sheeting/shoring or dewatering is required, the Contractor must provide a registered Professional Engineer to provide excavation, sheeting, shoring, and dewatering plans and inspection of excavations and soil/groundwater conditions throughout construction. The Engineer must be responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Engineer, with the concurrence of the contractor and the Contracting Officer, must update the excavation, sheeting, shoring, and dewatering plans as construction progresses to reflect actual site conditions and must submit the updated plan and a written report (with professional seal) at least monthly informing the Contractor and the Contracting Officer of the status of the plan and an accounting of Contractor adherence to the plan; specifically addressing any present or potential problems. The Engineer must be available to meet with the Contracting Officer at any time throughout the contract duration.

G20 SITE IMPROVEMENTS

Provide site improvements as required to make a useable facility that meets functional and operational requirements, incorporates all applicable anti-terrorism, force protection and physical security requirements and blends into the existing environment.

Provide site improvements in conformance with applicable requirements of the Uniform Federal Accessibility Standards.

1. Pavements: For work in and adjacent to existing pavements, the Contractor is required to match the existing adjacent finish elevation, materials, paving section and texture, and pavement markings, unless otherwise indicated in Part 3 or directed by the Contracting Officer.

   Provide pavement design and pavement section materials in accordance with UFC 3-201-01, Civil Engineering. Provide pavement markings in accordance with the SHS. Design materials for life expectancy of at least 3 years under an average daily traffic count per lane of approximately 9000 vehicles. Water based paints must have durability rating of at least 4 when determined in the wheel path area. Provide a half-rate initial marking application on bituminous pavements. Provide the remaining application at the end of the normal curing period.

2. Traffic Control: If the site work involves interference with normal vehicular and or pedestrian traffic, the Contractor must coordinate with the authority having jurisdiction, propose and obtain approval for traffic control measures that may be required in performance of the work required by the contract.

3. Performance Verification And Acceptance Testing:

   a. Subgrade Preparation: If required by the Designer of Record, perform proof rolling. Proof rolling must be performed in the presence of the Contracting Officer. Rutting or pumping of material must be undercut as directed by the Contracting Officer and replaced with satisfactory soil materials as defined by the Geotechnical Engineer.

   b. Base Course Performance Verification: At a minimum, Contractor must perform visual performance verification. Surface must be smooth with no ruts, sloped or crowned to not pond water.

   c. Bituminous Concrete Pavement Performance Verification: At a minimum, Contractor must perform visual performance verification. Finished surface must be uniform in texture and appearance, free of defects such as cracks and creases, and be sloped or crowned so as to not pond water.

   d. Portland Cement Concrete Pavement Performance Verification: At a minimum, Contractor must perform visual performance verification. Finished surface must be uniform in texture and appearance, free of defects such as cracks and spalls, and be sloped or crowned so as to not pond water.

   e. Concrete Joint Performance Verification: Joint sealer that fails to cure

      properly, or fails to bond to joint walls, or reverts to uncured state or fails in cohesion, or shows excessive air voids, blisters, or has surface defects, swells, or other deficiencies, or is not recessed within indicated tolerances will be rejected. Remove rejected sealer, re-clean and reseal joints.

4. Bases and Subbases:

   a. Prepare subgrade in accordance with Section G10, Site Preparation. Geotextiles may be used for separation or reinforcement in accordance with manufacturer's instructions. Provide base course under paved areas in accordance with the State Highway specifications (SHS) in the state where the project is located.

   b. Place base course in accordance with the SHS for that particular base course and in layers of equal thickness with no compacted layer more than 6 inches (150 mm) thick. Compact base course at optimum moisture content to 100 percent ASTM D 1557 maximum dry density.

   c. Where SHS are not available or applicable, the Designer of Record must utilize the applicable UFGS Specification Sections referenced under paragraph 1.1.2 entitled "Government Standards" for the project specification. Submit these specifications in edited form as a part of the design submittal for the project.

5. Curbs and Gutters: Provide concrete curbs and gutters in accordance with the SHS and standards or as specified in UFC 3-201-01, Civil Engineering, whichever is more stringent. Where the SHS do not include concrete materials for curbs and gutters, provide concrete in accordance with the applicable standard mix of the SHS for a minimum compressive strength at 28 days of 3500 psi (25 MPa) concrete.

6. Paved Surfaces: Where SHS are not available or applicable, the Designer of Record must utilize the applicable UFGS Specification Sections for the project specification. Submit these specifications in edited form as a part of the design submittal for the project.

7. Pavement Mix:

   a. Bituminous Concrete Pavement: Provide bituminous concrete pavement in accordance with the applicable standard mix of the SHS based on the pavement design and vehicle loading indicated in this RFP.

   b. Bituminous Concrete Placement: Provide bituminous concrete placement, including minimum temperature during placement, joints, and maximum lift thickness in accordance with the SHS. Compact bituminous concrete in accordance with the SHS, modified to 96 percent of maximum laboratory density.

   c. Portland Cement Concrete Pavement: If reinforced, provide the welded wire fabric in conformance to ASTM A185. Provide bar reinforcement in conformance to ASTM A615/A615M, Grade 400 (Grade 60). Provide concrete in accordance with the applicable standard mix of the SHS for the design strength required by UFC 3-201-01, Civil Engineering, plus any allowable deviations. Unless noted otherwise in Part 3 or Part 6, provide a minimum compressive strength at 28 days of 3500 psi (25 MPa) concrete. If required for applicable sustainability goal, provide Portland cement concrete pavement with a Solar Reflectance Index (SRI) greater than or equal to 29.

8. Joints for Portland Cement Concrete Pavement: Provide joints in accordance with SHS and the applicable portions of UFC 3-250-01, Pavement Design for Roads and Parking Areas. Install joints in a manner and at such time to prevent random or uncontrolled cracking. Joints must form a regular rectangular pattern. Wherever curved pavement edges occur, make joints to intersect tangents to curve at right angles.

    a. Expansion Joints: Provide thickened edge expansion joints at the intersection of two rigid pavements. Use preformed joint filler, ASTM D1751. Filler must be compatible with joint sealer material. Securely hold preformed joint filler in position during concreting operations.

    b. Isolation Joints: Provide thickened edge isolation joints by placing a 1/2inch (12 mm) preformed joint filler (ASTM D 1751) around each structure that extends into or through the pavement before concrete is placed at that location.

    c. Contraction Joints: Saw joint lines within specified tolerance, straight, and extend for width of transverse joint, and for entire length of longitudinal joint.

    d. Construction Joints: If an emergency stop occurs remove the concrete back to location of transverse joint and install a construction joint.

    e. Joint Sealants: ASTM D5893/D5893M; provide single component cold-applied silicone. Silicone sealant must be self-leveling and non-acid curing.

    f. Preformed Compression Seals: Use preformed compression seals in areas where silicone joint sealant does not perform, such as areas subject to water inundation, blasts, or constant/repeated fuel spillage. ASTM D 2628. ASTM D 2835, for lubricant.

9. Prime Coat: Use prime coat in accordance with the SHS. Prime coat must be emulsified asphalt materials.

10. Tack Coat: Tack coat is required for bituminous pavement overlays and on vertical cut faces of pavement patches. Provide tack coat in accordance with the SHS.

11. Pavement Patches: Provide pavement patches for existing pavements where required for installation of utility trenches. Sawcut 12 inches beyond edge of trench. Provide thicknesses of pavement materials equal to or greater than the existing pavement section. For spalls or repairs of existing concrete pavement, perform repairs in conformance with UFC 3-270-03, Concrete Crack and Partial Depth Spall Repair, and UFC 3-270-04, Concrete Repair. Spall repair materials must be either Rapid Setting Cementitious Concrete (RSCC), epoxy concrete, or polymer-modified Portland Cement (non-sag mortar) products specially formulated for spall repairs, with a proven record (in service at least three years) of satisfactory use under loading and environmental conditions similar to those at the location of intended use. Provide a manufacturer's data sheet and certificate supporting the satisfactory use to the Contracting Officer with the design. A product manufacturer's representative is required to be present during the initial two days of product application to verify that manufacturer's instructions for use are adhered to by the contractor. Give the Contracting Officer 7 days notice prior to the initial application in order to be present.

12. Markings and Signage: Provide pavement markings in accordance with the SHS. Design materials for life expectancy of at least 3 years under an average daily traffic count per lane of approximately 9000 vehicles. Water based paints must have durability rating of at least 4 when determined in the wheel path area. Provide a half-rate initial marking application on bituminous pavements. Provide the remaining application at the end of the normal curing period. Provide signage in accordance with the HN Road Code.

13. Guardrails and Barriers: Provide guard (guide) rails in accordance with the SHS. Where the SHS do not include materials for guardrails, provide guardrails in accordance with the applicable portions of the AASHTO Roadside Design Guide.

14. Bollards: For bollards to prevent damage, provide minimum 4 feet height, 4 inch diameter steel pipe filled with concrete, painted, and embedded in a portland cement concrete foundation. For bollards located at building entries or other highvisibility areas provide decorative bollards matching the design of the facility or consistent with the Base Exterior Architecture Plan (BEAP) and the Installation Appearance Plan.

15. Resurfacing: Adjust rims of existing utility structures to match proposed grades after resurfacing.

    a. Slurry Seal: ASTM D 3910 and in accordance with the SHS.

    b. Bituminous Concrete Overlay: Remove old pavement by cold milling to depths required to provide new surface and leave underlying materials intact. Clean the pavement of excessive dirt, clay or other foreign matter with power brooms and hand brooms immediately prior to the milling operation. Repair or replace damaged utility structures, valve boxes, or pavement that is torn, cracked, gouged, rutted, broken or undercut at no addition expense to the government. Provide bituminous concrete overlay produced from hot or cold recycling of the milled material or from virgin materials in accordance with the applicable provisions of UFC 3-201-01, Civil Engineering, and the standard mix of the SHS based on the pavement design and vehicle loading as indicated in this RFP.

c. Crack Sealing: Use fiber reinforced crack sealer for sealing cracks in asphalt pavement after milling and prior to resurfacing. Provide crack sealing conforming to the following requirements in UFC 3-270-01, Asphalt Maintenance and Repair, and UFC 3-270-02, Asphalt Crack Repair.

16. Parking Lots: Refer to Pavements above.

17. Permeable Pavements: Provide permeable concrete pavers of solid interlocking paving units complying with ASTM C936, resistant to freezing and thawing when tested according to ASTM C67, and made from normal-weight aggregates. If required for applicable sustainability goal, provide permeable concrete pavers with a Solar Reflectance Index (SRI) greater than or equal to 29. Provide pervious concrete in accordance with UFGS Specification Section 32 13 43, Pervious Concrete Paving. Do not use asphalt-surfaced porous pavement.

a. Bases and Subbases: Refer to Bases and Subbases paragraph above.

b. Curbs and Gutters: Refer to Curbs and Gutters paragraph above.

c. Paved Surfaces: Refer to Paved Surfaces paragraph above.

d. Markings and Signage: Refer to Markings and Signage paragraph above. Provide water-based paints only.& Mark neatly to denote traffic lanes and parking spaces; mark in accordance with the requirements of UFC 3-20101, Civil Engineering.

e. Guardrails and Barriers: Refer to Guardrails and Barriers paragraph above.

f. Wheelstops: Provide precast concrete wheelstops.

18. Pedestrian Paving: Locate new sidewalks such that they maintain continuity of pedestrian traffic to and from the existing sidewalks adjacent to the site(s).

a. Bases & Subbases: Provide as required by local standards or geotechnical report; refer to Bases and Subbases paragraph above.

b. Paved Surfaces:

c. Concrete Sidewalks: Provide sidewalks of Portland cement concrete pavement with 4 inches (100 mm) thick minimum or permeable pavement. Provide concrete and permeable pavement in accordance with Section

G201003 and G2020, respectively. For PCC sidewalks, provide a broomed finish. Provide sidewalks of at least 5 feet (1.5 meters) wide, except that sidewalks connecting entry points of housing units to the housing unit's parking are required to be at least 36 inches (900 mm) wide. In housing

areas, offset sidewalks paralleling streets to maintain a minimum grassed separation of 5 feet (1.5 meters) from the back face of the curb to the closest edge of the sidewalk Unless indicated otherwise, provide a transverse slope of 1/48. Limit variation in cross section to 0.25 inch in 5 feet (6 mm in 1.50 m). Submit samples boards per ESR G2050 and PTS G2050 and finish schedule on final plans. Provide contraction joints spaced at intervals equivalent to the width of the sidewalk. Provide 0.5 inch (13 mm) thick transverse expansion joints at changes in direction where sidewalk abuts curb, steps, rigid pavement, or other similar structures; space expansion joints every 50 feet (15 m) maximum. Provide isolation joints by placing a 1/2-inch (12 mm) preformed expansion joint filler around each structure that extends into or through the sidewalk before concrete is placed at that location.

d. Concrete Pavers as Sidewalk: Concrete pavers shall meet ASTM C936. Install in accordance with manufacturers recommendations.

19. Chain Link Fencing and Gates: Chain link fence fabric must be at least 9 gauge (3 mm) steel wire mesh material (before any coating) with mesh openings not larger than 2 inches (51 mm). Do not use aluminum fabric, posts or accessories. Install fence in accordance with ASTM F567 and the manufacturer's written installation instructions. Provide rails in accordance with FS RR-F-191/3, Class 1, steel pipe, Grade A. Provide gates in accordance with FS RR-F-191/2 with posts and fabric as specified for fence. Provide posts and braces in accordance with FS RR-F-191/3, Class 1, steel pipe, Grade A. Each gate, terminal and end post will be braced with truss rods. Provide fencing accessories in accordance with FS RR-F-191/4. If PVC coating is required, provide accessories with PVC color coating similar to that specified for chain-link fabric or framework.

20. Security Fencing: Provide security fencing systems in accordance with UFC 4022-03, Security Fences and Gates, and this RFP. Provide chain link fence in accordance with paragraphs above, except as noted otherwise. Ensure that the fabric has twisted and barbed selvage at the top and bottom. Do not provide top rails. Locate all posts and structural supports on the inner side of the fencing. Install outriggers facing outward except when the fence must be mounted directly on the property line. Provide signage at a minimum of 200 foot (61 m) intervals along the entire perimeter. Provide protective measures to prevent access through culverts, storm drains, sewers, air intakes, exhaust tunnels and utility openings or across drainage ditches or swales as required in UFC 4-022-03. Do not cover, block or lace openings in perimeter fencing and security fencing with material

which would prevent a clear view of personnel, vehicles or material in the outer or inner vicinity of the fence line.

21. Fence Grounding: Grounding and bonding of the fencing must be in accordance with the National Electric Safety Code (NESC) - IEEE C2 and UFC 4-022-03. Ground fencing on either side of every gate and at other locations when the fencing is near and parallel to high tension power lines. Grounding is also

    required at intervals of 1000 feet (305 meters) to 1500 feet (457 meters) when the fencing runs through isolated areas and at lesser distances depending on the proximity of the fencing to public roads, highways and buildings where the fencing is around or within any explosive storage, production, operating or handling areas.

22. Enclosures for Utility Equipment: Where fencing is used to provide an enclosure for utility equipment, ensure a minimum clearance is provided no less than 3 feet (900 mm) around the equipment to permit maintenance access and ventilation. Provide stone, gravel or concrete paving within the enclosure.

G2040 EXTERIOR FURNISHINGS

All site furnishings must be permanently attached to concrete pads. Site furnishings must conform to the Base Exterior Architecture Plan (BEAP) or Installation Appearance Plan (IAP) for each Activity. If no product guidance is given, coordinate material, finish and color with architecture (fiberglass and aluminum are not acceptable) and provide to the greatest extent possible, materials with industrial recycled content, preferably from regionally local manufacturers. At a minimum, provide a trash and ash receptacle at each entry and gathering/smoking area.

G2050 LANDSCAPING

G30 SITE CIVIL/MECHANICAL UTILITIES

Develop the site to provide water, fire protection, sanitary sewer, storm drainage, heating, cooling and fuel distribution services that meet the requirements of each utility provider and each applicable regulatory agency that governs and issues permits for the construction and operation of these systems. Site design must also comply with Department of Defense requirements concerning Low Impact Development (LID) per UFC 3-210-10, Low Impact Development, as well as state or local stormwater management regulations, and applicable project sustainability goals. Submit sealed calculations with narrative to the Government for civil and environmental review documenting all assumptions and which criteria governs the design.

Coordinate with the local utility providers and pay any fees or charges required to connect to their utility. Identify and obtain all permits to comply with all federal, state, and local regulatory requirements associated with this work. Coordinate all reports, submittals, and permit applications through the Contracting Officer. Perform work in accordance with the obtained permits.

Provide all required fittings, connections and accessories required for a complete and usable system. All equipment must be installed per the criteria indicated in this RFP and the manufacturer's recommendations. Where the word "should" is used in the manufacturer's recommendations, substitute the word "must".

G3010 WATER SUPPLY

1. Water System Design and Construction: Provide the new water system and connections to the existing water system in accordance with UFC 3-230-01, Water Storage, Distribution, and Transmission; the utility provider's requirements; or the state waterworks' regulations; whichever is more stringent.

2. Notifications: Notify the utility provider of the additional demand generated by the proposed facility. Provide a copy of all correspondence with the utility provider to the Government's Civil/Mechanical Reviewer.

3. Performance Verification And Acceptance Testing: Provide testing on water mains and service lines in accordance with the state waterworks' regulations and the following:

    a. Ductile iron and other materials: AWWA C600.

    b. PVC: AWWA C605.

    Whichever is more stringent. Do not begin testing on any section of a pipeline where concrete thrust blocks have been provided until at least 5 days after placing of the concrete.

4. Disinfection: Disinfect new water piping and existing water piping affected by Contractor's operations in accordance with the state waterworks' regulations and AWWA C651.

5. Water Distribution Mains: For underground applications, water mains 12 inches (300 mm) in diameter and less must be ductile iron or PVC. Water mains deeper than 10 feet (3.0 m) or larger than 12 inches (300 mm) in diameter must be ductile iron. For aboveground applications, water mains shall be flanged ductile iron pipe.

    a. Ductile Iron Pressure Pipe:
        1) Pipe: AWWA C151, Pressure Class 350.

2) Fittings: AWWA C110 or AWWA C153.
3) Interior Lining: AWWA C104.
4) Exterior Protection (if required): AWWA C105, polyethylene encasement.

b. PVC Pressure Pipe:
1) Pipe: AWWA C900, Pressure Class 150.
2) Fittings: Ductile Iron (AWWA C110 or AWWA C153).

c. Flanged Ductile Iron Pipe:
1) Pipe: AWWA C115 and its appendices.
2) Fittings: AWWA C110 or AWWA C153.

3) Lining: AWWA C104.

6. Installation of Water Distribution Mains:

a. Ductile Iron: AWWA C600.

b. PVC: AWWA C605.

c. Provide nondetectable warning tape and a continuous length of tracer wire for the full length of each run of nonmetallic piping below grade. Warning tape to be color coded with warning and identification of utility type imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (utility type) LINE BELOW" or similar wording. Color to be blue for potable water systems and purple for nonpotable, reclaimed water, and irrigation lines. Terminate tracer wire above grade at valve boxes and at exterior of building.

7. Connections to Existing Water Lines: Make connections to existing water lines after approval from the system owner is obtained and with a minimum interruption of service on the existing line. Make connections to existing lines under pressure in accordance with the recommended procedures of the manufacturer of the pipe being tapped.

8. Water Service Lines: Water service lines less than 4 inches (100 mm) in diameter must be copper tubing or PVC. Water service lines 4 inches (100 mm) and 6 inches (150 mm) in diameter must be ductile iron pipe and PVC pressure pipe; see paragraph "Water Distribution Mains" for additional requirements for ductile iron and PVC piping.

a. Copper Tubing
1) Pipe: ASTM B 88/B 88M, Type K.
2) Fittings for Solder-Type Joint: ANSI B16.8 or ASME B16.22.
3) Fittings for Compression-Type Joint: ASME B16.26, flared tube type.

b. PVC Pressure Pipe
   1) Pipe: ASTM D1785, Schedule 40 or ASTM D 2241, with SDR rating for 160 psi (1.1 MPa) pressure rating.
   2) Fittings: ASTM D 2466.
   3) Joints: Elastomeric gaskets for pressure rating; solvent cement joints, ASTM D 2564.

c. Service Connections: Connect service lines 2-inch (50 mm) diameter or less to the main by a corporation stop and install a gate valve on service line below the frostline.
   1) Ductile-iron water mains: AWWA C600.
   2) PVC water mains: UBPPA UNI-PUB-8 and the recommendations of AWWA M23, Chapter 9, "Service Connections."

9. Installation of Water Service Lines: Install pipe, fittings and accessories in accordance with manufacturer's instructions.

   a. Metallic Piping: applicable requirements of AWWA C600.

   b. PVC: ASTM D 2774 and ASTM D 2855.

10. Corrosion Protection: Provide insulating joints to prevent contact between dissimilar metals at the joint between adjacent sections of piping in accordance with the pipe manufacturer's recommendations. Ensure that there is no metal-tometal contact between dissimilar metals after the joint has been assembled. To prevent the possibility of bi-metallic corrosion, service lines of dissimilar metal to the water mains and the attendant corporation stops must be wrapped with polyethylene or suitable dielectric tape for a minimum clear distance of 3 feet (900 mm) from the main.

11. Valves: Install valves with the same diameter and have the same joint ends as the mains to which they are installed. Each type of valve must be of one manufacturer.

   a. Gate Valves: Install valves at all new points of connection. At a minimum, locate valves to ensure that no more than two fire hydrants will be out of service in the event of a single break in a water main. Locate valves outside of pavement and heavy traffic areas whenever possible.

   b. Gate Valves 3-inch (75 mm) and Larger in Diameter:
      1) Valves (20-inch and smaller in diameter): AWWA C509 or AWWA C515, nonrising stem and of one manufacturer.
      2) Valves (greater than 20-inch in diameter): AWWA C500.
      3) Valves for Indicator Post: AWWA C509 or AWWA C500, as indicated above, with indicator post flange in accordance with applicable requirements of UL 262.
      4) Interior Coating: AWWA C550.

c. Gate Valves Smaller than 3-inch (75 mm) in Diameter: MSS SP-80, Class 150, solid wedge. Provide valves with flanged or threaded end connections, with unions on both sides of the valve and a handwheel operator.

d. Valve Box: Provide a cast iron, adjustable, valve box for each gate valve on buried piping. Provide valve boxes of a size suitable for the valve on which it is to be used with a minimum diameter of 5-1/4 inches (130 mm). Provide a round head and cast the word "WATER" on the lid.

e. Check Valves: Provide check valves sized 2-inches (50 mm) to 24-inches (600 mm) as swing-check type (AWWA C508) and with a protective epoxy interior coating conforming to AWWA C550. For underground applications, provide check valve in a valve vault.

f. Air Release, Air/Vacuum, and Combination Air Valves: AWWA C512 and AWWA M51.

g. Corporation Stops: If service lines 2-inch diameter or less are tapping water mains, provide corporation stops. The corporation stops must be ground key type, bronze, ASTM B61 or ASTM B62.

h. Installation of Valves: Make and assemble joints to valves as specified for making and assembling the same type of joints between pipe and fittings.

12. Water Meters: Provide water meter and remote reading as required by the utility provider and in accordance with AWWA standards.

13. Backflow Prevention: Provide backflow prevention and cross connection control in accordance with AWWA M-14 and governing local/state plumbing codes and waterworks' regulations.

14. Fire Hydrants: Fire hydrants must be of one manufacturer and in accordance with UFC 3-600-01, Fire Protection Engineering. Coordinate with the project's fire protection designer of record. Provide protection for fire hydrants located in areas subject to vehicle damage. Fire hydrants must have National Standard threads on hose and pumper connections. Provide a 6 inch (150 mm) inlet, two 2.5 inch (62 mm) hose connections and one pumper connection sized to accommodate local fire department equipment requirements. Paint hydrants with at least one coat of primer and two coats of enamel paint. Barrel and bonnet colors shall be in accordance with UFC 3-600-01. Stencil hydrant number and main size on the hydrant barrel using black stencil paint.

a. Dry Barrel Fire Hydrants: AWWA C502 with frangible sections.

b. Wet Barrel Fire Hydrants AWWA C503 or UL 246, "Wet Barrel" design, with breakable features.

c. Installation: Install hydrants with the pumper connection facing the adjacent paved surface. If there are two, paved adjacent surfaces, contact the Contracting Officer for further direction.

15. Thrust Restraint: Provide thrust restraint for all piping, valves, fittings, and other appurtenances of the water distribution system. Provide thrust restraint using restrained joints in accordance with pipe manufacturer's recommendations, AWWA C600 and if for fire service main, NFPA 24.

16. Fire Protection Water Distribution: Refer to applicable portions of this Section and Section D40, Fire Protection Systems. Water main piping, service lines, fittings, valves, accessories and all other materials must meet the American Water Works Association (AWWA) standards for a minimum system working pressure of 200 psi (1380 kPa).

    a. Detector Checks: UL 312; detector check includes bypass meter, piping, gate valves, check valve and connections to detector check valve. Set valve to allow minimal water flow through bypass meter when major water flow is required.

    b. Fire Department Connections: UL 405.

    c. Indicator Posts: UL 789.

G3020 SANITARY SEWER

1. Sanitary System Design and Construction: Provide the new sanitary sewer system and connections to the existing sanitary sewer collection system in accordance with UFC 3-240-01, Wastewater Collection; the utility provider□s requirements; or the state sewerage regulations; whichever is more stringent.

2. Notifications: Notify the utility provider of the additional wastewater flow generated by the proposed facility. Provide a copy of all correspondence with the utility provider to the Government Reviewer.

3. Wastewater Pump Station: Where required, provide a duplex, grinder pump station in accordance with the utility provider's requirements. Provide pump station wet well of fiberglass construction. Provide adjacent valve vault of precast concrete construction.

   Provide automatic control to start and stop the pump system. Provide automatic level control by floats in accordance with the preferences of the system owner to fill and prevent overflow of the wet well. Provide an emergency pump connection.

   Provide a telephone dialer in the control panel capable of accepting up to 8 telephone numbers. At the control panel provide an alarm horn and light with battery backup. Alarms shall include high liquid wet well level; loss of main power; no flow as sensed by current sensor; and pump failure via overload or motor heat sensor trip. Provide seal failure indicator lights and elapsed time meters.

Provide electrical connections for a portable emergency generator hook-up sized to start up and maintain the total rated running capacity of the station, including the pumps, controls, lighting, and other auxiliary equipment.

4. Performance Verification And Acceptance Testing:

a. Sanitary Sewer Distribution System Performance Verification: Provide testing on sewer mains and laterals in accordance with the state sewerage regulations. At a minimum, perform the following:

1) Visual Test: Remove manhole colvers and conducts a visual inspecation as follows:

a) Inspect for visible leaks in lines or manholes.b) Inspect condition of grout in the interior joints of the manholes.c) Inspect manhole frames and covers for proper type and installation.d) Inspect to see if lines are free of debris.e)

Inspect manhole benches and inverts.f) Check alignment and grade of gravity lines by laser or by introducing sufficient water into the line to verify the absence of sags, as directed by the Contracting Officer.g) Mirror test: Check each straight run of pipeline for gross deficiencies by holding a light in a manhole; it shall show a full circle of light through the pipeline when viewed from the adjoining end of line.

2) Leakage Tests: Test lines for leakage by either infiltration tests or exfiltration tests, or by low-pressure air tests in accordance with the following:

a) Exfiltration Tests: ASTM C 969M (ASTM C 969) and perform calculations in accordance with its Appendix.b) Lowpressure Air Tests: Pipelines: ASTM C 924M (ASTM C 924) and perform calculations in accordance with its Appendix. PVC plastic pipelines: UBPPA UNI-B-6 and perform calculations in accordance with its Appendix.3) Cross Connection Tests: Cross connection tests must be observed by the Contracting Officer and the utility provider's inspector.a) Perform a tracer study from the project sewer connection to the first manhole downstream on the active sewer system. Use a nontoxic, non-staining sewer tracing dye. Testing must continue until the dye visually confirms the design connection is appropriate.

During the test, the contractor must monitor the storm drainage system (structures and outfalls) downstream from the project for any sign of cross connection.b) Perform a smoke test on the project sewer to verify that project storm drainage inlets or drains have not been connected to the sanitary sewer.

b Sanitary Sewer Manholes Verification Testing: Provide testing on sanitary sewer manholes in accordance with the state sewerage regulations. At minimum, perform hydraulic testing in accordance with ASTM C 969M (ASTM C 969).

c. Wastewater Pump Station Verification Testing: Test the wastewater pump station in accordance with the state sewerage regulations. Conduct testing on discharge piping and force main in accordance with tests for water distribution mains; see G30, paragraph 1.3.2. Test pumps, controls, and alarms, in operation, under design conditions to ensure proper operation of all equipment.

d. TV Inspection for Sanitary Sewer: Post-installation TV inspection will be performed on all segments of the installed system when the total footage of sanitary sewer mains installed in the contract is in excess of 300 feet.

Complete the post-installation TV inspection to confirm that the completed lines are free of defects. For video recordings include an audio track recorded by the inspection technician during the actual inspection work describing the parameters of the line being inspected. The minimum information to be included is the pipe material, pipe size, starting and stopping manholes and descriptions of any features as they occur. Video recording playback must be at the same speed that it was recorded. Permanently label CDs / DVDs according to their contents; CDs / DVDs will become the property of the Government. Provide all TV inspections of sanitary sewer mains in accordance with the Pipeline Assessment and Certification Program as sponsored by the National Association of Sewer Service Companies (NASSCO). Prior to initiating CCTV inspection, provide copies of PACP Certification of the operators that will be performing the work. Complete pipe segments and manhole work, including pipe penetrations, manhole benches, main line and manhole visual inspection, pressure testing, deflection and leakage tests on a section of line (manhole to manhole) prior to performing TV. Complete post-installation TV inspection in the presence of the Contracting Officer or his designated representative. The importance of accurate measurements is emphasized. The meter device must be accurate to one tenth of a foot. Utilize the full capabilities of the camera equipment to document the completion and the conformance of the work to the Contract

Documents. Provide a full 360Ã□Â° view of the pipe, joints and service connections. Move the camera through the line in either direction at a moderate rate, stopping when necessary to permit proper documentation of the sewer's condition. The maximum speed must be no greater than 30 feet per minute. Use manual winches, power winches, TV cable and powered rewinds or other devices that do not obstruct the camera view or interfere with the proper documentation of the sewer conditions to move the camera through the sewer line. Once video recording has commenced, the recording must be continuous, without interruption, until the termination manhole is reached. Provide a color video showing the completed work. Prepare and submit Television Inspection Logs providing location of service connections along with the location of any discrepancies. Keep computer printed location records (Television Inspection Logs) and clearly show the location and orientation in relation to an adjacent manhole for each point observed during the TV inspection. Record features of significance such as locations and orientations of service connections, pipe deflections, leaks, rolled or dislodged gaskets, sags or bellies in the line, or wide joints. Document noted defects and lateral connections as color digital files and color hard copy prints. Photo logs must accompany each photo submitted. Prior to submission of the TV inspection video, Television Inspection Logs, and digital photographs to the Contracting Officer, review the submittal items to ensure that they meet the quality criteria set forth in this specification. A copy of such video along with the Television Inspection Logs and Digital photographs must be supplied to the Contracting Officer within five (5) business days of completion of the video-inspection. In the event that the video, Television Inspection Logs or digital photographs are deemed of poor quality or substandard by the Contracting Officer, the videos, and / or Television Logs, or digital photographs will be returned and a re-inspection provided by the Contractor, at no additional cost to the Government.

5. Gravity Sewer Piping: Gravity sanitary sewer mains and laterals must be Ductile Iron, PVC or Polypropylene sewer pipe and fittings. Use Ductile Iron under roadways or at depths greater than 10 feet (3.0 m). PVC and Polypropylene may only be used under roadways or at depths greater than 10 feet (3.0 m) when written approval is received by the Government's Civil Reviewer or indicated in another part of the RFP.

   a. PVC Gravity Sewer Pipe

      1) Piping and Fittings: ASTM D3034 or ASTM F679, SDR 35. 2) Joints: ASTM D3212 and ASTM F477.

   b. Ductile Iron Gravity Sewer Pipe

1) Piping: ASTM A746. Provide required Thickness Class based on design information and methods in ASTM A746.
2) Fittings: AWWA C110 or AWWA C153.
3) Joints: AWWA C111.
4) Interior Coating: AWWA C104.
5) Exterior Protection (if required): AWWA C105, polyethylene encasement.

c. Dual Wall and Triple Wall Polypropylene Sewer Pipe 12" to 60" 1) Piping and Fittings: ASTM F2736 and ASTMF2764/F2764M. 2) Joints: ASTM D3212 and ASTM F477.

6. Connections to Existing Lines: Obtain approval from the Contracting Officer before making a connection to an existing line. Conduct work so that there is minimum interruption of service on existing line and provide a new manhole at the connection point.

7. Gravity Sewer Installation: Install pipe, fittings and accessories in accordance with manufacturer's instructions.

a. PVC and Dual and Triple Wall Polypropylene: ASTM D2321. Do not use ASTM D2321 Class IV or V materials for bedding, haunching or initial backfill materials.

b. Ductile Iron: AWWA C600.

c. Provide nondetectable warning tape and a continuous length of tracer wire for the full length of each run of nonmetallic piping below grade. Warning tape to be color coded with warning and identification of utility type imprinted in bold black letters continuously over the entire tape length.

Warning and identification to read, "CAUTION, BURIED (utility type) LINE BELOW" or similar wording. Color to be green for sewer systems. Terminate tracer wire above grade at valve boxes and at exterior of building.

8. Piping for Gravity Sewer Cleanouts: Install cast iron pipe and fittings in accordance with the recommendations of the pipe manufacturer.

a. Cast-Iron Soil Pipe for Cleanouts 1) Pipe: ASTM A 74, service. 2) Joints: ASTM C 564 compression-type rubber gaskets. 3) Exterior Protection (if required): AWWA C105, polyethylene encasement.

9. Sanitary Sewer Manholes and Cleanouts: Provide all materials, equipment, labor, testing, and miscellaneous related items for the sanitary manholes in accordance with the following:

a. Set manhole rim elevations flush with finished surface of paved areas or 1 inch (25 mm) above finished grade in unpaved areas.

b. Resilient connectors for making joints between manhole and pipes entering manhole must conform to ASTM C 923/C 923M.

c. Provide drop manholes when a gravity sewer pipe enters a manhole at an elevation of 24 inches (610 mm) or more above the manhole invert.

d. Precast Concrete Manholes: ASTM C 478/C 478M; base and first riser must be monolithic. Precast manhole sections must have:
1) ASTM C 990/C 990M butyl gaskets; 2) ASTM C 443/C 443M rubber O-ring joints; or 3) ASTM C 443, Type B gaskets.

e. Cast-in-Place Concrete Manholes: Reinforced concrete; designed according to ASTM C 890 for A-16 (AASHTO HS20-44), heavy-traffic, structural loading. Provide concrete work in accordance with ACI 301/301M and ACI 350-01; provide a minimum compressive strength of 4000 psi (28 MPa).

f. Manhole Frames and Covers: Frame and cover must be cast gray iron, ASTM A48/A48M, Class 35B, cast ductile iron, ASTM A536, Grade 6545-12, or reinforced concrete, ASTM C478 ASTM C478M. Frame and cover must be designed to accommodate the imposed live load. Stamp or cast the words "Sanitary Sewer" into covers so that it is plainly visible.

g. Manhole Steps:
1) Zinc-coated steel: 29 CFR 1910.27.
2) Plastic or rubber coating pressure molded to steel: ASTM D 4101, copolymer polypropylene; or ASTM C 443/C 443M, except shore A durometer hardness must be 70 plus or minus 5.

3) Aluminum steps or rungs will not be permitted. Steps are not required in manholes less than 4 feet (1.2 m) deep.

10. Manhole Construction: Where a new manhole is constructed on an existing line, remove existing pipe as necessary to construct the manhole. Cut existing pipe so that pipe ends are approximately flush with the interior face of manhole wall, but not protruding into the manhole. For changes in direction of the sewer and entering branches into the manhole, make a circular curve in the manhole invert of as large a radius as manhole size will permit. For cast-in-place concrete, no parging will be permitted on interior manhole walls.

11. Connections to Existing Manholes: Center pipe connections to existing manholes on the manhole. Holes for the new pipe must be of sufficient diameter to allow packing cement mortar around the entire periphery of the pipe but no larger than

1.5 times the diameter of the pipe. Cut the manhole in a manner that will cause the least damage to the walls.

12. Cleanouts: Construct cleanouts of cast iron soil pipe and fittings; see paragraph, "Piping for Gravity Sewer Cleanouts" above.

13. Lift Stations and Pumping Stations: If a pump station is allowed, provide all materials, equipment, labor, testing and miscellaneous related items for a packaged lift or pump station system for the facility in compliance with the UFC 3-240-01, Wastewater Collection; the state sewerage regulations; and the utility provider's requirements.

    a. Submersible Pumps: Provide pumps capable of handling raw wastewater and passing spheres of at least 3 inches (75 mm) in diameter. The pump's suction and discharge openings must be at least 4 inches (100 mm) in diameter. Provide submersible type sewage pumps, with guide rail system. Include ASTM A48/A48M, Class 25, nonclog, cast-iron impeller; and hermetically sealed motor with moisture-sensing probe, mechanical seals, and waterproof power cable. Construct the guide rail system of stainless steel. Provide a stainless steel lifting chain for raising and lowering the pump in the basin.

    b. Grinder Pumps: Provide grinder-type sewage pumps, with guide rail system. Include stainless steel or bronze impeller and hermetically sealed motor with moisture-sensing probe, mechanical seals, and waterproof power cable. Construct the guide rail system of stainless steel. Provide a stainless steel lifting chain for raising and lowering the pump in the basin.

    c. Suction Lift Pumps: Provide pumps capable of handling raw wastewater and passing spheres of at least 3 inches (75 mm) in diameter. The pump's suction and discharge openings must be at least 4 inches (100 mm) in diameter. Provide dry-chamber-mounting, vacuum-primed, nonclog sewage pumps located in dry compartment above wet pit. Include ASTM A48/A48M, Class 25, nonclog, cast iron impeller; mechanical or stuffing box seals; pedestal mounted motor; and suction piping extending to

       bottom of wet pit. Provide suction-lift pumps capable of automatic rapid self priming and re-priming at the "lead pump on" elevation. Suction piping must not exceed 25 feet (7.6 meters) in total length. Priming lift at the "lead pump on" elevation must include a safety factor of at least 4 feet (1.2 meters) from the maximum allowable priming lift for the specific equipment at design operating conditions. The combined total of dynamic suction-lift at the "pump off" elevation and the required net positive suction head at design operating conditions must not exceed 22 feet (6.7 meters).

d. Pump Motors: Provide pump motor sized to accommodate pump operation along the entire impeller curve.

e. Station Piping Within Wet Well and Valve Vault:
Piping Less than 4-Inch (100 mm) in Diameter
1) PVC Pressure Pipe

a. Pipe: ASTM D 1785, Schedule 80.b. Fittings: Schedule 80 socket fittings, ASTM D 2467; Schedule 80 threaded fittings, ASTM D 2464.Piping 4 inch (100 mm) Diameter and Larger
2) Flanged Ductile Iron Pipe

a. Pipe: AWWA C115 and its appendices.b. Fittings:AWWA C110 or AWWA C153.c. Lining: AWWA C104.

14. Force Mains:

a. Force Mains for Submersible and Suction Lift Pumps: Force mains must be at least 4 inches (100 mm) in diameter and must be either ductile iron or PVC pressure pipe.
1) Ductile Iron Pressure Pipe

a. Pipe: AWWA C151, Pressure Class 350.

b. Fittings: AWWA C110 or AWWA C153.

c. Interior Lining: AWWA C104.d. Exterior Protection (if required): AWWA C105, polyethylene encasement.2) PVC Pressure Pipe

a. Pipe: AWWA C900, Pressure Class 150. AWWA C905.b. Fittings: Ductile Iron (AWWA C110 or AWWA C153).b. Force Mains for Grinder Pumps: Force mains less than 4 inches (100 mm) in diameter must be PVC pressure pipe: 1) PVC Pressure Pipe

a. Pipe: ASTM D 1785, Schedule 40 or ASTM D 2241, with SDR rating for 160 psi (1.1 MPa) pressure rating.b. Fittings: ASTM D 2466.c. Joints: Elastomeric gaskets for pressure rating; solvent cement joints, ASTM D 2564.15. Piping Accessories:

a. Insulating Joints: Provide between pipes of dissimilar metals a rubber gasket or other approved type of insulating joint or dielectric coupling to effectively prevent metal-to-metal contact between adjacent sections of piping.

b. Accessories: Provide flanges, connecting pieces, transition glands, transition sleeves, and other adapters as required.

c. Flexible Flanged Coupling: Provide flexible flanged coupling applicable for sewage as indicated. Use flexible flanged coupling designed for a working pressure of 350 psi (2400 kPa).

16. Valves: Provide suitable shutoff and check valves on the discharge line of each pump. Locate the check valve between the shutoff valve and the pump. Locate valves in accordance with state sewerage regulations. Check valves must be suitable for the material being handled and placed on the horizontal portion of the discharge piping except for ball check valves, which may be placed in the vertical run. Provide valves capable of withstanding normal pressure and water hammer. Use valves from one manufacturer.

a. Shut Off Valves
Shut Off Valves Less than 4 Inch (100 mm) in Diameter PVC ball valves.
1) Shut Off Valves 4 Inch (100 mm) and Larger in Diameter
AWWA C509 or AWWA C515, nonrising stem, and flanged. Provide valves with handwheels that open by counterclockwise rotation of the valve stem. Provide epoxy coating in accordance with AWWA C550.

b. Check Valves
1) Check Valves Less than 4-Inch (100 mm) in Diameter
Neoprene ball check valve with integral hydraulic sealing flange, designed for a hydraulic working pressure of 175 psi (1200 kPa).

Check Valves 4-Inch (100 mm) and Larger in Diameter
AWWA C508, flanged. Provide a nonclog, swing check valve rated for not less than 175 psig (1200 kPa) working pressure capable of passing 3inch (75 mm) diameter solids.

2) Air Relief Valves: Provide air relief valves at high points in the force main to prevent air locking in accordance with AWWA M51. Provide vacuum relief valves, where required, to relieve negative pressures on force mains.

17. Identification Tags and Plates: Provide valves with tags or plates numbered and stamped for their usage. Use plates and tags of brass or nonferrous material and mounted or attached to the valve.

18. Thrust Restraint: Provide thrust restraint for force mains, valves and other features of the wastewater distribution system. Provide thrust restraint using restrained joints in accordance with pipe manufacturer's recommendations, AWWA C600 and if for fire service main, NFPA 24.

19. Station Control System: Provide alarms for all pumping and lift stations; at minimum provide alarms for high level, power failure, pump failure, unauthorized entry or any cause of station malfunction. Provide alarms as required by the pump manufacturer to obtain warranty. If required, provide a telemetry system in accordance with state sewer collection and treatment regulations and system owner's requirements to relay alarms to a facility that is manned 24 hours a day.

20. Station Accessories:

    a. Ventilation: Provide covered wet wells with provisions for air displacement venting to the outside. Provide galvanized ASTM A 53/A 53M pipe with insect screening. Provide adequate ventilation for all pump stations.

    b. Metering: Provide devices for measuring wastewater flow at all pumping stations. Provide indicating, totalizing and recording flow measurement at pumping stations with a 1200 gpm (76 l/s) or greater design peak hourly flow. For smaller stations, provide elapsed time meters in conjunction with pumping rate tests.

    c. Pipe and Valve Supports: Use schedule 40 galvanized steel piping conforming to ASTM A 53/A 53M for pipe and valve supports. Provide either ANSI B16.3 or ANSI B16.11 galvanized threaded fittings.

    d. Miscellaneous Metals: Use stainless steel bolts, nuts, washers, anchors, and supports for installation of equipment.

G3030 STORM SEWER

1. Storm System Design and Construction: Provide the new storm sewer system and connections to the existing storm sewer system in accordance with UFC 3-201-01, Civil Engineering; the utility provider's requirements; UFC 3-21010, Low Impact Development; the state stormwater management laws and regulations; and applicable sustainability requirements; whichever is more stringent.

   The Contractor must make necessary adjustments to the drainage design in order to avoid disruption to existing utilities and to protect existing trees to remain.

   Confirm that the existing receiving system has adequate capacity to receive the additional stormwater flow generated by the project.

2. Storm Sewer System Performance Verification: At a minimum, perform the following:

    a. Visual Test: Remove drainage structure covers and conduct a visual inspection as follows:

1) Inspect for visible leaks in lines or structures.2) Inspect condition of grout in the interior joints of the structures.3) Inspect structure frames and covers for proper type and installation.4) Inspect to see if lines are free of debris.5) Inspect structure inverts.6) Check alignment and grade of gravity lines by laser or by introducing sufficient water into the line to verify the absence of sags, as directed by the Contracting Officer.7) Mirror test: Check each straight run of pipeline for gross deficiencies by holding a light in a structure; it must show a full circle of light through the pipeline when viewed from the adjoining end of the line.8) TV Inspection for Storm Sewer Under Pavements: Post-installation TV inspection will be performed on all segments of the installed system when the total footage of storm sewer lines installed in the contract is in excess of 300 feet. Complete the post-installation TV inspection to confirm that the completed lines are free of defects. For video recordings include an audio track recorded by the inspection technician during the actual inspection work describing the parameters of the line being inspected. The minimum information to be included is the pipe material, pipe size, starting and stopping manholes and descriptions of any features as they occur. Video recording playback must be at the same speed that it was recorded. Permanently label CDs / DVDs according to their contents; CDs / DVDs will become the property of the Government. Provide all TV inspections of storm sewer lines in accordance with the Pipeline Assessment and Certification Program as sponsored by the National Association of Sewer Service Companies (NASSCO). Prior to initiating CCTV inspection, provide copies of PACP Certification of the operators that will be performing the work. Complete pipe segments and manhole work, including pipe penetrations, manhole benches, main line and manhole visual inspection, pressure testing, and deflection test on a section of line (manhole to manhole) prior to performing TV. Complete postinstallation TV inspection in the presence of the Contracting Officer or his designated representative. The importance of accurate measurements is emphasized. The meter device must be accurate to one tenth of a foot. Utilize the full capabilities of the camera equipment to document the completion and the conformance of the work to the Contract Documents. Provide a full 360Ã□Â° view of the pipe, joints and service connections. Move the camera through the line in either direction at a moderate rate, stopping when necessary to permit proper documentation of the sewer's condition. The maximum speed must be no greater than 30 feet per minute. Use manual wenches, power winches, TV cable and powered rewinds or other devices that do not obstruct the camera view or interfere with the proper documentation of the sewer conditions to move the camera through the sewer line. Once video recording has commenced, the recording must be continuous, without interruption, until the termination manhole is reached. Provide a color video showing the completed work. Prepare and submit Television Inspection Logs providing location of service connections along with the location of any discrepancies. Keep computer printed location records (Television Inspection Logs) and clearly show the location and orientation

in relation to an adjacent manhole for each point observed during the TV inspection. Record features of significance such as locations and orientations of service connections, pipe deflections, leaks, rolled or dislodged gaskets, sags or bellies in the line, or wide joints. Document noted defects and lateral connections as color digital files and color hard copy prints. Photo logs must accompany each photo submitted. Prior to submission of the TV inspection video, Television Inspection Logs, and digital photographs to the Contracting Officer, review the submittal items to insure that they meet the quality criteria set forth in this specification. A copy of such video along with the Television Inspection Logs and Digital photographs must be supplied to the Contracting Officer within five (5) business days of completion of the video -inspection. In the event that the video, Television Inspection Logs or digital photographs are deemed of poor quality or substandard by the Contracting Officer, the videos, and / or Television Logs, or digital photographs will be returned and a re-inspection provided by the Contractor, at no additional cost to the Government.

3. Piping: Storm sewer piping 12 inches (300 mm) and larger in diameter shall be reinforced concrete, dutile iron or corrugated steel; PVC, corrugated aluminum, polyethylene and polypropylene pipe may only be used when written approval is received by the Government's Civil Reviewer or indicated in another part of the RFP. Subsurface drainage piping shall be perforated PVC or HDPE.

4. Piping Materials:

   a. PVC Pipe
      1) Piping and Fittings: ASTM D3034, SDR 35. 2) Joints: ASTM D3212 and ASTM F477.

   b. Ductile Iron Pipe
      1) Piping: ASTM A746. Provide required Thickness Class based on design information and methods in ASTM A746.
      2) Fittings: AWWA C110 or AWWA C153.
      3) Joints: AWWA C111.
      4) Interior Coating: AWWA C104.
      5) Exterior Protection (if required): AWWA C105, polyethylene encasement.

   c. Reinforced Concrete Pipe:
      1) Circular Pipe: ASTM C76/C76M. Provide required Class based on design information and methods in ASTM C76/C76M. Class III minimum.
      2) Elliptical Pipe: ASTM C507/C507M. Provide required Class based on design information and methods in ASTM C76/C76M.
      3) Joints:

a) ASTM C990/C990M butyl gaskets; b) ASTM C 443/C 443M rubber O-ring joints; or

c) AASHTO M 198, Type B preformed plastic gaskets.

d) Corrugated Aluminum Pipe:
1) Piping: ASTM B745/B745M.
2) Joints: Coupling bands conforming to ASTM B745/B745M. 3) Coating: Fully bituminous coated for all applications in accordance with ASTM A849. For applications where piping is part of a piped storm sewer system (not a culvert), provide pipe fully bituminous coated, invert (half) paved with concrete lining in accordance with ASTM A849.

e) Corrugated Steel Pipe:
1) Piping: ASTM A760/A760M.
2) Joints: Coupling bands conforming to ASTM A760/A760M. 3) Coating: Fully bituminous coated for all applications in accordance with ASTM A849. For applications where piping is part of a piped storm sewer system (not a culvert), provide pipe fully bituminous coated, invert (half) paved with concrete lining in accordance with ASTM A849.

f) Polyetheylene (PE) Pipe
1) Piping 12" to 60" and Fittings: ASTM 2648/F2648M and AASHTO M 294 Type S, corrugated.
2) Joints: ASTM F477 and ASTM D3212

g) Dual and Triple Wall Polypropylene (PP) Pipe
1) Piping 12" to 60" and Fittings: ASTM F2736, ASTM F2764/F2764M, ASTM F2881 and AASHTO M 330 Type S or D 2) Joints: ASTM F477 and ASTM D3212

h) Perforated PVC Pipe: ASTM D 2729.

i) Perforated PE Pipe
1) Piping and Fittings: AASHTO M 294, Type SP, corrugated. 2) Joints: AASHTO M 294, Soiltight.

5. Installation: Install piping in accordance with manufacturer's recommendations and the following standards:

a. PVC, PE and Dual and Triple Wall PP: ASTM D 2321. Do not use ASTM D 2321 Class IV or V materials for bedding, haunching or initial backfill materials.

b. Ductile Iron: AWWA C600.

c. Reinforced Concrete: ACPA 01-102 and 01-103.

d. Corrugated Aluminum: ASTM B 788/B 788M.

e. Corrugated Steel: ASTM A 798/A 798M.

f. Perforated PVC and Perforated PE: ASTM D 2321. Do not use ASTM D

2321 Class IV or V materials for bedding, haunching or initial backfill materials.

Provide nondetectable warning tape and a continuous length of tracer wire for the full length of each run of nonmetallic piping below grade. Warning tape to be color coded with warning and identification of utility type imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (utility type) LINE BELOW" or similar wording. Color to be green for sewer systems. Terminate tracer wire above grade at valve boxes and at exterior of building.

6. Piping for Cleanouts: Materials

a. Cast-Iron Soil Pipe for Cleanouts 1) Pipe: ASTM A 74, service. 2) Joints: ASTM C 564 compression-type rubber gaskets. 3) Exterior Protection (if required): AWWA C105, polyethylene encasement.

b. Installation: Install cast iron pipe and fittings in accordance with the recommendations of the pipe manufacturer.

7. Storm Sewer Structures: Provide all materials, equipment, labor, testing, and miscellaneous related items for the drainage structures in accordance with the following:

a. Set structure rim elevations flush with finished surface of paved areas or 1 inch (25 mm) above finished grade in unpaved areas.

b. Provide resilient connectors for making joints between manhole and pipes entering manhole in conformance with ASTM C 923/C 923M.

c. Provide precast or cast-in-place concrete drainage structures, except castin-place concrete is required for airfield drainage structures, headwalls and gutters.

8. Precast Concrete Inlets: Provide work and materials in accordance with applicable requirements of the State Highway Specifications (SHS) and standards where the project is located.

9. Cast-In-Place Concrete Drainage Structures: Provide work and materials in accordance with drainage structures indicated in the State Highway Specifications (SHS) and standards where the project is located. For airfield drainage structures, provide work and materials in accordance with FAA ACA 150/537010B.

10. Drainage Structure Frames And Covers: Frame and cover for gratings shall be cast gray iron, ASTM A48/A48M, Class 35B; cast ductile iron, ASTM A536, Grade 65-45-12; or cast aluminum, ASTM B26/B26M, Alloy 356.OT6. Frame

    and cover must be designed to accommodate the imposed live loads. Stamp or cast the words "Storm Sewer" into covers so that it is plainly visible. For airfield drainage structures, fabricate frames and covers of standard commercial grade steel welded by qualified welders in accordance with AWS D1.1/D1.1M. Provide covers of rolled steel floor plate having an approved anti-slip surface. Steel frames and covers must be hot dipped galvanized after fabrication. At the contractor's option, ductile iron covers and frames may be used for airfield drainage structures if designed for a minimum proof load of 100,000 pounds (45,000 kg) in lieu of the steel frames and covers. Covers must be of the same material as the frames (i.e. ductile iron frame with ductile iron cover, galvanized steel frame with galvanized steel cover). Perform proof loading in accordance with ASTM A 48/A 48M. Physically stamp proof loads into the cover. Provide the Contracting Officer copies of previous proof load test results performed on the same frames and covers as proposed for this contract. Modify the top of the structure to accept the ductile iron structure in lieu of the steel structure indicated. The finished structure must be level and non-rocking, with the top flush with the surrounding pavement.

11. Drainage Structure Steps:

    a. Zinc-coated steel: 29 CFR 1910.27.

    b. Plastic or rubber coating pressure molded to steel: ASTM D 4101, copolymer polypropylene; or ASTM C 443/C 443M, except shore A durometer hardness shall be 70 plus or minus 5.

    c. Aluminum steps or rungs will not be permitted. Steps are not required in structures less than 4 feet (1.2 m) deep.

12. Drainage Structure Construction: Where a new structure is constructed on an existing line, remove existing pipe as necessary to construct the structure. Cut existing pipe so that pipe ends are approximately flush with the interior face of structure wall, but not protruding into the structure.

13. Connections to Existing Structures: Center pipe connections to existing structures on the structure. Holes for the new pipe must be of sufficient diameter to allow packing cement mortar around the entire periphery of the pipe but no larger than 1.5 times the diameter of the pipe. Cut the structure in a manner that will cause the least damage to the walls.

14. Cleanouts: Construct cleanouts of cast iron soil pipe and fittings; see paragraph 6 above.

15. Lift Stations: A stormwater pump station(s) will not be allowed.

16. Culverts: Culverts 12 inches (300 mm) and larger in diameter shall be reinforced concrete or corrugated steel; PVC, corrugated aluminum, polyethylene and polypropylene pipe may only be used when written approval is received by the Government's Civil Reviewer or indicated in another part of the RFP. See G303001, paragraphs above for material and installation requirements. Flared

    end sections must be the same material as pipe material. Provide erosion control in accordance with the State Erosion and Sedimentation Control Standards or EPA guidance where State Standards are unavailable.

17. Headwalls: Provide cast-in-place concrete headwalls in accordance with the State Highway Specification (SHS) and standards where the project is located.

18. Erosion & Sediment Control Measures: Refer to Section G10.

19. Stormwater Management:

    a. Stormwater Collection And Storage: Provide permanent stormwater management (i.e., detention and retention ponds, LID and other drainage features) to control stormwater runoff in accordance with UFC 3-201-01, Civil Engineering, UFC 3-210-10, Low Impact Development, FC 1-30009N Navy and Marine Corps Design Procedures, State and local stormwater management Laws and Regulations and applicable project sustainability goals. Integrate permanent stormwater management features into the site design in accordance with UFC 3-201-01, Civil Engineering. Parking areas, roads, walks, courtyards, training areas and similar site features may not be used to detain or retain stormwater. Manage stormwater within detention or retention ponds and the LID features indicated in Part 3 of this RFP. Prevent upstream and downstream property damage.

20. OIL/WATER SEPARATOR: Provide an oil/water separator to remove free oil from oil-in-water mixtures originating from proposed facility operations. Provide grit protection upstream of the oil/water separator. Provide an oil/water separator utilizing coalescing media and conforming to the applicable guidelines of the American Petroleum Institute (API). Provide materials or a coating system which will protect the separator from the oil-in-water mixture, atmosphere, and in-situ soil conditions. Use a separator with a completely removable cover.

G3060 FUEL DISTRIBUTION

Gas Distribution System: Refer to Section D20 for requirements.

G40 SITE ELECTRICAL UTILITIES

G4010 ELECTRICAL DISTRIBUTION

1. Electrical Utilities Design and Construction: Site electrical utilities include all exterior electrical work, including the connection to the primary distribution system. This also includes telephone and cable television supplies.

   Provide electrical overhead and underground, distribution systems in accordance with IEEE C2 (National Electrical Safety Code), CEI 0-16, CEI 0-21, NFPA 70, local utilities company requirements, and local Activity guidelines.

2. Coordination With Local Utilities Company and Local Activity: Service meters for electrical services must be provided and installed in conformance with the local utilities company requirements and local activity guidelines.

3. Substations: When secondary unit substations are required, the Designer of Record must utilize UFGS Section 26 11 13, Secondary Unit Substation, and UFGS Section 26 23 00, Low Voltage Switchgear, or UFGS 26 24 13, Switchboards, for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

4. Transformers: When transformers are required, the Designer of Record must utilize UFGS Section 26 12 19.10, Three-Phase Pad Mounted Transformers, UFGS Section 26 12 21, Single-Phase Pad Mounted Transformers, or UFGS Section 33 71 01, Overhead Transmission and Distribution, for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

5. Switches, Controls and Devices: When switches or control devices are required, the Designer of Record must utilize UFGS Section 26 13 00, SF6 Insulated Pad Mounted Switchgear, or UFGS Section 33 71 01, Overhead Transmission and Distribution, for the project specification, and must submit the edited specification section as a part of the design submittal for the project.

G4020 EXTERIOR LIGHTING FIXTURES AND CONTROLS

1. Utilize broad spectrum (white light) sources such as metal halide, induction, Light Emitting Diode (LED), and fluorescent to provide good visibility at low light levels, unless lighting is required to match existing sources. The IESNA 10th Edition Handbook has developed a methodology to apply white light.

2. Comply with ANSI/ASHRAE/IES 90.1 for all exterior lighting applications and controls.

3. Comply with EPACT 2005, the exterior lighting power density must be below ASHRAE by 30% if considered a building load and 20% if considered a nonbuilding load.

4. Provide surge protective device (SPD) at panelboards that include circuits feeding exterior lighting systems.

5. Coordinate the design and luminaire selection with the landscape designer. Such coordination should include the location of poles which may conflict with tree locations.

6. When exterior lighting is required the designer of record must utilize UFGS Section 26 51 00 for the project specification section as part of the design submittal for the project and must submit the edited specification section as a part of the design submittal for the project. Provide "dark-sky" compliant exterior light fixtures and design to minimize light trespass and light pollution.

G4030 SITE COMMUNICATION & SECURITY

1. Telephone Distribution System: Provide all telephone distribution systems in accordance with EIA/TIA Standards, NFPA 70, and the cognizant telephone company requirements.

2. Cable Television System: Provide all cable television systems in accordance with NFPA 70, and the cognizant cable television company requirements and BICSI recommendations.

PART 5 – PRESCRIPTIVE SPECIFICATIONS

SECTION 01 32 16.00 20

SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES

08/18 CHANGE 1: 08/20

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 33 29.05 20 SUSTAINABILITY REPORTING FOR DESIGN-BUILD. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Baseline Design Schedule; G

Baseline Construction Schedule; G

SD-07 Certificates

Monthly Updates; G

1.2 APPROVAL

Prior to the start of work, prepare and submit to the COR a Design and Construction Schedule in the form of a Network Analysis Schedule (NAS) in accordance with the terms in Contract Clause FAR 52.236-15 Schedules for Construction Contracts, except as modified in this contract.

The approval of a Schedule is a condition precedent to:

a. The Contractor starting work on the demolition or construction stage(s) of the contract.

b. Processing Contractor's invoice(s) for construction activities/items of work.

c. Review of any schedule updates.

Submittal of the Baseline Design and Construction Schedule, and subsequent schedule updates, is understood to be the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents, represents the Contractor's plan on how the work will be accomplished, and accurately reflects the work that has been accomplished and how it was sequenced (as-built logic).

1.3 SCHEDULE FORMAT

1.3.1 Network Analysis Schedule (NAS)

Use the critical path method (CPM) to schedule and control project activities. Prepare and maintain project schedules using Primavera P6. Importing data into the scheduling program using data conversion techniques or third party software is cause for rejection of the submitted schedule.

Within 15 calendar days after approval of the Initial Schedule or approval of the final design for a design build project, submit to the Contracting Officer a final NAS schedule.

1.3.1.1 Activity Requirements

a. At a minimum, identify the following in the schedule:

(1) Design and Construction time for major systems and components

(2) Each activity assigned with its appropriate Responsibility Code

(3) Each activity assigned with its appropriate Phase and Area Codes

(4) Major submittals and submittal processing time

(5) Major equipment lead time

b. Build the Schedule as follows:

(1) Show design periods, submittals, Government review periods, material/equipment delivery, utility outages, on-site construction, inspection, testing, and closeout activities. Government and Contractor on-site work activities must be driven by calendars that reflect Saturdays, Sundays and all Holidays as non-work days for 5day work week calendars.

(2) With the exception of the Contract Award and End Contract milestone activities, use of open-ended activities is not allowed; each activity must have predecessor and successor ties. No activity must have open start or open finish (dangling) logic. Minimize redundant logic ties.Once an activity exists on the schedule it must not be deleted or renamed to change the scope of the activity and must not be removed from the schedule logic without approval from the COR. While an activity cannot be deleted, where said activity is no longer applicable to the schedule but must remain within the logic stream for historical record, it can be changed to a milestone. Document any such change in the milestone's "Notebook," including a date and explanation for the change. The ID number for a deleted activity must not be re-used for another activity.

(3) Assign each activity its appropriate Responsibility Code and Area Code, indicating location and responsibility to accomplish the work indicated by the activity, Phase Code, and Work Location Code. Include anticipated tasks to be assigned Government responsibility.

(4) Date/time constraints or lags, other than those required by the contract, are not allowed unless approved by the COR. Include as the last activity in the contract schedule, a milestone activity named "Contract Completion Date".

(5) Include the following Contract Milestones:

(a) Include as the first activity on the schedule a start milestone titled "Contract Award", which must have a Mandatory Start constraint equal to the Contract Award Date.

(b) Include Interim or Phased Completion Milestones required by the Contract or as approved by the COR.

(c) Include Facility Turnover Planning Meeting Milestones.

(d) Include an unconstrained finish milestone on the schedule titled "Substantial Completion". Substantial Completion is defined as the point in time the Government would consider the project ready for beneficial occupancy wherein by mutual agreement of the Government and Contractor. Government use of the facility is allowed while construction access continues in order to complete remaining items (e.g. punch list and other close out submittals).

(e) Include an unconstrained finish milestone on the schedule titled "Projected Completion". Projected Completion is defined as the point in time the Government would consider the project complete. This milestone must have the Contract Completion Date (CCD) milestone as its only successor.

(f) Include as the last activity on the schedule a finish milestone titled "Contract Completion (CCD)" with constraint type "Must Finish No Later Than". Calculation of schedule updates must be such that if the finish of the "Projected Completion" milestone falls after the contract completion date, then negative float will be calculated on the longest path and if the finish of the "Projected Completion" milestone falls before the contract completion date, the float calculation must reflect positive float on the longest path.This milestone must be set to 5:00 pm.

(6) Provide lead time for major equipment.

1.3.1.2 Responsibility Code

Assign each activity its appropriate Responsibility Code indicating responsibility to accomplish the work indicated by the activity, Phase Code and Work Location Code.

1.3.1.3 Primavera P6 Settings and Parameters

Use the following Primavera P6 settings and parameters in preparing the Baseline Schedule. Deviation from these settings and parameters, without prior consent of the COR, is cause for rejection of schedule submission.

a. General: Define or establish Calendars and Activity Codes at the "Project" level, not the "Global" level.

b. Admin Drop-Down Menu, Admin Preferences, Time Periods Tab:

(1) Set time periods for P6 to 8.0 Hours/Day, 40.0 Hours/Week, 172.0 Hours/Month and 2000.0 Hours/Year.

(2) Use assigned calendar to specify the number of work hours for each time period: Must be checked.

c. Admin Drop-Down Menu, Admin Preferences, Earned Value Tab: Earned Value Calculation: Use "Budgeted values with current dates".

d. Project Level, Dates Tab: Set "Must Finish By" date to "Contract Completion Date", and set "Must Finish By" time to 05:00pm.

e. Project Level, Defaults Tab:

(1) Duration Type: Set to "Fixed Duration & Units".

(2) Percent Complete Type: Set to "Physical".

(3) Activity Type: Set to "Task Dependent".

(4) Calendar: Set to "Standard 5 Day Workweek". Calendar must reflect Saturday, Sunday and all Federal holidays as non-work days. Alternative calendars may be used with COR approval.

f. Project Level, Calculations Tab:

(1) Activity percent complete based on activity steps: Must be Checked.

(2) Reset Remaining Duration and Units to Original: Must be Checked.

(3) Subtract Actual from At Completion: Must be Checked.

(4) Recalculate Actual units and Cost when duration percent complete changes: Must be Checked.

(5) Link Actual to Date and Actual This Period Units and Cost: Must be Checked.

(6) Price/Unit: Set to "$1/h".

(7) Update units when costs change on resource assignments: Must be Unchecked.

g. Project Level, Settings Tab:

(1) Define Critical Activities: Check "Longest Path".

h. The NAS must have a minimum of 30 construction activities. No on-site construction activity may have durations in excess of 20 working days.

1.4 SCHEDULE MONTHLY UPDATES

Update the Design and Construction Schedule at monthly intervals or when the schedule has been revised. Keep the updated schedule current, reflecting actual activity progress and plan for completing the remaining work. Submit copies of purchase orders and confirmation of delivery dates as directed by the COR.

a. Narrative Report: Identify and justify the following:

(1) Progress made in each area of the project;

(2) Longest Path: Include printed copy on 11 by 17 inch paper, landscape setting;

(3) Date/time constraint(s), other than those required by the contract;

(4) Listing of changes made between the previous schedule and current updated schedule including: added or removed activities, original and remaining durations for activities that have not started, logic (sequence, constraint, lag/lead), milestones, planned sequence of operations, longest path, calendars or calendar assignments, and cost loading.

(5) Any decrease in previously reported activity Earned Amount;

(6) Pending items and status thereof, including permits, changes orders, and time extensions;

(7) Status of Contract Completion Date and interim milestones;

(8) Current and anticipated delays (describe cause of delay and corrective actions(s) and mitigation measures to minimize);

(9) Description of current and future schedule problem areas.

For each entry in the narrative report, cite the respective Activity ID and Activity Name, the date and reason for the change, and description of the change.

1.5 TIME IMPACT ANALYSIS

Provide notice of delay in accordance with FAR 52.249-10. Additionally submit a Time Impact Analysis (TIA) with each cost and time proposal for a proposed change. TIA must illustrate the influence of each change or delay on the Contract Completion Date or milestones. No time extensions will be granted nor delay damages paid unless a delay occurs which consumes all available Project Float, and extends the Projected Finish beyond the Contract Completion Date.

a. Each TIA must be in both narrative and schedule form. The narrative must define the scope and conditions of the change; provide start and finish dates of impact, successor and predecessor activity to impact period, responsible party, describe how it originated, and how it impacts the schedule. The schedule submission must consist of three native files:

(1) Fragnet used to define the scope of the changed condition.

(2) Most recent accepted schedule update as of the time of the proposal or claim submission that has been updated to show all activity progress as of the time of the impact start date.

(3) The impacted schedule that has the fragnet inserted in the updated schedule and the schedule "run" so that the new completion date is determined.

b. For claimed as-built project delay, the inserted fragnet TIA method must be modified to account for as-built events known to occur after the data date of schedule update used.

c. TIAs must include any mitigation, and must determine the apportionment of the overall delay assignable to each individual delay. Apportionment must provide identification of delay type and classification of delay by compensable and non-compensable events. The associated narrative must clearly describe analysis methodology used, and the findings in a chronological listing beginning with the earliest delay event.

(1) Identify and classify types of delays as follows:

(a) Force major delay (e.g. weather delay): Any delay event caused by something or someone other than the Government (including its agents) or the Contractor, or the risk of which has not been assigned solely to the Government or the Contractor. If the force majeure delay is on the critical path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, classified as a non-compensable event.

(b) A Contractor-delay: Any delay event caused by the Contractor, or the risk of which has been assigned solely to the Contractor. If the contractor-delay is on the critical path, in absence of other types of concurrent delays, Contractor is not granted extension of contract time, and classified as a non-compensable event. Where absent other types of delays, and having impact to project completion, provide a Corrective Action Plan, identifying plan to mitigate delay, to the COR.

(c) A Government-delay: Any delay event caused by the Government, or the risk of which has been assigned solely to the Government. If the Government-delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, and classified as a compensable event.

(2) Use functional theory to analyze concurrent delays, where: Separate delay issues delay project completion, do not necessarily occur at same time, rather occur within

same monthly schedule update period at minimum, or within same as-built period under review. If a combination of functionally concurrent delay types occurs, it is considered Concurrent Delay, which is defined in the following combinations:

(a) Government-delay concurrent with Contractor-delay: Excusable time extension, classified non-compensable event.

(b) Government-delay concurrent with force majeure delay: Excusable time extension, classified non-compensable event.

(c) Contractor-delay concurrent with force majeure delay: Excusable time extension, classified non-compensable event.

(3) A pacing delay, reacting to another delay (parent delay) equally or more critical than paced activity, must be identified prior to pacing. COR will notify Contractor prior to pacing. Contractor must notify COR prior to pacing. Notification must include identification of parent delay issue, estimated parent delay time period, paced activity(s) identity, and pacing reason(s). Pacing Concurrency is defined as follows:

(a) Government-delay concurrent with Contractor-pacing: Excusable time extension, classified compensable event.

(b) Contractor-delay concurrent with Government-pacing.

(c) Inexcusable time extension, classified non-compensable event.

1.6 3-WEEK LOOK AHEAD SCHEDULE

Prepare and issue a 3-Week Look Ahead schedule to provide a more detailed day-to-day plan of upcoming work identified on the Construction Schedule. Key the work plans to activity numbers when a NAS is required and update each week to show the planned work for the current and following two-week period. Additionally, include upcoming outages, closures, preparatory meetings, and initial meetings. Identify critical path activities on the Three-Week Look Ahead Schedule. The detail work plans are to be bar chart type schedules, maintained separately from the Construction Schedule on an electronic spreadsheet program and printed on 11 by 17 inch sheets as directed by the COR. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools and equipment required to complete the work. Deliver one hard copy and one electronic file of the 3-Week Look Ahead Schedule to the COR no later than 8 a.m. each Monday, and review during the weekly CQC Coordination or Production Meeting.

1.7 CORRESPONDENCE AND TEST REPORTS

Correspondence (e.g., letters, Requests for Information (RFIs), e-mails, meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) must reference Schedule Activities that are being addressed. Test reports (e.g., concrete, soil compaction, weld, pressure) must reference Schedule Activities that are being addressed.

1.8 ADDITIONAL SCHEDULING REQUIREMENTS

Any references to additional scheduling requirements, including systems to be inspected, tested and commissioned, that are located throughout the remainder of the Contract Documents, are subject to all requirements of this section.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 33 10.05 20

DESIGN SUBMITTAL PROCEDURES

05/17 CHANGE 5: 03/19

PART 1 GENERAL

1.1 SUMMARY

This section includes requirements for Contractor-originated design documents and design submittals.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-200-01	DoD Building Code (General Building Requirements)
UFC 1-200-02	High Performance and Sustainable Building Requirements
UFC 1-300-08	Criteria for Transfer and Acceptance of DoD Real Property
FC 1-300-09N	Navy and Marine Corps Design
UFC 3-560-01	Electrical Safety, O&M
UFC 3-600-01	Fire Protection Engineering
UFC 4-010-06	Cybersecurity of Facility-Related Control Systems

1.3 GENERAL DESIGN REQUIREMENTS

Contractor-originated design documents must provide a project design that complies with the SOW, FC 1-300-09N, UFC 1-200-01, the Core UFCs, and other UFC's listed above.

1.4 SUBMITTALS

Submit design submittals, including shop drawings used as design drawings, to the Government for approval. The use of a "G" following a submittal indicates that a Government approval action is required. Submit the following in accordance with this section and Section 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES. Submittals with an "S" are for inclusion in the Sustainability eNotebook in conformance to Section 01 33 29 REQUIREMENTS AND REPORTING for "S" submittal requirements.

Submit the following in accordance with this section and Section 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Consolidated Documents; G

Submittal Register; G

SD-05 Design Data

Design Drawings; G

Specifications; G

Basis of Design; G

Design Submittals; G

Sustainability Notebook; G

BIM Project Execution Plan (PxP); G

Design Model; G

Visual Review Report; G

Clash Detection Report; G

1.5 DESIGN QUALITY CONTROL

1.5.1 Contractor Reviewing and Certifying Authority

The QC organization is responsible for reviewing and certifying that design submittals are in compliance with the contract requirements.

1.5.2 Government Approving Authority

The COR is the approving authority for design submittals.

1.5.3 Designer of Record Certifying Authority

The Designer of Record (DOR), as registered and defined in FC 1-300-09N, is the design certifying authority. The DOR accepts responsibility for design of work in each respective design discipline, by stamping and approving final construction drawings submitted to the Government approval authority. The DOR shall perform the duties of the Design Safety Coordinator "Coordinatore per la Sicurezza in fase di Progettazione" as per technical attachment to the Joint Mixed Commission (JMC) Instruction.

1.5.4 Contractor Construction Actions

Upon submission of sealed and signed design documents certified by the DOR, Design Quality Control Manager (DQCM) and the Quality Control Manager (QCM), the Contractor may proceed with material and equipment purchases, fabrication and construction of any elements covered by that submittal, except as specified in the following paragraph. Exception to Contractor Construction Actions:

The Government will approve the following final submittals before the Contractor shall be allowed to proceed with construction:

a. Any design submittal that includes or will be impacted by a design change to the contract. Final Government approval of the design change is required before construction can begin on the work included in that design submittal.

1.5.5 Contractor's Responsibilities

a. Designate a lead licensed architect or engineer to be in responsible charge to coordinate the design effort of the entire project. This lead architect or engineer must coordinate all design segments of the project to assure consistency of design between design disciplines.
b. With the DOR, verify available site information. In addition, provide additional field investigations and verification of existing site conditions as may be required to support the development of design and construction of the project.
c. Indicate on the transmittal form accompanying submittal which design submittals are being submitted as shop drawings.
d. Advise COR of variations, as required by paragraph VARIATIONS.
e. Provide an updated, cumulative submittal register with each design package that identifies the design and construction submittals required by that design package and previous submittals.
f. Refer to Section 01 33 29 REQUIREMENTS AND REPORTING for Contractor's responsibilities for Guiding Principle Validation and Third Party Certification.
g. Refer to Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI) for Contractor's eOMSI responsibilities.

1.5.6 QC Organization Responsibilities

a. The QCM and, if Commissioning is required in Part 3, the CxA, must certify design submittals for compliance with the contract documents. The DOR stamp on drawings indicates approval from the DOR.
b. QC organization must certify submittals forwarded by the DOR to the COR with the following certifying statement:

“I hereby certify that the (equipment) (material) (article) shown and marked in this submittal is that proposed to be incorporated with Contract Number (insert contract number here), is in compliance with the contract documents, and is submitted for Government approval.

Certified by Design Quality Control Manager (DQCM) ...., Date ....

Certified by Quality Control Manager (QCM) ...., Date ....”

c. Sign certifying statement. The persons signing certifying statements must be the QC organization members designated in the approved QC Plan. The signatures must be digital or scanned placed as an image in the .pdf files.

d. Update submittal register as submittal actions occur and maintain the submittal register at project site until final approval of all work by the COR.

e. Retain a copy of approved submittals at project site.

1.5.7 Government Responsibilities

a. Note date on which submittal was received from QCM, on each submittal.

b. Perform a quality assurance (QA) review of submittals. Government will notify Contractor when comments for that design package are posted and ready for Contractor evaluation and resolution.

Submittals will be returned with one of the following notations:

a. Submittals may be marked "approved." Submittals marked "approved" indicate a quality assurance (QA) review has been performed. Government review or approval of any portion of the proposal or final design does not relieve the Contractor from responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents. Furthermore, Government review or approval of a submittal is not to be construed as a complete check.

b. Submittals marked "not reviewed" indicate submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and certified by Contractor, or is not complete. Submittal will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

c. Submittals marked "revise and resubmit" or "disapproved" indicate submittal is incomplete or does not comply with design concept or requirements of the contract documents and must be resubmitted with appropriate changes. If work has been started on the unacceptable portion of the design submittal, the Contractor must propose corrective action. No further work is allowed to proceed until the issue is resolved in a manner satisfactory to the Government.

1.6 DESIGN DOCUMENTS

Provide design documents that include basis of design, design drawings, and design specifications, reports, design safety plan “Piano di Sicurezza e Coordinamento” facility maintenance handbook “Fascicolo dell’Opera”, and submittal register in accordance with FC 1-300-09N, Navy and Marine Corps Design Procedures and applicable HN Norms.

Select product, material, and system during the project design and indicate these choices on the design documents. Accomplish this by submitting design drawings and specifications that include proprietary submittal information such as manufacturers name, product names, model numbers, product data, manufactures information, provided optional features, appropriate connections, fabrication, layout, and product specific drawings. Adherence to SOW submittal requirements and provision of DOR approved construction submittal information on the design submittals eliminates the need for follow-on traditional construction submittals after the final design is approved.

Submit proprietary information to describe the construction submittal information in the design documents for all products, materials, and systems submittals listed below:

a. Fire Protection components.

b. Roof and building enclosure components.

c. Interior doors, hardware, cabinets, fireproofing/firestopping, components.

d. Major mechanical and electrical equipment such as chillers, transformers, and generators.

e. Telecommunications.

f. Interior finishes.

1.7 DESIGN DRAWINGS

Prepare, organize, and present design drawings in accordance with the requirements of FC 1300-09N, Navy and Marine Corps Design Procedures. Submit all CAD files for the final drawings on CD-ROM or DVD disks in AutoCAD format. Drawing files must be full files, uncompressed and unzipped. GIS Map and related data are considered part of the Design Drawings. They must be developed in accordance with SDSFIE Army Adaption and Army QAP. Submit all GIS files on CD-ROM or DVD disks or DoD SAFE delivery in ArcGIS 10.4.1 or ArcGIS PRO 3.0.3 (.SHP, .GDB, .MXD). GIS data Maps must be full files, uncompressed and unzipped.

1.7.1 Design Drawings Used as Shop Drawings

Design drawings may be prepared more like shop drawings to minimize construction submittals after final design is approved. If the Contractor chooses or is required to include the construction submittal information on the design documents, indicate proprietary information on the design drawings as necessary to describe the products, materials, or systems that are to be used on the project. Construction submittal information included directly in the design drawings must be approved by the DOR. All design documents must be professionally signed in accordance with FC 1-300-09N, Navy and Marine Corps Design Procedures.

1.7.2 Drawing Format For Design Drawings Used as Shop Drawings

The Contractor-originated drawings will be used as the basis for the record drawings. Shop drawings included as design documents must comply with the same drawing requirements such as drawing form, sheet size, layering, lettering, and title block used in design drawings.

1.7.3 Identification of Design Drawings Used as Shop Drawings

The Contractor's transmittal letter and submittal register must indicate which design drawings are being submitted as shop drawings.

1.7.4 Seals and Signatures on Document

All final Contractor-originated design drawings must be signed, dated, and bear the seal of the registered architect or the registered engineer of the respective discipline in accordance with FC 1-300-09N. This seal must be the seal of the DOR for that drawing, and who is professionally registered for work in that discipline. A principal or authorized licensed or certified employee must electronically sign and date final drawings and cover sheet in accordance with FC 1-300-09N. The design drawing coversheets must be sealed and signed by the lead licensed architect or engineer of the project design team. Indicate the Contractor's company name and address on the drawing coversheets of each design submittal. Application of the electronic seal and signature accepts responsibility for the work shown thereon.

1.7.5 Certification of Compliance with Italian Norms:

The final design submittal shall include the following statement, to be signed and stamped by a professional engineer:

> Having reviewed the project No. # ……-……………-……, "……………", I declare that I have ascertained that the structures and technological system included therein comply with the applicable Italian norms.
>
> DATE ………… SIGNATURE …………
>
> Esaminato il progetto No. # ……-……………-……, "……………", dichiaro di aver accertato che strutture ed impianti technolgici in esso conetenuti rispondono alla vigente normativa italiana.
>
> DATA ………… FIRMA …………

1.7.6 Units of Measure

Utilize metric units of measure on the design documents.

1.8 BUILDING INFORMATION MANAGEMENT/MODELING (BIM)

Include BIM submittals as required by and complying with FC 1-300-09N:

a. BIM Project Execution Plan (PxP)

b. Design Model

c. Visual Review Report

d. Clash Detection Report

e. Record Model (With Record Documents)

1.9 SPECIFICATIONS

Provide a Contractor-originated design specification that in conjunction with the drawings, demonstrates compliance with requirements of the SOW. The specified products, materials, systems, and equipment that are approved by the DOR; submitted to the Government by the Contractor; and reviewed by the COR must be used to construct the project. UFGS sections contained in SOW Part 2 become a part of the Contractor-originated Division 01 specification without modification. Specification Sections contained in SOW Part 5 become a part of the Contractor-originated specification without modification.

1.9.1 Specifications Components and Format

The Contractor must prepare design specifications that include a UFGS specification for each product, material, or system on the project. If the Contractor chooses or is required above to combine design and construction submittal information on the design documents, provide a UFGS specification and also proprietary information such as catalog cuts and manufacturers data that demonstrates compliance with the SOW. Organize the specifications using Construction Specification Institute (CSI) MasterFormat™ Provide project specifications to include the following:

a. Specification cover sheet with the professional seal and signature of the lead licensed architect or engineer of the project design team. Indicate the Contractor's company name and address on the specification coversheet.

b. Table of contents for entire specification.

c. Individual UFGS specification sections for each product, material, and system required by the SOW. Edit UFGS sections in accordance with SOW Part 4, PTS Section Z-10, Design Submittals.

d. If proprietary information is provided or required, include a coversheets for the product, material, or system information that is being proprietarily specified. This information is to follow the related UFGS specification.

e. If proprietary information is provided or required, include highlighted and annotated Catalog Cuts, Manufacturer's Product Data, Tests, Certificates, Manufactures information and letters for each product, material, or system that is being proprietary specified.

f. Coordinated submittal register for all products, materials and systems with each design submittal. Provide a cumulative register that identifies the design and construction submittals required by each design package along with previous design submittals. The DOR must assist in developing the submittal register by determining which submittal items are required to be approved by the DOR. Complete all fields in the final submittal register in order to obtain Government approval of the final design.

g. Coordinated submittal register to include separate but simultaneous delivery and approval of design or data required to fulfill sustainability requirements by Section 01 33 29 REQUIREMENTS AND REPORTING.

1.9.2 Specifications Section Source Priority

Choose UFGS sections that describe the products, materials, and systems that are used on the project. Use current UFGS sections that are available on the Whole Building Design Guide website available at this website: http://www.wbdg.org/ffc/dod and give priority to the Unified Tri-Service UFGS sections (no spec number suffix) and UFGS that are prepared by Army (.00 10 suffix), with the exception of controls spec. Use BACnet specifications for control systems. Only use a UFGS section prepared by another DoD Component (.00 20, and .00 30 suffix), if an applicable Army prepared specification section does not exist. Do not use NASA (.00 40 suffix) specifications. If no applicable UFGS technical specification exists to meet your project requirements, consult with the Army Component for guidance and create a new UFGS specification in accordance with UFC 1-300-02, Unified Facilities Guide Specifications (UFGS) Format Standard.

1.9.3 Fire Protection Specifications

Specifications pertaining to spray-applied fire proofing and fire stopping, exterior fire alarm reporting systems, interior fire alarm detection systems, and fire suppression systems, including fire pumps and standpipe systems must be either prepared by, or reviewed and approved by the Fire Protection DOR.

1.9.4 Identification of Manufacturer's Product Data Used with Specifications

Provide complete and legible catalog cut sheets, product data, installation instructions, operation and maintenance instructions, warranty, and certifications for products and equipment for which final material and equipment choices have been made. Indicate, by prominent notation, each product that is being submitted including optional manufacturer's features, and indicate where the product data shows compliance with the SOW.

1.9.5 Specification Software

Submit the final specification source files in both MS Word and SpecsIntact.

1.10 BASIS OF DESIGN

Prepare, organize, and present basis of design in accordance with the requirements of FC 1300-09N. The basis of design must be a presentation of facts to demonstrate the concept of the project is fully understood and the design is based on sound engineering principles. Provide design analyses for each discipline and include the following:

a. Basis of design that includes:

(1) An introductory description of the project concepts that addresses the salient points of the design;

(2) An orderly and comprehensive documentation of criteria and rationale for system selection; and

(3) The identification of any necessary licenses and permits that are anticipated to be required as a part of the design or construction process.

b. Code and criteria search must identify all applicable codes and criteria and highlight specific requirements within these codes and criteria for critical issues in the facility design.

c. Calculations as specified and as needed to support this design.

d. Sustainability Notebook: Maintain up to date documentation of analysis and calculations relative to sustainable design requirement. Refer to Section 01 33 29, REQUIREMENTS AND REPORTING and FC 1-300-09N for requirements.

e. Provide an exterior enclosure vapor pressure analysis, hygrothermal analysis, and written/graphic descriptions for each unique wall and roof assembly used as part of exterior enclosure barriers.

f. Section titled "Antiterrorism" that documents the antiterrorism features.

g. Fall Protection Analysis.

h. eOMSI Facility Data Workbook (FDW)

i. BIM PxP

j. Ergonomic Analysis

k. Section titled "Cybersecurity" that documents the cybersecurity design and construction of facility-related control system requirements.

1.10.1 Basis of Design Format

The basis of design for each design discipline must include a cover page indicating the project title and locations, contract number, table of contents, tabbed separations for quick reference, and bound in separate volumes for each design discipline.

1.10.2 Design Calculations

Place the signature and seal of the designer responsible for the work on the cover page of the calculations for the respective design discipline.

1.10.3 Fall Protection Analysis

Eliminate fall hazards in the facility or if not feasible provide control measures to protect personnel conducting maintenance work after completion of the project. Identify fall hazards in the Basis of Design with the Design Development and Prefinal submittals. The analysis must describe how fall hazards are considered, eliminated, prevented or controlled to prevent maintenance personnel from exposure to fall hazards while performing work at heights. Refer to SOW Part 5, Section 01 35 26, GOVERNMENTAL SAFETY REQUIREMENTS for fall hazard protection requirements.

1.10.4 Cybersecurity Basis of Design

Provide a single submittal indicating criteria and describe requirements for integrating cybersecurity in the design and construction of the facility-related control systems in accordance with UFC 4-010-06. The basis of design must describe specific guidance for control systems with the assigned Confidentiality, Integrity and Availability (C-I-A) impact ratings and shall list the security controls with recommendations and justifications for future tailoring of the security control set.

1.11 RECORD DOCUMENTS

Coordinate and provide record documents in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS and Part 2 General Requirements.

1.12 REAL PROPERTY RECORD

Coordinate and provide record documents in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS and Part 2 General Requirements.

1.13 RECORD MODEL

Provide Record Model in accordance with FC 1-300-09N.

PART 2 PRODUCTS

2.1 CONSOLIDATED DOCUMENTS

Within 30 days after contract award, provide two electronic and hard copies of consolidated documents incorporating the Contractor's Proposal and all SOW amendments and revisions that are contained in the contract award. Identify the changes to the SOW with the "Redlining" or "Track Changes" feature of SpecsIntact or MS Word to highlight the pre-award amendments to the contract. Identify the amendment source at each addition and deletion by annotation, such as footnote or reference in parenthesis.

2.2 DESIGN SUBMITTALS

Complete the Contractor-originated design submittals as defined by this contract, and coordinate with the approved design network analysis schedule. Refer to Section 01 33 29 REQUIREMENTS AND REPORTING for sustainable design submittals.

2.2.1 Design Submittal Packages

The contractor shall be responsible for providing the complete submittal package for each submission. If one item is missing from any submittal, it shall not be accepted and the Government review time will not begin until all items are presented. If changes are to be made to any document the Contractor shall re-submit the entire portion of that package. The Contractor shall not submit single sheets of the drawings, or partial specification section. When required by the Government, Contractor resubmittal of design package, due to nonconformance to the contract, is not a delay in the contract. Buildings/Tasks/Work Orders may be combined into one submittal package for each submission.

2.2.2 Required Design Submittals

Provide comprehensive, multi-discipline design packages that include design documentation for project elements, fully developed to the design stage indicated, and in accordance with FC 1-300-09N, except where specified otherwise.

2.2.3 Review Copies of Design Submittal Packages

Provide the following Design Submittal Packages. Provide comprehensive, multidiscipline design packages that include design documentation for project elements, fully developed to the design stage indicated. Provide the same quantities of copies for resubmittals.

2.2.3.1 Site Design

The Site Design typically includes the following components:

a. Master Site Plan
b. Demolition
c. Site work including Environmental
d. Geotechnical
e. Parking lot
f. Vertical and horizontal signing

2.2.3.2 Building Design

The Building Design typically includes the following components:

a. Foundation
b. Structural
c. Building Enclosures
d. Remaining Work
e. Furniture/Equipment

2.2.4 Design Submittal Review Schedule

For construction scheduling purposes add additional time to the identified minimum review time periods to allow for the following scheduling conditions:

a. Submittals received after noon will be considered received on the following business day.
b. Federal holidays, including the period between Christmas and New Year's Day, will be considered non-working days for Government personnel in reviewing design submittals and attending design related meetings.
c. Postpone delivery if Government personnel to receive the submittal are unavailable. Assure in advance of the submittal delivery it can be received.
d. Postpone delivery when heightened security restricts access to the Base. Coordinate heightened security requirements in advance with the COR.

e. Government Design Review Comments and Request for Information (RFI) will be returned to the Contractor within 14 calendar days from the date of receipt. Contractor shall account in the schedule adequate time for the DOR to review and respond to all Government Comments before the Design Review Conference is held.

f. Period of review for a resubmittal is the same as the initial submittal. Review time for resubmittals caused by non-conformance, do not result in a change in contract duration or cost.

2.3 IDENTIFICATION OF DESIGN SUBMITTALS

Provide a title sheet to clearly identify each submittal, the completion status, and the date. The title sheet must be unique to a particular design submittal. Submit the project title sheet with design status and date for the design submittals.

2.3.1 Construction Document Validation

All CAD design documents used to construct the facility must bear a visible and legible AutoCAD generated plotstamp in the lower right hand margin of each drawing. The plotstamp information on the jobsite construction documents must match the plotstamp information contained on the following development stages of the design documents:

a. The Final Design Submittal professionally signed by the DOR and submitted for Government approval.

b. The Final Design Submittal drawings that have approved by the Government. This development stage may be combined with "c." below, if issued at the same time.

c. The Final Design drawings that have been included in the contract by modification.

d. The Final Design drawings which include subsequent revisions to the design documents that have been included in the contract by modifications.

Issue new drawings for construction which bear the current plotstamp once a new development stage of the design documents has been accomplished. Design documents which do not bear a plotstamp that matches the corresponding plotstamp exhibited on the design documents described above, are not allowed to be used for the construction of the project. The plotstamp must bear the date and time of the plot, at a minimum. Maintain a plotstamp record at the jobsite that lists the applicable plotstamp information for each drawing through each stage of development described above.

PART 3 EXECUTION

3.1 CONTRACTOR'S RESOLUTION OF COMMENTS

Provide written responses to all written comments by the Government. During the design review process, it is intended that all comments made on the previous review be addressed in the next submission. Reviewers will not “closeout” or “concur” with any evaluations or responses made by the Contractor until the next scheduled submission is received and reviewed. This will ensure that the reviewer is given ample time to ensure concerns have been addressed. Additionally, if any comments are left open and unaddressed during the Final Design submission, the Government will not accept the project, regardless of contract timelines or deadlines. Resubmittal of an unacceptable design submittal must be a complete package that includes all the required, specified components of that design submittal. When required by the Government, Contractor resubmittal of design package, due to nonconformance to the contract, is not a delay in the contract.

3.4 GOVERNMENT REVIEW OR APPROVAL

Government review or approval of any portion of the proposal or final design does not relieve the Contractor from responsibility for errors or omissions with respect thereto.

-- End of Section --

SECTION 01 35 26

GOVERNMENTALSAFETYREQUIREMENTS

11/20 CHANGE 3: 02/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.22	(2007; R 2017) Safety Requirements for RopeGuided and Non-Guided Workers' Hoists
ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ASSP Z244.1	(2016) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ASSP Z359.0	(2018) Definitions and Nomenclature Used for Fall Protection and Fall Arrest
ASSP Z359.1	(2020) The Fall Protection Code
ASSP Z359.2	(2017) Minimum Requirements for a Comprehensive Managed Fall Protection Program
ASSP Z359.3	(2019) Safety Requirements for Lanyards and Positioning Lanyards
ASSP Z359.4	(2013) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
ASSP Z359.6	(2016) Specifications and Design Requirements for Active Fall Protection Systems
ASSP Z359.7	(2019) Qualification and Verification Testing of Fall Protection Products
ASSP Z359.11	(2014) Safety Requirements for Full Body Harnesses
ASSP Z359.12	(2019) Connecting Components for Personal Fall Arrest Systems

ASSP Z359.13	(2013) Personal Energy Absorbers and Energy Absorbing Lanyards
ASSP Z359.14	(2014) Safety Requirements for Self-Retracting Devices for Personal Fall Arrest and Rescue Systems
ASSP Z359.15	(2014) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems
ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems
ASSP Z359.18	(2017) Safety Requirements for Anchorage Connectors for Active Fall Protection Systems
ASSP Z490.1	(2016) Criteria for Accepted Practices in Safety, Health, and Environmental Training

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2019) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Z535.2	(2011; R 2017) Environmental and Facility Safety Signs

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
NFPA 51B	(2019; TIA 20-1) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2020; TIA 22-1; ERTA 1 2022) National Electrical Code
NFPA 70E	(2021) Standard for Electrical Safety in the Workplace
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222	(2018H; Add 1 2019) Structural Standard for Antenna Supporting Structures and Antennas and Small Wind Turbine Support Structures
TIA-1019	(2012; R 2016) Standard for Installation, Alteration and Maintenance of Antenna Supporting Structures and Antennas

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety - Safety and Health Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20	Standards for Protection Against Radiation
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.146	Permit-required Confined Spaces
29 CFR 1910.147	The Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1910.333	Selection and Use of Work Practices
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)
29 CFR 1919	Gear Certification
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.16	Rules of Construction
29 CFR 1926.450	Scaffolds
29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
29	CFR 1926.1400 Cranes and Derricks in Construction
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 02-01-056 (2014)	Inspection Procedures for Accessing Communication Towers by Hoist

CPL 2.100 (1995)	Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

ITALIAN CONSTRUCTION SAFETY LAW REQUIREMENTS

81/2008	Legislative Decree 9April 2008 n. 81 and subsequent amendments and integrations

1.2 DEFINITIONS

1.2.1 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are dangerous to personnel, and who has authorization to take prompt corrective measures with regards to such hazards.

1.2.2 Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person requirements as defined EM 385-1-1 Appendix Q, with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146, and designated in writing to be responsible for the immediate supervision, implementation and monitoring of the confined space program, who through training, knowledge and experience in confined space entry is capable of identifying, evaluating and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.3 Competent Person, Cranes and Rigging

The CP, Cranes and Rigging, as defined in EM 385-1-1 Appendix Q, is a person meeting the competent person requirements, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the Crane and Rigging Program, who through training, knowledge and experience in crane and rigging is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.4 Competent Person, Excavation/Trenching

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the excavation/trenching program, who through training, knowledge and experience in excavation/trenching is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and in accordance with ASSP Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the fall protection program, who through training, knowledge and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1 Appendix Q, and designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and also has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented including experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1 Section 22.B.02.

1.2.7 Competent Person (CP) Trainer

A competent person trainer as defined in EM 385-1-1 Appendix Q, who is qualified in the training material presented, and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

1.2.8 High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.9 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.10 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

1.2.11 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even when provided by a physician or registered personnel.

1.2.12 Near Miss

A Near Miss is a mishap resulting in no personal injury and zero property damage, but

given a shift in time or position, damage or injury may have occurred (e.g., a worker falls off a scaffold and is not injured; a crane swings around to move the load and narrowly misses a parked vehicle).

1.2.13 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.14 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.2.15 Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the definition requirements of EM 385-1-1 Appendix Q, and ASSP Z359.2 standard, having a recognized degree or professional certificate and with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

1.2.16 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

a. Death, regardless of the time between the injury and death, or the length of the illness;

b. Days away from work (any time lost after day of injury/illness onset);

c. Restricted work;

d. Transfer to another job;

e. Medical treatment beyond first aid;

f. Loss of consciousness; or

g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above

1.2.17 Government Property and Equipment

"USACE" property and equipment specified in USACE EM 385-1-1 should be interpreted as Government property and equipment.

1.2.18 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document an LHE mishap using the Crane High Hazard working group mishap reporting form.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

APP - Design; G

APP - Construction; G

"Fascicolo dell'Opera";

SD-06 Test Reports

Monthly Exposure Reports

Notifications and Reports

Accident Reports; G

LHE Inspection Reports

SD-07 Certificates

Contractor Safety Self-Evaluation Checklist

Crane Operators/Riggers

Standard Lift Plan; G

Critical Lift Plan; G

Activity Hazard Analysis (AHA)

Confined Space Entry Permit

Hot Work Permit

Certificate of Compliance

License Certificates

Radiography Operation Planning Work Sheet; G

Portable Gauge Operations Planning Worksheet; G

Machinery & Mechanized Equipment Certification Form

1.4 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report and attach to the monthly billing request. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher.

1.5 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, Legislative Decree 9April 2008 n. 81 and host nation laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.6 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.6.1 Personnel Qualifications

1.6.1.1 Site Safety and Health Officer (SSHO)

Provide an SSHO that meets the requirements of EM 385-1-1 Section 1. The SSHO must ensure that the requirements of 29 CFR 1926.16 are met for the project. Provide a Safety oversight team that includes a minimum of one person at each project site to function as the Site Safety and Health Officer (SSHO). The SSHO or an equally-qualified Alternate SSHO must be at the work site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with EM 385-1-1 Section 01.A.17, and all associated sub-paragraphs and/or qualifications in accordance with Legislative Decree 9 April 2008 n. 81. If the SSHO is off-site for a period longer than 24 hours, an equally-qualified alternate SSHO must be provided and must fulfill the same roles and responsibilities as the primary SSHO. When the SSHO is temporarily (up to 24 hours) off-site, a Designated Representative (DR), as identified in the AHA may be used in lieu of an Alternate SSHO, and must be on the project site at all times when work is being performed. Note that the DR is a collateral duty safety position, with safety duties in addition to their full time occupation.

1.6.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties

The SSHO may serve as the Quality Control Manager unless specified in Part 3. The SSHO may not serve as the Superintendent.

1.6.1.2 Competent Person Qualifications

Provide Competent Persons in accordance with EM 385-1-1, Appendix Q and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection). The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the COR for information in consultation with the Safety Office.

1.6.1.2.1 Competent Person for Confined Space Entry

Provide a Confined Space (CP) Competent Person who meets the requirements of EM 3851-1, Appendix Q, and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1, Section 34. When work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, OSHA 29 CFR 1915, Subpart B, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable, 29 CFR 1910.146 for general industry.

1.6.1.2.2 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements of EM 385-1-1, Section 22.B.02 and herein.

1.6.1.2.3 Competent Person for Fall Protection

Provide a Competent Person for Fall Protection who meets the requirements of EM 385-1-1, Section 21.C.04, 21.B.03, and herein.

1.6.1.3 Qualified Trainer Requirements

Individuals qualified to instruct the 40 hour contract safety awareness course, or portions thereof, must meet the definition of a Competent Person Trainer, and, at a minimum, possess a working knowledge of the following subject areas: EM 385-1-1, Electrical Standards, Lockout/Tagout, Fall Protection, Confined Space Entry for Construction; Excavation, Trenching and Soil Mechanics, and Scaffolds in accordance with 29 CFR 1926.450, Subpart L.

Instructors are required to:

a. Prepare class presentations that cover construction-related safety requirements.

b. Ensure that all attendees attend all sessions by using a class roster signed daily by each attendee. Maintain copies of the roster for at least five years. This is a certification class and must be attended by 100 percent. In cases of emergency where

an attendee cannot make it to a session, the attendee can make it up in another class session for the same subject.

c. Update training course materials whenever an update of the EM 385-1-1 becomes available.

d. Provide a written exam of at least 50 questions. Students are required to answer 80 percent correctly to pass.

e. Request, review and incorporate student feedback into a continuous course improvement program.

1.6.1.4 Crane Operators/Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in EM 385-1-1, Section 15.B for Riggers and Section 16.B for Crane Operators and Signal Persons. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 22,680 kg 50,000 pounds or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.6.2 Personnel Duties

1.6.2.1 Duties of the Site Safety and Health Officer (SSHO)

The SSHO must:

a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and actual dates of corrections. Attach safety inspection logs to the Contractors' daily production report.

b. Conduct mishap investigations and complete required accident reports. Report mishaps and near misses.

c. Use and maintain OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and subcontractors, and make available to the COR upon request. Post and maintain the Form 300A on the site Safety Bulletin Board.

d. Maintain applicable safety reference material on the job site.

e. Attend the pre-construction meeting conference, pre-work meetings including preparatory meetings, and periodic in-progress meetings.

f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement and enforce them.

g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.

h. Ensure subcontractor compliance with safety and health requirements.

i. Maintain a list of hazardous chemicals on site and their material Safety Data Sheets (SDS).

j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and be prepared to discuss details during QC Meetings.

k. Provide and keep a record of site safety orientation and indoctrination for Contractor employees, subcontractor employees, and site visitors.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above or any other required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.6.3 Meetings

1.6.3.1 Preconstruction Conference

a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).

b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the COR as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays.

c. Deficiencies in the submitted APP, identified during the COR review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the COR.

1.6.3.2 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, foremen, or CDSOs must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the COR on request. Notify the COR of all scheduled meetings 7 calendar days in advance.

1.7 ACCIDENT PREVENTION PLAN (APP)

Provide a site-specific Accident Prevention Plan (APP), including Activity Hazard Analyses (AHA), in accordance with EM 385-1-1 Appendix A, for the design team to follow during site visits and investigations. For subsequent visits, update the plan if there are changes in the personnel who will be attending, or the tasks to be performed. Submit the APP for review and acceptance by the Government at least 15 calendar days prior to the start of the design field work. Field work may not begin until the design APP is accepted by the COR. If the design scope includes borings or other subsurface investigations, include in the APP the type of field investigation and verification techniques, such as visual, local utility locating service scanning and third party/subcontractor scanning, potholing, or hand digging within two feet of a known utility that will be required. Mark underground utilities before starting any ground-disturbing actions. Notify the COR 15 days prior to the start of soil borings or sub-surface investigations. Prior to the start of construction incorporate the Design APP into the Construction APP so that one site specific APP exists for the project and submit to the COR for acceptance.

1.7.1 APP - Construction

Submit the APP to the COR 15 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the COR, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the COR, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the COR, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the COR within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ASSP A10.34), and the environment.

1.7.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

b. Qualifications of competent and of qualified persons. As a minimum, designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

1.7.3 Plans

Provide plans in the APP in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

1.7.3.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other federal, state and local regulatory requirements identified in this Contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by Contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.7.3.2 Standard Lift Plan (SLP)

Plan lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP in accordance with EM 385-1-1, Section 16.A.03, using Form 16-2 for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed and accepted by all personnel involved in the lift in conjunction with the associated AHA. Signature on the AHA constitutes acceptance of the plan. Maintain the SLP on the LHE for the current lift(s) being made. Maintain historical SLPs for a minimum of three months.

1.7.3.3 Critical Lift Plan - Crane or Load Handling Equipment

Provide a Critical Lift Plan as required by EM 385-1-1, Section 16.H.01, using Form 16-3. In addition, Critical Lift Plans are required for the following:

a. Lifts over 50 percent of the capacity of barge mounted mobile crane's hoist.

b. When working around energized power lines where the work will get closer than the minimum clearance distance in EM 385-1-1 Table 16-1.

c. For lifts with anticipated binding conditions.

d. When erecting cranes.

1.7.3.3.1 Critical Lift Plan Planning and Schedule

Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the COR 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with EM 385-1-1, Section 16.F.03.

1.7.3.3.2 Lifts of Personnel

In addition to the requirements of EM 385-1-1, Section 16.H.02, for lifts of personnel, demonstrate compliance with the requirements of 29 CFR 1926.1400 and EM 385-1-1, Section 16.T.

1.7.3.4 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan

Multi-purpose machines, material handling equipment, and construction equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. Demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.7.3.5 Fall Protection and Prevention (FP&P) Plan

The plan must be in accordance with the requirements of EM 385-1-1, Section 21.D and ASSP Z359.2, be site specific, and address all fall hazards in the work place and during different phases of construction. Address how to protect and prevent workers from falling to lower levels when they are exposed to fall hazards above 1.8 m 6 feet. A competent person or qualified person for fall protection must prepare and sign the plan documentation. Include fall protection and prevention systems, equipment and methods employed for every phase of work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. Review and revise, as necessary, the Fall Protection and Prevention Plan documentation as conditions change, but at a minimum every six months, for lengthy projects, reflecting any changes during the course of construction due to changes in personnel, equipment, systems or work habits. Keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site for the duration of the project. Include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP).

1.7.3.6 Rescue and Evacuation Plan

Provide a Rescue and Evacuation Plan in accordance with EM 385-1-1 Section 21.N and ASSP Z359.2, and include in the FP&P Plan and as part of the APP. Include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

1.7.3.7 Hazardous Energy Control Program (HECP)

Develop a HECP in accordance with EM 385-1-1 Section 12, 29 CFR 1910.147, 29 CFR 1910.333, 29 CFR 1915.89, ASSP Z244.1, and ASSP A10.44. Submit this HECP as part of the Accident Prevention Plan (APP). Conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. Document this meeting and inspection in accordance with EM 385-1-1, Section 12.A.02. Ensure that each employee is familiar with and complies with these procedures.

1.7.3.8 Excavation Plan

Identify the safety and health aspects of excavation, and provide and prepare the plan in accordance with EM 385-1-1, Section 25.A and Section 31 00 00 EARTHWORK.

1.7.3.9 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium and chromium, and prepare in accordance with Section 02 83 00 LEAD REMEDIATION.

1.7.3.10 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00 ASBESTOS REMEDIATION. Follow the Asbestos Notification Standard Operating Procedure (SOP) developed by the DPW Environmental to provide proper notification to Ufficio Generale Prevenzion, Vigilanza Antinfortunistica e Tutela Ambientale (UGPREVATA). This is to allow the competent Italian Military Authority to verify the application of the rules and regulations related to the safety on the workplace, and in particular to the asbestos removal or abatement works that are planned and executed inside the footprint of the USAG-Italy Vicenza Military Community (VMC).

1.7.3.11 Site Safety and Health Plan

Identify the safety and health aspects, and prepare in accordance with Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.

1.7.3.12 Polychlorinated Biphenyls (PCB) Plan

Identify the safety and health aspects of Polychlorinated Biphenyls work, and prepare in accordance with Sections 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs) and 02 61 23 REMOVAL AND DISPOSAL OF PCB CONTAMINATED SOILS.

1.7.3.13 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION and referenced sources.

1.8 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the COR. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1, Section 1 or as directed by the COR. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented. AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.8.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.8.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.9 DISPLAY OF SAFETY INFORMATION

1.9.1 Safety Bulletin Board

Prior to commencement of work, erect a safety bulletin board at the job site. Where size, duration, or logistics of project do not facilitate a bulletin board, an alternative method, acceptable to the COR, that is accessible and includes all mandatory information for employee and visitor review, may be deemed as meeting the requirement for a bulletin board. Include and maintain information on safety bulletin board as required by EM 385-1-1, Section 01.A.07. Additional items required to be posted include:

a. Confined space entry permit.

b. Hot work permit.

1.9.2 Safety and Occupational Health (SOH) Deficiency Tracking System

Establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. Use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and updated daily, and provide the following information:

a. Date deficiency identified.

b. Description of deficiency.

c. Name of person responsible for correcting deficiency.

d. Projected resolution date.

e. Date actually resolved.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide emergency medical treatment.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Mishap Notification

Notify the COR as soon as practical, but no more than twenty-four hours, after any mishaps, including recordable accidents, incidents, and near misses, as defined in EM 385-1-1 Appendix Q, any report of injury, illness, or any property damage. For LHE or rigging mishaps, notify the COR as soon as practical but not more than four hours after mishap. The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical mishaps, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving. These mishaps must be investigated in depth to identify all causes and to recommend hazard control measures. Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any mishap.

1.12.2 Accident Reports

a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. Complete the applicable USACE Accident Report Form 3394, and provide the report to the COR within 5 calendar days of the accident.

b. Near Misses: For Army projects, report all "Near Misses" to the GDA, using local mishap reporting procedures, within 24 hrs. Near miss reports are considered positive and proactive Contractor safety management actions.

c. Conduct an accident investigation for any load handling equipment accident (including rigging accidents) to establish the root cause(s) of the accident. Complete the LHE Accident Report (Crane and Rigging Accident Report) form and provide the report to the COR within 30 calendar days of the accident. Do not proceed with crane operations until cause is determined and corrective actions have been implemented to the satisfaction of the COR.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.13 HOT WORK

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. cutting, welding, brazing, or soldering) or operating other flame-producing/spark producing devices. Contractors must schedule an appointment with Fire Prevention Office personnel (0444-715504) to inspect the area and issue the Hot work permit. Contractor is responsible to notify the Fire Department and the MP Desk/Dispatch Center (0444-71-5307) prior to any work that may affect operation of the fire protection system within the facility. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two 9 kg 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour after completion of the task or as specified on the hot work permit. When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Fire Department phone number. Ensure that contractors are familiar with on-post fire and emergency reporting requirements and phone numbers (0444-71-7117). Personnel should receive basic fire safety training. All fires must be reported even those that have been extinguished. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE FIRE DEPARTMENT IMMEDIATELY.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres. Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Section 06.H

1.14 RADIATION SAFETY REQUIREMENTS

Submit License Certificates, employee training records, and Leak Test Reports for radiation materials and equipment to the COR and Radiation Safety Office (RSO for all specialized and licensed material and equipment proposed for use on the construction project (excludes portable machine sources of ionizing radiation including moisture density and X-Ray Fluorescence (XRF)). Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property. Protect workers from radiation exposure in accordance with 10 CFR 20, ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.14.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray Radiography Operation Planning Work Sheet to COR 14 days prior to commencement of operations involving radioactive materials or radiation generating devices. For portable machine sources of ionizing radiation, including moisture density and XRF, use and submit the Portable Gauge Operations Planning Worksheet instead. The COR will review the submitted worksheet and provide questions and comments. Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.14.2 Site Access and Security

Coordinate site access and security requirements with the COR for all radiological materials and equipment containing ionizing radiation that are proposed for use on a government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Installation. The Government authorized representative will meet the Contractor at a designated location outside the Installation, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Installation, to the job site, and off the Installation. For portable machine sources of ionizing radiation, including moisture density and XRF, the Government authorized representative will meet the Contractor at the job site. Provide a copy of all calibration records, and utilization records for radiological operations performed on the site.

1.14.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the COR, RSO, and Base Security Department Emergency Number.

1.14.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with EM 385-1-1, Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet. Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the COR.

1.14.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in the Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or oneway doors, make no assumptions as to building occupancy. Where necessary, COR will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.14.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.14.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must not be initiated until after 5 p.m. on weekdays.

1.14.8 Transmitter Requirements

Adhere to the base policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.15 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with Section 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.15.1 Entry Procedures

Prohibit entry into a confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. Comply with EM 385-1-1, Section 34 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

1.15.2 Forced Air Ventilation

Forced air ventilation is required for all confined space entry operations and the minimum air exchange requirements must be maintained to ensure exposure to any hazardous atmosphere is kept below its action level.

1.15.3 Sewer Wet Wells

Sewer wet wells require continuous atmosphere monitoring with audible alarm for toxic gas detection.

1.15.4 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.16 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

a. Secure outside equipment and materials and place materials that could be damaged in protected areas.

b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.

c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, LgD 81/2008 and other related submittals and activity fire and safety regulations. The most stringent standard prevails. PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

a. Hard Hat

b. Long Pants

c. Appropriate Safety Shoes

d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, landline telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, leadbased paint, and hexavalent chromium, are prohibited. The COR, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further COR approval. Notify the Industrial Hygienist prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the COR immediately. Within 14 calendar days determine if the material is hazardous through lab test confirmation. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least 14 days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Section 11.A.02 (Isolation). Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Section 12.D. In accordance with EM 385-1-1, Section 12.A.01, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy. These activities include, but are not limited to, a review of HECP and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1, Section 11.A.02 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when deenergizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1, Section 12.A. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the COR, and the Installation representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1, Section 12.C. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1 Section 12, 29 CFR 1910.333, 29 CFR 1915.89, ASSP A10.44, NFPA 70E, and paragraph HAZARDOUS ENERGY CONTROL PROGRAM (HECP).

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the COR, and required by EM 385-1-1, Section 12.A.02. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Sections 05.I and 11.B. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Section 12.E.06.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, Section 12.E.08, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government confirming that steps of deenergization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with ASSP Z359.2 and EM 385-1-1, Sections 21.A and 21.D.

3.5.1 Training

Institute a fall protection training program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards and using personal fall protection equipment. Provide training by a competent person for fall protection in accordance with EM 385-1-1, Section 21.C. Document training and practical application of the competent person in accordance with EM 385-1-1, Section 21.C.04 and ASSP Z359.2 in the AHA.

3.5.2 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Section 21. Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 Section 21.I, 29 CFR 1926.500 Subpart M,ASSP Z359.0, ASSP Z359.1, ASSP Z359.2, ASSP Z359.3, ASSP Z359.4, ASSP Z359.6, ASSP Z359.7, ASSP Z359.11, ASSP Z359.12, ASSP Z359.13, ASSP Z359.14, ASSP Z359.15, ASSP Z359.16 and ASSP Z359.18.

3.5.2.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1, Sections 21.O through 21.O.06. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.2.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal Dring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 1633 kg 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 1.8 m 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma

Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Section 21.I.06.

3.5.3 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weightbearing capabilities for the projected loading.

a. Low Sloped Roofs:

(1) For work within 1.8 m 6 feet from unprotected edge of a roof having a slope less than or equal to 4:12 (vertical to horizontal), protect personnel from falling by the use of conventional fall protection systems (personal fall arrest/restraint systems, guardrails, or safety nets) in accordance with EM 385-1-1, Section 21 and 29 CFR 1926.500. A safety monitoring system is not adequate fall protection and is not authorized.

(2) For work greater than 1.8 m 6 feet from the unprotected roof edge, addition to the use of conventional fall protection systems the use of a warning line system is also permitted, in accordance with 29 CFR 1926.500 and EM 385-1-1, Section 21.L.

b. Steep-Sloped Roofs: Work on a roof having a slope greater than 4:12 (vertical to horizontal) requires a personal fall arrest system, guardrails with toe-boards, or safety nets. This requirement also applies to residential or housing type construction.

3.5.4 Horizontal Lifelines (HLL)

Provide HLL in accordance with EM 385-1-1, Section 21.I.08.d.2. Commercially manufactured horizontal lifelines (HLL) must be designed, installed, certified and used, under the supervision of a qualified person, for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 (29 CFR 1926.500). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

3.5.5 Guardrails and Safety Nets

Design, install and use guardrails and safety nets in accordance with EM 385-1-1, Section 21.F.01 and 29 CFR 1926 Subpart M.

3.5.6 Rescue and Evacuation Plan and Procedures

When personal fall arrest systems are used, ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. Prepare a Rescue and Evacuation Plan and include a detailed discussion of the following: methods of rescue; methods of self-rescue or assisted-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility. Include the Rescue and Evacuation Plan within the Activity Hazard Analysis

(AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must be in accordance with the requirements of EM 385-1-1, ASSP Z359.2, and ASSP Z359.4.

3.6 WORK PLATFORMS

3.6.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of

any scaffold braces or supports not specifically designed for access is prohibited. Comply with the following requirements:

a. Scaffold platforms greater than 6 m 20 feet in height must be accessed by use of a scaffold stair system.

b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 6 m 20 feet maximum in height.

c. An adequate gate is required.

d. Employees performing scaffold erection and dismantling must be qualified.

e. Scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.

f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.

g. Special care must be given to ensure scaffold systems are not overloaded.

h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.

i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (51 mm x 254 mm x 203 mm 2 in x 10 in x 8 in minimum) or other adequate firm foundation.

j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than 1.83 meters 6 feet.

k. Delineate fall protection requirements when working above 1.83 meters 6 feet or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

3.6.2 Elevated Aerial Work Platforms (AWPs)

Workers must be anchored to the basket or bucket in accordance with manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the CP). Lanyards used must be sufficiently short to prohibit worker from climbing out of basket. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying off to an adjacent pole or structure is not permitted unless a safe device for 100 percent tie-off is used for the transfer. Use of AWPs must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of AWPs must be designated as qualified operators by the Prime Contractor. Maintain proof of qualifications on site for review and include in the AHA.

3.7 EQUIPMENT

3.7.1 Material Handling Equipment (MHE)

a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.

b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.

c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.7.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systemsengineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the activity but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

a. Equip cranes and derricks as specified in EM 385-1-1, Section 16.

b. Notify the COR 15 working days in advance of any LHE entering the activity, in accordance with EM 385-1-1, Section 16.A.02, so that necessary quality assurance spot checks can be coordinated. Contractor's operator must remain with the crane during the spot check. Rigging gear must be in accordance with OSHA, ASME B30.9 Standards and host country safety standards.

c. Comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. Perform erection under the supervision of a designated person (as defined in ASME B30.5). Perform all testing in accordance with the manufacturer's recommended procedures.

d. Comply with ASME B30.5 for mobile and locomotive cranes, ASME B30.22 for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.8 for floating cranes and floating derricks, ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices and ASME B30.26 for rigging hardware.

e. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 Section 11, and ASME B30.5 or ASME B30.22 as applicable.

f. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the COR. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.

g. Inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.

h. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.

i. Use cribbing when performing lifts on outriggers.

j. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.

k. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching or crushing personnel.

l. Maintain inspection records in accordance by EM 385-1-1, Section 16.D, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the COR.

m. Maintain written reports of operational and load testing in accordance with EM 385-1-1, Section 16.F, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the COR.

n. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).

o. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 9 m/s 20 mph, the operator, rigger and lift supervisor must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.

p. Follow FAA guidelines when required based on project location.

3.7.3 Machinery and Mechanized Equipment

a. Proof of qualifications for operator must be kept on the project site for review.

b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

3.7.4 Base Mounted Drum Hoists

a. Operation of base mounted drum hoists must be in accordance with EM 385-1-1 and ASSP A10.22.

b. Rigging gear must be in accordance with applicable ASME/OSHA standards.

c. When used on telecommunication towers, base mounted drum hoists must be in accordance with TIA-1019, TIA-222, ASME B30.7, 29 CFR 1926.552, and 29 CFR 1926.553.

d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in accordance with EM 385-1-1 Section 16.B.05 and trained, at a minimum, in accordance with EM 385-1-1 Section 16.U and 16.T. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.

e. Material and personnel must not be hoisted simultaneously.

f. Personnel cage must be marked with the capacity (in number of persons) and load limit in kg pounds.

g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.8 EXCAVATIONS

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.8.1 Utility Locations

Provide a third party, independent, private utility locating company to positively identify underground utilities in the work area in addition to any station locating service and coordinated with the station utility department.

3.8.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within one meter 3 feet of the underground system.

3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.9 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Sections 11 and 12.

3.9.1 Conduct of Electrical Work

As delineated in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the COR. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method. When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.147.

3.9.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA.

Journeyman/Apprentice ratio must be in accordance with State, Local and Host Nation requirements applicable to where work is being performed.

3.9.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E. All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.9.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1. Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

3.9.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

08/15, CHANGE 2: 08/21

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data G

Warranty Management Plan;

Warranty Information;

Warranty Tags;

Spare Parts Data;

SD-08 Manufacturer's Instructions G

SD-10 Operation and Maintenance Data G

O&M Database;

Training Plan;

Training Outline;

Training Content;

SD-11 Closeout Submittals G

Training Video Recording;

Validation of Training Completion;

1.2 OPERATION AND MAINTENANCE DATA

Submit Operation and Maintenance (O&M) Data for the provided equipment, product, or system, defining the importance of system interactions, troubleshooting, and long-term preventive operation and maintenance. Compile, prepare, and aggregate O&M data to include clarifying and updating the original sequences of operation to as-built conditions. Organize and present information in sufficient detail to clearly explain O&M requirements at the system, equipment, component, and subassembly level. Include an index preceding each submittal. Submit in accordance with this section and Section 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES and Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES.

1.2.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.2.2 Package Content

Provide data package content in accordance with paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES. Provide a Data Package 1, 2, 3, 4 or 5 as required in Part 3. Use data package 3, 4, or 5 for commissioned and/or tested items. Comply with the data package requirements specified in the individual technical sections, including the content of the packages and addressing each product, component, and system designated for data package submission, except as follows.

1.2.3 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the COR for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

1.2.4 Commissioning Authority Review and Approval

If a Commissioning Authority (CxA) is required in Part 3, submit the commissioned systems and equipment submittals to the CxA to review for completeness and applicability. Obtain validation from the CxA that the systems and equipment provided meet the requirements of the Contract documents and design intent, particularly as they relate to functionality, energy performance, water performance, maintainability, sustainability, system cost, indoor environmental quality, and local environmental impacts. The CxA communicates deficiencies to the COR. Submit the O&M manuals to the COR or to the Quality Management Specialist upon a successful review of the corrections, and with the CxA recommendation for approval and acceptance of these O&M manuals. This work is in addition to the normal review procedures for O&M data.

1.3 O&M DATABASE

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program. As a minimum, provide list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.4 OPERATION AND MAINTENANCE MANUAL FILE FORMAT

Assemble data packages into electronic Operation and Maintenance Manuals. Assemble each manual into a composite electronically indexed file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) or via DoD SAFE as appropriate, so that each one contains operation, maintenance and record files, and training videos. Include a complete electronically linked operation and maintenance directory.

1.4.1 Organization

Bookmark Product and Drawing Information documents using the current version of CSI MasterFormat numbering system, and arrange submittals using the specification sections as a structure. Use CSI MasterFormat and UFGS numbers along with descriptive bookmarked titles that explain the content of the information that is being bookmarked.

1.4.2 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

a. Building Number

b. Project Title

c. Activity and Location

d. Construction Contract Number

e. Prepared For: (Contracting Agency)

f. Prepared By: (Name, title, phone number and email address)

g. Include the disk content on the disk label

h. Date

i. Virus scanning program used

1.5 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES

The following are a detailed description of the data package items listed in paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES.

1.5.1 Operating Instructions

Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.5.1.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS. Provide recommended safeguards for each identified hazard.

1.5.1.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.5.1.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.5.1.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

1.5.1.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.5.1.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.5.1.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.5.1.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

1.5.1.9 Equipment Software

Provide full as-built print out of equipment software program, and provide also CD/DVD copy of software and configuration applied to the equipment (lights dimmer software, VFD programming setup, etc.).

1.5.1.10 Additional Requirements for HVAC Control Systems

Provide Data Package 5 and the following for control systems:

a. Narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.

b. Full as-built sequence of operations.

c. Copies of checkout tests and calibrations performed by the Contractor (not Cx tests).

d. Full points list. Provide a listing of rooms with the following information for each room:

(1) Floor

(2) Room number

(3) Room name

(4) Air handler unit ID

(5) Reference drawing number

(6) Air terminal unit tag ID

(7) Heating or cooling valve tag ID

(8) Minimum cfm

(9) Maximum cfm

e. Full print out of all schedules and set points after testing and acceptance of the system.

f. Full as-built print out of software program and CD/DVD copy of software and configuration applied.

g. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.5.2 Preventive Maintenance

Provide the following information for preventive and scheduled maintenance to minimize repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.5.2.1 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

a. A table showing recommended lubricants for specific temperature ranges and applications.

b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities.

c. A Lubrication Schedule showing service interval frequency.

1.5.2.2 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including craft requirements by type of craft. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

a. Define the anticipated time required to perform each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.

b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.5.3 Repair

Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.5.3.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.5.3.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.5.3.3 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards.

1.5.3.4 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.5.3.5 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.5.3.6 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.5.4 Appendices

Provide information required below and information not specified in the preceding paragraphs but pertinent to the maintenance or operation of the product or equipment. Include the following:

1.5.4.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction. Submit the warranty management plan in electronic format together with all other Close-out documents, as required in Section 01 78 00. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. The construction warranty period must begin on the Beneficial Occupancy Date (BOD) and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance with the Contractor, KO and the Customer Representative. The warranty management plan must include, but is not limited to, the following:

a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or suppliers involved.

b. For each warranty, the name, address, telephone number, and e-mail of each of the guarantor's representatives nearest to the project location.

c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.

d. As-Built Record of Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:

(1) Name of item.
(2) Model and serial numbers.
(3) Location where installed.
(4) Name and phone numbers of manufacturers or suppliers.

(5) Names, addresses and telephone numbers of sources of spare parts.
(6) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.
(7) Cross-reference to warranty certificates as applicable.
(8) Starting point and duration of warranty period.
(9) Summary of maintenance procedures required to continue the warranty in force.
(10) Cross-reference to specific pertinent Operation and Maintenance manuals.
(11) Organization, names and phone numbers of persons to call for warranty service.
(12) Typical response time and repair time expected for various warranted equipment.

e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.

f. Procedure and status of tagging of equipment covered by warranties longer than one year.

g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.5.4.2 Contractor's Response to Construction Warranty Service Requirements

Following written notification by the COR, respond to construction warranty service requirements in accordance with the "Construction Warranty Service Priority List" and the three categories of priorities listed below. Submit a report on any warranty item that has been repaired during the warranty period. Include within the report the cause of the problem, date reported, corrective action taken, and when the repair was completed. If the Contractor does not perform the construction warranty within the timeframe specified, the Government will perform the work and back charge the construction warranty payment item established.

a. First Priority Code 1. Perform onsite inspection to evaluate situation, and determine course of action within 4 hours, initiate work within 6 hours and work continuously to completion or relief.

b. Second Priority Code 2. Perform onsite inspection to evaluate situation, and determine course of action within 8 hours, initiate work within 24 hours and work continuously to completion or relief.

c. Third Priority Code 3. All other work to be initiated within 3 work days and work continuously to completion or relief.

d. The "Construction Warranty Service Priority List" is as follows:

Code 1-Life Safety Systems
(1) Fire suppression systems.
(2) Fire alarm system(s) in place in the building.

Code 1-Air Conditioning Systems
(1) Recreational support.
(3) Air conditioning leak in part of building, if causing damage.
(4) Air conditioning system not cooling properly.

Code 1-Doors
(1) Overhead doors not operational, causing a security, fire, or safety problem.
(2) Interior, exterior personnel doors or hardware, not functioning properly, causing a security, fire, or safety problem.

Code 3-Doors
(1) Overhead doors not operational.
(2) Interior/exterior personnel doors or hardware not functioning properly.

Code 1-Electrical
(1) Power failure (entire area or any building operational after 1600 hours).
(2) Security lights
(3) Smoke detectors

Code 2-Electrical
(1) Power failure (no power to a room or part of building).
(2) Receptacle and lights (in a room or part of building).

Code 3-Electrical
(1) Street lights.

Code 1-Gas
(1) Leaks and breaks.
(2) No gas to family housing unit or cantonment area.

Code 1-Heat
(1) Area power failure affecting heat.
(2) Heater in unit not working.

Code 2-Kitchen Equipment
(1) Dishwasher not operating properly.
(2) All other equipment hampering preparation of a meal.

Code 1-Plumbing
(1) Hot water heater failure.
(2) Leaking water supply pipes.

Code 2-Plumbing
(1) Flush valves not operating properly.
(2) Fixture drain, supply line to commode, or any water pipe leaking.
(3) Commode leaking at base.

Code 3-Plumbing
(1) Leaky faucets

Code 3-Interior
(1) Floors damaged.
(2) Paint chipping or peeling.
(3) Casework.

Code 1-Roof Leaks
Temporary repairs will be made where major damage to property is occurring.

Code 2-Roof Leaks
Where major damage to property is not occurring, check for location of leak during rain and complete repairs on a Code 2 basis.

Code 2-Water (Exterior) No water to facility.

Code 2-Water (Hot)
No hot water in portion of building listed.

Code 3-All other work not listed above.

1.5.4.3 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Include warranty information for primary components of the system.

1.5.4.4 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid.

1.5.4.5 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the COR. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit within the O&M Manual record copy of the warranty tag showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	
Construction Contractor	
Address	
Telephone number	
Warranty contact	
Address	
Telephone number	
Warranty response time priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

1.5.4.6 SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

a. Indicate manufacturer's name, part number, and stock level required for test and balance, pre-commissioning, maintenance and repair activities. List those items that may be standard to the normal maintenance of the system.

b. At acceptance of Commissioning, if required in Part 3, ensure the required stock level is supplied as indicated in subparagraph a for maintenance and repair activities through the facilities warranty period.

c. Provide 2% of spare parts for each item of inventory. Provision of spare parts does not relieve the Contractor of responsibilities listed under the contract guarantee provisions.

1.5.4.7 O&M submittal data

Provide SD-10 Operation and Maintenance Data submittals.

1.5.4.8 Parts Identification

Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master parts catalog.

1.5.4.9 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.5.4.10 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.5.4.11 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.5.4.12 Contractor Information

Provide a list that includes the name, address, and telephone number of the Prime Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.6 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in Part 3 and in individual technical sections. The information required in each type of data package follows:

1.6.1 Data Package 1

a. Safety precautions and hazards
b. Cleaning recommendations
c. Maintenance and repair procedures
d. Warranty information
e. Extended warranty information
f. Contractor information
g. Spare parts and supply list

1.6.2 Data Package 2

a. Safety precautions and hazards
b. Normal operations
c. Environmental conditions
d. Lubrication data
e. Preventive maintenance plan, schedule, and procedures
f. Cleaning recommendations
g. Maintenance and repair procedures
h. Removal and replacement instructions
i. Spare parts and supply list
j. Parts identification
k. Warranty information
l. Extended warranty information
h. Contractor information

1.6.3 Data Package 3

a. Safety precautions and hazards
b. Operator prestart
c. Startup, shutdown, and post-shutdown procedures
d. Normal operations
e. Emergency operations
f. Environmental conditions
g. Operating log
h. Equipment software
i. Lubrication data
j. Preventive maintenance plan, schedule, and procedures
k. Cleaning recommendations
l. Troubleshooting guides and diagnostic techniques
m. Wiring diagrams and control diagrams

n. Maintenance and repair procedures

o. Removal and replacement instructions
p. Spare parts and supply list
q. Warranty Management Plan
r. O&M submittal data
s. Parts identification
t. Warranty information
u. Extended warranty information
v. Testing equipment and special tool information
w. Testing and performance data
x. Contractor information
y. Field test reports

1.6.4 Data Package 4

a. Safety precautions and hazards
b. Operator prestart
c. Startup, shutdown, and post-shutdown procedures
d. Normal operations
e. Emergency operations
f. Operator service requirements
g. Environmental conditions
h. Operating log
i. Equipment software
j. Lubrication data
k. Preventive maintenance plan, schedule, and procedures
l. Cleaning recommendations
m. Troubleshooting guides and diagnostic techniques
n. Wiring diagrams and control diagrams
o. Repair procedures
p. Removal and replacement instructions
q. Spare parts and supply list
r. Repair work-hours
s. Warranty Management Plan
t. O&M submittal data
u. Parts identification
v. Warranty information
w. Extended warranty information
x. Personnel training requirements
y. Testing equipment and special tool information
z. Testing and performance data aa. Contractor information bb. Field test reports

1.6.5 Data Package 5

a. Safety precautions and hazards

b. Operator prestart
c. Start-up, shutdown, and post-shutdown procedures
d. Normal operations
e. Environmental conditions

f. Preventive maintenance plan, schedule, and procedures
g. Troubleshooting guides and diagnostic techniques
h. Wiring and control diagrams
i. Maintenance and repair procedures
j. Removal and replacement instructions
k. Spare parts and supply list
l. Warranty Management Plan
m. Manufacturer's instructions
n. O&M submittal data
o. Parts identification
p. Equipment software
q. Testing equipment and special tool information
r. Warranty information
s. Extended warranty information
t. Testing and performance data
u. Contractor information
v. Field test reports
w. Additional requirements for HVAC control systems

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the submitted Operation and Maintenance Manual. Training must include classroom or field lectures based on the system operating requirements. The location of classroom training requires approval by the COR.

3.1.1 Training Plan

Submit a written training plan to the COR for approval at least 20 calendar days prior to the scheduled training. Training plan must be approved by the QCM and, if Commissioning is required in Part 3, by the CxA prior to forwarding to the COR. Also, coordinate the training schedule with the COR, QCM and CxA. Include within the plan the following elements:

a. Equipment included in training.

b. Intended audience.

c. Location of training.

d. Dates of training.

e. Objectives.

f. Outline of the information to be presented and subjects covered including description.

g. Start and finish times and duration of training on each subject.

h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts).

i. Instructor names and instructor qualifications for each subject.

j. List of texts and other materials to be furnished by the Contractor that are required to support training.

k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The QCM and the CxA, if Commissioning is required in Part 3, are responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.

b. Relevant health and safety issues.

c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.

d. Design intent.

e. Use of O&M Manual Files.

f. Review of control drawings and schematics.

g. Interactions with other systems.

h. Special maintenance and replacement sources.

i. Tenant interaction issues.

3.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF) and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the COR two copies of the training session(s) in DVD video recording format. Provide one copy of the training session(s) to the O&M Manual Preparer for inclusion into the electronic O&M Manual's documentation. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the COR for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the QCM and, if Commissioning is required in Part 3, with the CxA in accordance with section 01 45 00.00 10 QUALITY CONTROL

-- End of Section --

SECTION 01 78 00

CLOSEOUTSUBMITTALS

05/19, CHANGE 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-1-2909 (2012) Engineering and Design -- Geospatial Data

and Systems

ERDC/ITL TR-19-6 (2019) A/E/C Graphics Standard, Release 2.1

ERDC/ITL TR-19-7 (2019) A/E/C CAD Standard - Release 6.1

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N (2014; with Change 4, 2018) Navy and Marine

Corps Design

UFC 1-300-08 (2009; with Change 2, 2011) Criteria for Transfer

and Acceptance of DoD Real Property

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site. These files serve as the basis for the creation of the record drawings.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings. Produce the record drawings from the Record Model(s) and do not include annotations indicating revisions.

1.2.3 Record Model

A model reflecting approved changes during construction including red-lines, requests for information (RFI's), and contract modifications. Include updated construction phase facility/site data for components.

1.2.4 Advanced Modeling

A subset of geospatial technologies as defined in EM 1110-1-2909 to include Building Information Modeling (BIM), Civil Information Modeling (CIM), Geographic Information Systems (GIS), and Computer-Aided Design (CAD). Advanced modeling is comprised of models and drawings that form a digital representation of the project, or part thereof, that are comprised of model elements with facility data.

1.2.5 USACE CAD/BIM Technology Center

The USACE CAD/BIM Technology Center hosts all standard content for USACE. This content can be accessed through the CAD/BIM Technology Center website, https://cadbimcenter.erdc.dren.mil/.

1.3 SOURCE DRAWING FILES

The full set of design drawings, in the source format, are the source drawing files for Record Drawing preparation.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only:

SD-03 Product Data G

Products (datasheets of the used materials and installed technical devices);

Sortable Submittal Register;

SD-06 Test Reports G

SD-07 Certificates G

Refer to Part 3 Statement of Work/Project Program

SD-11 Closeout Submittals G

As-Built Drawings;

Record Drawings;

Geo-Data-Base Files;

Excel Spreadsheet “Fire Detection and Suppression System Schedule.xlsx” (FDSS);

Building Evacuation Maps;

Searchable electronic copy of the photo documentation of QC Daily Reports;

High Performance and Sustainable Building (HPSB) Checklist;

Record Model;

1.5 QUALITY CONTROL

Additions and corrections to the contract drawings must be equal in quality and detail to that of the originals. Line colors, line weights, lettering, layering conventions, and symbols must be the same as the original line colors, line weights, lettering, layering conventions, and symbols.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 GEO-DATA-BASE FILES

a. Source Data: GIS data shall be collected using differential GPS Field by means of nonrecreational GPS equipment, such as Trimble or Magellan. To ensure high quality/precision GPS data, conduct data collection when positional dilution of precision (PDOP) is 4 or less. In addition, the raw GPS data must undergo differential post-processing utilizing a permanent GPS base station with a high integrity index value (i.e., 70-100). The target accuracy for the resulting differentially corrected GPS field survey data is meter or sub-meter. Submit all GIS files in CD-ROM or DVD disks or DoD SAFE delivery in ArcGIS 10.4.1 or ArcGIS PRO
3.0.3 (.SHP, .GDB, .MXD),

b. Submit all GIS files in CD-ROM or DVD disks in ArcGIS 10.4.1 (or previous version) compatible format: at least one pre-final submittal is requested for review before the final delivery. GIS data will be submitted for review to all concerned SME, including the IGI&S Manager. No data will be accepted without IGI&S Manager QA/QC review and formal acceptance;

c. Installation geospatial data shall be provided in a Personal Geodatabase compliant with the "Spatial Data Standards for Facilities, Infrastructure, and Environment" (SDSFIE), version 4.0 Army Adaptation (https://www.sdsfieonline.org/);

d. Features shall be stored in GIS Personal Geodatabase according and compliant with "Army IGI&S Program - Geospatial Data Layer – Quality Assurance Plan (QAP)" version 4.0 (type of geometry, type of data, geographic representation, positional accuracy, attribute accuracy, completeness, logical consistency, lineage.). All data shall be topologically sound and geometrically correct. This includes no null or empty features, no non-simple features and no duplicate features. All data shall meet the basic topology rule set for installation geospatial data. Exceptions to the topology rules are possible. In case of an exception, a justification must be provided in the data layer documentation. The following rules shall be obeyed: a. Point features:

1. Must be located inside polygons of parent feature class;

b. Line features:
1. Must not self-overlap;
2. Must not self-intersect;
3. Must be single part;
4. Must not have pseudo-nodes;
5. Must not have dangles;

c. Polygon features:
1. Must not overlap;
2. Must not have gaps

If no QAP exists for the data layers developed, Contractor shall comply also with the last point of this paragraph (see below);

The bounding coordinates to capture the north, south, east, and west-most spatial extents of the geospatial data layers will be based on Universal Transverse Mercator (UTM) Zone, meters. At a minimum, feature types within the database should have a spatial reference with a precision of 1,000. The horizontal datum to be utilized is World Geodetic System 1984 (WGS84), and the vertical datum shall be Mean Sea Level (MSL) as determined by the Earth Gravitational Model 2008 (EGM2008) Geoid. For mapping purposes, WGS 1984 UTM Zone 32N, meter, is the defined projection;

For each applicable Data Layer, Attributes Table shall be populated, at the minimum, with the "Required Populated Attributes" listed in the specific Chapter (usually 1.3 “Army Adaptation Attribute Table” and related table) of the QAPs, version 4.4 or newer. If additional Attribute Field are requested as part of the Contract, they will be specified in a specific section of the present Contract. If the additional Fields are not part of the SDSFIE 4.0 Army Adaption schema, they will be provided separately (i.e. with a spreadsheet properly joined with the Data Layer through a convenient key link; i.e. a separate business table with the custom data may be created, using the data layer primary key as the join attribute);

Each data layer shall be accompanied by metadata conforming to the "Metadata Requirements for Army IGI&S Geospatial Data". Populate the metadata for each feature type to meet the minimum requirements contained in the "Metadata Requirements for Army IGI&S Geospatial Data".

Digital copy of the applicable QAPS, of the "Metadata Requirements for Army IGI&S Geospatial Data" and of all other needed Documents can be asked to IGI&S Manager or to the COR. Empty schema of SDSFIE 4.0 Army Adaption can be requested to IGI&S

Manager. Only if the nature of the Contract requires it, existing USAG ITALY GIS data will be provided to the Contractor as a basis for his/her work. The Contractor will be supplied with a full SDSFIE 4.0 Army Adaptation schema GDB, with all the Data Layers empty except the ones required (if there are) to allow the Contractor to perform the task;

If no QAP exists for the data layers developed, the following rules must be obeyed, in addition to the directions given above, and a specific intermediate (or at least pre-final) submission shall be submitted to IGI&S Manager and involved SMEs and, only after approval, the Data Layers will be accepted. Data Layers without QAPs but included in the SDSFIE 4.0 Army Adaption, will be included in the Geodatabase. Data Layers without QAPs and not part of the SDSFIE 4.0 Army Adaptation, will be delivered as single shapefiles or, if they are more than one, they can be stored and submitted together in a Geodatabase, separate from the official SDSFIE 4.0 Army Adaption, but still compliant with all the technical directions listed in this paragraph. Additional Minimum Requirements for Data Layers without an Established Quality Assurance Plan. If a geospatial data layer contains a discriminator per SDSFIE v4.0, the discriminator must be populated. All features shall be attributed with the Installation Code from the Headquarters Installation Information System (HQIIS). The FGDC National Standard for Spatial Data Accuracy (NSSDA) shall be used to evaluate and report the positional accuracy of all data layers submitted.

3.2 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF design drawings for As-Built Drawings. Maintain the As-Builts throughout construction as red-lined hard copies on site. Submit As-Built Drawings for COR's review before the date of the Final Inspection.

3.2.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built Drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

a. Use base colors of red, green, and blue. Color code for changes as follows:
   (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.
   (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
   (3) Additions (Green) - Added items, lettering in notes and leaders.

b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.

c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.

d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.

e. Use legible lettering and precise and clear values when marking prints. Clarify ambiguities concerning the nature and application of change involved.

f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.

g. For deletions, cross out all features, data and captions that relate to that revision.

h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.

i. The revision delta size must be about 8 mm 5/16 inch unless the area where the delta is to be placed is crowded. Use a smaller size delta for crowded areas.

j. Place a revision delta at the location of each deletion.

k. For new details or sections which are added to a drawing, place a revision delta by the detail or section title.

l. For minor changes, place a revision delta by the area changed on the drawing (each location).

m. For major changes to a drawing, place a revision delta by the title of the affected plan, section, or detail at each location.

n. For changes to schedules or drawings, place a revision delta either by the schedule heading or by the change in the schedule.

o. Indicate one of the following when attaching a print or sketch to a markup print:
   (1) Add an entire drawing to contract drawings
   (2) Change the contract drawing to show
   (3) Provided for reference only to further detail the initial design.

p. Incorporate all shop and fabrication drawings into the markup drawings.

3.2.2 As-Built Drawings Content

Keep these working as-built markup drawings current on a weekly basis and at least one set available on the jobsite at all times. Changes from the design drawings which are made during construction or additional information which might be uncovered in the course of construction must be accurately and neatly recorded as they occur by means of details and notes. For failure to maintain the working As-Built Drawings as specified herein, the KO will withhold 10 percent of the progress payment until approval of updated drawings. Show on the As-Built Drawings, but not limited to, the following information:

a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the Record Drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.

b. If the source drawings show utility lines not involved in this Contract, resulting from previous Record Drawings given to the Contractor for information only, clearly distinguish these unverified utility lines by marking them with different colors or line types.

c. The location and dimensions of any changes within the building structure.

d. Layout and schematic drawings of electrical circuits and piping.

e. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.

f. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.

g. The topography, invert elevations and grades of drainage installed or affected as part of the project construction.

h. Changes or Revisions which result from the final inspection.

i. Where contract drawings or specifications present options, show only the option selected for construction on the working as-built markup drawings.

j. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.

k. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.

l. Changes in location of equipment and architectural features.

m. Modifications (include within change order price the cost to change working as-built markup drawings to reflect modifications) and compliance with FC 1-300-09N procedures.

n. Actual location of anchors, construction and control joints, etc., in concrete.

o. Unusual or uncharted obstructions that are encountered in the contract work area during construction.

p. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.2.3 As-Built Re-Submission Requirements

If elements of an As-Built submittal are rejected by the COR, provide the following for each re-submission, in addition to any information required in Sections 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES and 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES:

a. Re-submit only the rejected elements in response to COR's comments.

b. Provide a copy of all COR's review comments.

c. Provide a disposition/response to each COR's review comment for a back-check of the re-submission deliverable.

3.3 RECORD DRAWINGS

Prepare and provide Record Drawings in accordance with FC 1-300-09N. Transfer the changes from the approved working as-built markup drawings to the original electronic CAD source drawings, and add such additional drawings as may be necessary. Do not include Previous Conditions, Demolition Plans and Construction Site Plans in the Record drawings.
Prepare the additional drawings using the specified electronic file format applying the same graphic standards specified for design drawings. The title block and drawing border to be used for any new Record Drawing must be identical to that used on the design drawings. If a sheet needs to be added between two sequential sheets, append a Supplemental Drawing Designator. Accomplish additions and corrections to the contract drawings using CAD files. Jointly review the working as-built markup drawings with printouts from PDF working Record Drawing files for accuracy and completeness. Provide all program files and hardware necessary to prepare final PDF Record Drawings.

3.3.1 CAD Drawing files

a. Prepare the CAD drawing files in AutoCAD Release 2018 format compatible with a Windows 10 operating system.

b. When final revisions have been completed by the Contractor's Quality Control Manager (QCM), show the wording "RECORD DRAWING AS-BUILTS" followed by the name of the Contractor in letters at least 5 mm 3/16 inch high on the cover sheet drawing. In addition, copy initials and dates from the COR reviewed As-Built Drawings to the title block of the Record Drawings. Date "RECORD DRAWING AS-BUILTS" drawing revisions in the revision block.

c. The CAD files must be complete in all details and identical in form and function to the design CAD drawings. Make all changes on the layer/level as the original item. Make any transactions or adjustments necessary to accomplish this. The Government reserves the right to reject any drawing files it deems incompatible with the customer's CAD system. Drawing files and storage media submitted will become the property of the Government upon final approval.

3.3.2 Final Record Drawing Package

Provide one digital copy of PDF and CAD Record Drawings on CDs or DVDs or via DoD SAFE to the Quality Management Specialist (QMS) or to the COR within14 calendar days from COR approval of the As-Built drawings. Prepare AutoCAD files for transmittal using eTransmit or properly sorting them into folders, including XRef and PlotCfgs folders. The size shall be in accordance with contract drawings. The digital Record drawings shall be signed by the QCM of the Prime Contractor either with a digital signature, or with a scanned signature placed as image on the pdf files. In the second case, the Contractor shall protect all the pdf drawings from any possible change with a password. Failure to submit final record PDF and CAD drawing files as specified will be cause for withholding any payment due under this contract and preventing the acceptance of the facility. The QMS will review final CAD and PDF Record Drawings for accuracy and communicate to the Contractor the required corrections, changes, additions, and deletions. Approval and acceptance of final Record Drawings must be accomplished by the QMS before final payment is made.

DoD SAFE is a web-based tool that provides authenticated DoD CAC users and guests (unauthenticated users) the capability to securely send and receive large files, including files that are too large to be transmitted via email. The unauthenticated users need to ask an authenticated DoD CAC user for a link, to be allowed to upload documentation on DoD SAFE.

3.3.3 Record Drawings Re-Submission Requirements

If elements of a Record Drawing submittal are rejected by the QMS, provide the following for each re-submission, in addition to any information required in Sections 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES and 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES:

a. Re-submit only the rejected elements in response to QMS's comments.

b. Provide a copy of all QMS's review comments.

c. Provide a disposition/response to each QMS's review comment for a back-check of the re-submission deliverable.

3.4 FDSS EXCEL SPREADSHEET

Fill in the Excel Spreadsheet "Fire Detection and Suppression System Schedule.xlsx" (FDSS) provided as Attachment with the devices installed through this Project, considering that if a device was already present but was broken and was replaced with a new one, this device must not be put on the spreadsheet.

3.5 SORTABLE SUBMITTAL REGISTER

Provide an excerpt from the construction submittal register including all installed products, materials and systems furnished under this contract. This digital register in Excel format shall be sortable and filterable, in order to facilitate the consultation of the Products datasheets.

3.6 OPERATION AND MAINTENANCE (O&M) MANUALS

Provide project electronic O&M manuals as specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit them to the QMS or to the COR together with all other Close-out documemts. No hard copy is required.

3.7 REAL PROPERTY RECORD

In addition to the requirements set forth in this section, refer to RFP PART 2, General Requirements, Section 01 33 00.05 20 1.3, CONSTRUCTION SUBMITTAL PROCEDURES, near the completion of the project, but a minimum of 45 days prior to Beneficial Occupancy Date (BOD), provide to the COR any project specific information necessary to complete the DD FORM 1354. For convenience, a blank fillable PDF DD FORM 1354 may be obtained at the following link: https://www.esd.whs.mil/Portals/54/Documents/DD/forms/dd/dd1354.pdf

PART 4 PROCEDURES

4.1 FINAL ACCEPTANCE AUTHORITY

The Quality Management Specialist of DPW possesses final acceptance authority for Record Drawings, O&M Manuals and all other Close-out documents required by this Construction Contract, with the exception of the DD1354, BIM, CIM, and Geo-Data-Base Files. The process for receiving and acceptance of the DD1354 is under the authority of the Master Planning Real Property Specialist and the Real Property Accountability Officer. The process for receiving and acceptance of the Geo-Data-Base Files is under the authority of the IGI&S Manager.

4.2 DD1354, CHECKLIST FOR DD1354 AND HPSB CHECKLIST SUBMISSON

For the submission of Interim DD Form 1354, Checklist for DD FORM 1354 and High Performance and Sustainable Building (HPSB) Checklist refer to Sections 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES.

4.3 GEO-DATA-BASE FILES SUBMISSION

GIS data will be submitted for review to the IGI&S Manager. No data will be accepted without IGI&S Manager QA/QC review and formal acceptance. Submit all GIS files in CDROM or DVD disks or DoD SAFE delivery in ArcGIS 10.4.1 or ArcGIS PRO 3.0.3 (.SHP, .GDB, .MXD) compatible format: at least one pre-final submittal is requested for review 14 calendar days before the final delivery. The final delivery must be provided 14 calendar days prior to the Final Inspection.

4.4 AS-BUILT DRAWINGS SUBMISSION

The COR will review the Contractor-originated As-Built Drawings, in order to verify that the As-Built Drawings submitted truly reflect what was constructed. Submit the As-Built Drawings for COR’s review at least 30 calendar days prior to Final Inspection, and within 14 calendar days the COR will return them to the Contractor marked with the required revisions, if any. When the entire As-Builts package is deemed acceptable by the COR, the Contractor can prepare the Final Record Drawing package.

4.5 FINAL DOCUMENTATION SUBMISSION EXCEPT 4.2, 4.3, 4.4, and 4.5

4.5.1 Digital Documentation

Provide the complete final documentation in digital form on CDs or DVDs or using DoD SAFE to the QMS or to the COR within 14 calendar days of the COR approval of the As-Built drawings, to allow time to assess and catalog the submittals. Deliver the final documentation properly sorted in folders ("Record Drawings", "Dich. di Conformità & Certificates", "O&M Manuals", "Photos", "Technical Submittals", "Tests", "Warranty", and any other folders that are needed for the specific Project). ".rar" compressed files are not acceptable. Deliver the final documentation together with one digital ENG FORM 4025-R signed by the Contractor's QCM, with one line item for each type of Submittal. Upon reception of the documentation, the QMS or the COR will sign a copy of the 4025 Form, entering "F" (Receipt acknowledged) in the last column of this copy. No partial deliveries will be accepted by the QMS or by the COR.

The QMS will review the delivered documentation for accuracy, and within 14 calendar days after the delivery of the documentation will return to the Contractor the 4025 Form with indication of the required corrections, changes, additions, and deletions, if any. If required, the Contractor shall submit the missing documents and/or resubmit only the inaccurate documents with a new digital 4025 Form.

4.5.2 Paper Documentation

Only part of the final documentation shall be provided in paper form, and only after receiving the QMS approval of the complete digital documentation:

a. Submit one hard copy of each of the following documents, only if the electronic copy already provided and accepted is not digitally signed:

   (1) "Dichiarazioni di Conformità" as per Italian Decree 37 of 2008, signed and stamped in original (but the mandatory Design consisting of some drawings and a technical report, and the optional Attachments shall be submitted only in digital form, password-protected to avoid any possible change);

   (2) "Certificato di Collaudo Statico", "Relazione a Strutture Ultimate" and any other Technical Reports drawn up by Professionals registered in the National Professional Rolls of Italy, signed and stamped in original.

b. Submit one hard copy of each of the following documents, only when they have been submitted to the Italian Fire Brigade Command (VVF) in paper form. If instead they have been sent via certified email to the VVF or uploaded to the SUAP Portal, submit them only in digital form (".p7m" files):

   (1) "Istanza di valutazione progetto" (PIN 1) as per "DPR 1 agosto 2011, n.151" and "DM 7 agosto 2012" with the original VVF stamp for receipt acknowledgment (but the whole attached mandatory documentation shall be submitted only in digital form, password-protected to avoid any possible change), and "Valutazione Progetto" issued from VVF signed and stamped in original;

   (2) "Segnalazione Certificata di Inizio Attività" (PIN 2 - SCIA) as per "DPR 1 agosto

2011, n.151" and "DM 7 agosto 2012", and related PIN Forms with the original VVF stamp for receipt acknowledgment (but the whole attached mandatory documentation shall be submitted only in digital form, password-protected to avoid any possible change);

(3) "Rinnovo periodico di conformità antincendio" (PIN 3) as per "DPR 1 agosto 2011, n.151" and "DM 7 agosto 2012", and related PIN Forms with the original VVF stamp for receipt acknowledgment (but the whole attached mandatory documentation shall be submitted only in digital form, password-protected to avoid any possible change).

4.6 SUBMITTAL REVIEW SCHEDULE

Add additional time to the identified review time period to allow for the following scheduling conditions:

a. Submittals received after noon will be considered received on the following business day.

b. Federal holidays, including the period between Christmas and New Year's Day, will be considered non-working days for Government personnel in reviewing As-Built Drawings and Final Documentation submittals.

c. Postpone delivery if Government personnel to receive the submittal are unavailable. Assure in advance of the submittal delivery it can be received.

d. Postpone delivery when heightened security restricts access to the Base. Coordinate heightened security requirements in advance with the COR.

e. Period of review for a resubmittal is the same as the initial submittal. Review time for resubmittals caused by non-conformance, do not result in a change in contract duration or cost.

4.7 FINAL ACCEPTANCE

Acceptance and Approval of final documentation, both in digital and in paper form, must be granted by the QMS before final payment is made by the COR.

-- End of Section --